

Technical Supplement to Five Maximal Incoming Classes Suffice: Exact Certificates and Structural Constraints for Seymour’s Second-Neighbourhood Conjecture

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Abstract

This supplement provides the original deletion and partner estimates, the complete residual exclusions used in the manuscript, the additional closure and transport arguments, and the separately scoped dense odd-order tools. It records the finite verification contract and the status of earlier computational branches. All second neighbourhoods are exact. No full solution of Seymour’s conjecture is claimed.

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S1 Coverage, dependencies, and scope of the supplementary results

This supplement supplies the longer proofs and auxiliary results underlying the main article. It also records how the earlier special cases are covered by the final statements. All graphs are finite and nonempty, and all second neighborhoods are defined by directed distance *exactly* two. In particular, first neighbors are excluded from second neighborhoods. Write $g_D(v) = d_D^+(v) - d_D^{++}(v)$; thus a Seymour vertex has $g_D(v) \leq 0$, and an all-deficient graph has positive gap everywhere.

The hypotheses are part of each result. The residual arguments use a strongly connected all-deficient graph that is minimal under deletion of *every nonempty set of arcs*, on its fixed vertex set. We call this the MIN setting. Single-arc criticality alone is not the stated hypothesis. Sink-component arguments extend the numerical missing-pair bound to arc-set-minimal graphs without strong connectivity. The incoming-pattern certificate theorems and the basic closure theorems apply to arbitrary oriented graphs and do not require MIN. The class-cover and cloning normal forms instead use their explicitly specified minimum-order or lexicographic extremal choices. An intermediate closure graph, or a graph obtained by vertex deletion, is never assumed to inherit all-deficiency or minimality without a separate argument.

Incoming-pattern certificates. The main article proves the transfer from representative cores, its tournament case, the at-most-five maximal incoming-class theorem, the iterated core reduction, and the equivalence between universal pattern feasibility and the second-neighborhood conjecture. The accompanying integer tables and independent verifiers certify every core of order at most five. This bound concerns core order, not the total graph order.

Cloning and extremal forms. The main article gives simultaneous original-snapshot row cloning, the minimum-order whole-class gap-one predecessor cover, and the specified extremal true-twin normal form. Gap-one vertices here must not be confused with residual-one vertices. Classical gap-one predecessor and cycle facts are distinguished from their class-level refinements.

Residual foundation. The supplement supplies the external-boundary deletion lemma, target packet, $\eta \geq 0$, $\sum \eta = 2M - n$, zero-residual protection, unit residual budgets, and the shared exceptional-partner remainder. These are the local inputs to the MIN statements in the article.

Completion and residual margin. Canonical convenient orientations, witness transfer, safe retention, two-positive-support exclusion and the fixed heavy-cover criterion lead to the one-exception theorem. The remaining mixed cases $(m, 2, 1)$, $(2, 2, 2)$, and $(2, 2, 1, 1)$ are proved here in full. Together they yield the article's parity-sensitive bound on missing pairs. All comparisons of completed gaps are made in the specified completion.

Closure and equality. The main article develops iterated forced and single weak protection steps, exact original feed neighborhoods, strict far-head surplus, and the original partner-cycle and seed-path conclusions. The supplement retains the one-round antecedents,

packet refinements, low-residual equality shapes, near-whole feed boundary bound, and the separately scoped additional safe operator.

Additional transfer tools. Complete supplementary arguments cover all-keeper and overlap budgets, multiple retained cross-witness arcs, flexible exception-clique feed reduction, arbitrary outgoing-star deletion identities, and the gap-two witness multiplicity consequences. These have their own local hypotheses and do not assert universal completion existence.

Dense odd boundary. At $n = 2\delta + 3$, the supplement includes the target-capacity core, minimum-degree predecessor bounds, even-order and odd-deletion identities, certified basins, heavy-set capacity, and the degree- and residual-dependent four-heavy restrictions. These are independent auxiliary consequences rather than a solution of the dense boundary.

Obstructions and verification. Cyclic amplification, exact weighted transport, its dual formulation, and explicit controls document limits of proposed shortcuts. The artifact index distinguishes exact integer certificate checking, finite local tests, solver-dependent infeasibility logs, and heuristic search outcomes. None of the control graphs is a counterexample to the second-neighborhood conjecture.

The short dependency chain for the numerical bound. In MIN, at least three vertices have positive residual, and the one-exception theorem requires at least two *actual* bad-direction failure sources with residual at least two. Consequently the only positive residual partitions with total at most six are

$$(2, 2, 1), \quad (3, 2, 1), \quad (2, 2, 2), \quad (2, 2, 1, 1).$$

The general $(m, 2, 1)$ theorem eliminates the first two, and the other two exclusions eliminate the remainder. Hence

$$R = 2M - n \geq 7, \quad M \geq \left\lceil \frac{n}{2} \right\rceil + 3 + \mathbf{1}_{\{n \text{ even}\}}.$$

The even-order residual is at least eight by parity. The sink-component extension adds the stated penalty for vertices outside a sink component; equality in the uniform bound forces strong connectivity. These bounds are assertions about arc-set-minimal counterexamples. Since arc deletion increases the number of missing pairs, they do not directly prove the conjecture for arbitrary denser graphs.

Earlier exclusions and their final disposition. The unrestricted one-exception theorem excludes all allocations $(m, 1^k)$, for every $m \geq 2$ and finite k , and also covers the pure-unit situation. It therefore subsumes the earlier finite unit-support thresholds, the standalone all-unit exclusion, and the cases $(m, 1, 1)$, $(m, 1, 1, 1)$, and $(2, 1, 1, 1, 1)$. The complete two-support theorem supersedes its smaller residual and restricted-order cases. The final numerical bound supersedes the successive exclusions of residuals zero through five and the older $n = 19, \delta = 8$ conclusions $M \geq 6$ and $M \geq 7$; in that MIN case the present bound is $M \geq 13$. Useful local lemmas from these arguments remain available, including full unit packets, unit-unit witness tails, simultaneous two-vertex priorities, and the matching-module deletion identities. They are not counted again as independent improvements of the final bound.

Zero gap and strong-feed equality are distinct branches. For the terminal FORCED/WEAK construction, a positive-original-gap feed always satisfies $t_D(f) \geq g_D(f) + 1$. If its completed gap is zero, the remaining partner set has size $m \geq 3$, contains an original directed cycle, and gives $t_D(f) \geq g_D(f) + m$. In MIN this implies $\eta_D(f) \geq m + 1 \geq 4$. Equality at residual four has the stated necessary shape $m = 3, k = s = 0, g_D = 1, t_D = 4$, with nonempty original $P(f)$ and a saturated full packet. The positive-indegree assertion belongs to the *terminal* partner graph; only cycle existence is lifted to the original graph. These partner vertices are not thereby shown to be actual failure sources or to have positive residual.

At strong Good equality with negative completed gap, the corrected partner-path theorem applies instead: residual two permits only original nongood-second partners, while residual three permits at most one far partner, reached by an original arc from a nongood-second partner. That far partner may be exceptional in the packet. Neither the all-second case nor strict Good surplus has been excluded. Thus the five remaining residual-seven partitions

$$(3, 3, 1), (3, 2, 2), (3, 2, 1, 1), (2, 2, 2, 1), (2, 2, 1, 1, 1)$$

are forced into negative completed gap, but remain unresolved.

The separate dense-boundary parameters. In the chapters restricted to $n = 2\delta + 3$, σ denotes target-capacity surplus and $K = \{\sigma > 0\}$ its core. It is not η , and a heavy target $\sigma \geq 2$ need not be an actual bad-direction failure source. The general core conclusions are $|K| \geq 6$ and $z \geq h + 8$, where z counts whole minimum-degree vertices and h counts gap-two vertices. The stronger $z \geq h + 10$ requires $\delta \geq 10$; the $h + 11$ conclusion has an additional degree-residual condition, for example $\delta > 40R + 12$. The predecessor inequalities and odd-deletion bounds retain their own hypotheses. In particular, a retained minimum-degree condition is not asserted to hold for every chosen odd set.

Corrections and evidence boundaries. The saved proofs do not use the rejected assertions that the external second boundary of a target packet always lies in its incoming set, that an exceptional far head cannot belong to the remainder, or that the released-direction completion family contains the canonical family as a set. The proved comparison is of maximum ordinary forward score. At a fixed tournament the normalized Fisher losing density coincides with the normalized reverse-expansion density; these do not provide independent choices of weights. Three inward-priority families need not be compatible. The enlarged convenient-tail closure preserves the stated feed invariants but does not inherit original partner-cycle lifting without another proof. Corrected or disproved shortcuts are recorded only as scope controls, not as theorems.

The imported Tahara excess manuscript is attributed related work, not an original result of this article. The archived local $\delta = 7$ packages concern two specified branches and have solver-log validation; their presence does not constitute a proof here of the whole $\delta = 7$ theorem or a formal unsatisfiability certificate. Search attainment is an upper bound on an extremal objective, and a timeout is unresolved. Complete source-level coverage and the retained artifact provenance are recorded in the accompanying inventory. The article and supplement establish the stated partial results; they neither prove the full conjecture nor exhibit a counterexample.

Part I

Foundations and convenient completions

S2 Definitions and the foundational inequality

All graphs are finite and nonempty. An oriented graph has no loops and no pair of oppositely directed arcs. Write $N^+(x)$ and $N^-(x)$ for its out- and inneighborhoods. The sets $N^{++}(x)$ and $N^{--}(x)$ consist, respectively, of vertices at directed distance exactly two from x and to x . In particular, direct neighbors are excluded. Their cardinalities are denoted by $d^+(x), d^-(x), d^{++}(x), d^{--}(x)$.

A Seymour vertex satisfies $d^{++}(x) \geq d^+(x)$. Call a graph *all-deficient* if it has no Seymour vertex, and set

$$g(x) = d^+(x) - d^{++}(x) > 0.$$

An all-deficient graph is *arc-minimal* here if deleting any nonempty set of arcs, with the vertex set fixed, destroys this property. This is stronger than merely assuming that each single arc is individually indispensable. Every all-deficient graph has an arc-minimal spanning subgraph.

If a counterexample exists, take an arc-minimal spanning subgraph and then a sink strongly connected component. That component is still all-deficient and inherits arc-minimality: an arc deletion confined to it cannot improve any vertex outside it. Thus the strongly connected class considered here is nonempty whenever the conjecture is false.

Let $\mathcal{M}(x)$ be the set of vertices distinct from and nonadjacent to x , let $\mu(x) = |\mathcal{M}(x)|$, and let M be the total number of missing unordered pairs. Thus $\sum_x \mu(x) = 2M$.

Theorem S2.1. *Let D be a strongly connected arc-minimal all-deficient oriented graph on n vertices. Then, for every vertex x ,*

$$d^{++}(x) - d^{--}(x) \leq \mu(x) - 1. \quad (\text{S2.1})$$

Consequently,

$$M \geq \left\lceil \frac{n}{2} \right\rceil, \quad |A(D)| \leq \left\lfloor \frac{n(n-2)}{2} \right\rfloor. \quad (\text{S2.2})$$

The restriction to arc-minimal graphs matters. Deleting arcs increases the number of missing pairs, so (S2.2) cannot be transferred backwards to an arbitrary denser counterexample. We do not claim that every oriented graph with fewer than $n/2$ missing pairs satisfies the conjecture.

S3 An external-boundary deletion lemma

For $S \subseteq V(D)$, put

$$\Gamma(S) = \bigcup_{v \in S} N^+(v), \quad \partial S = \Gamma(S) \setminus S, \quad \partial^2 S = \Gamma(\partial S) \setminus (S \cup \partial S).$$

These are external neighborhoods. Seacrest's notation [10] permits intersection with the starting set, so the following adapted statement and proof are made explicit. The proof is included to make the external convention explicit.

Lemma S3.1. *In an arc-minimal all-deficient graph, every S with $\partial S \neq \emptyset$ satisfies $|\partial^2 S| < |\partial S|$.*

Proof. Choose an inclusion-maximal nonempty $T \subseteq \partial S$ with $|\partial T \setminus S| < |T|$, or $T = \emptyset$ if none exists. If $T = \partial S$, the assertion follows. Otherwise delete all arcs $S \rightarrow \partial S \setminus T$. Any resulting Seymour vertex v lies in S .

Partition its deleted outneighbors as $A \dot{\cup} B$, where B becomes second neighbors and A does not. Then $B \subseteq \partial T$. Put

$$L = \partial(A \cup B) \setminus (S \cup T \cup \partial T).$$

Every member of L is a lost second neighbor of v : retained intermediates lie in $S \cup T$, and neither can reach L after deletion. Thus the change in deficiency implies $|L| - |B| < |A| + |B|$. For $T' = T \cup A \cup B$,

$$|\partial T' \setminus S| \leq |\partial T \setminus S| + |L| - |B| < |T| + |A| + |B| = |T'|.$$

The deleted outneighbor set is nonempty and disjoint from T . This contradicts maximality. The empty- T case uses $|\partial T \setminus S| \leq |T| = 0$. $\square$

In the proof, L excludes $A \cup B$ through the definition of the external boundary. This exclusion prevents original first neighbors from being mistaken for lost second neighbors.

Lemma S3.2 (Single-arc consequence; cf. [1]). *In an arc-minimal all-deficient graph, $g(x) \in \{1, 2\}$ for every x .*

Proof. Delete one arc $x \rightarrow y$. All other vertices retain their outneighbors, and their second neighborhoods can only shrink, so the resulting Seymour vertex must be x . Its outdegree decreases by one and its second outdegree increases by at most one. Hence $g(x) \leq 2$. $\square$

Lemma S3.2 is used for the refinements in Section S5; it is not needed for the main inequality.

S4 A target bundle and the local inequality

Fix x and define

$$W = N^-(x), \quad T = V(D) \setminus (\{x\} \cup W), \quad P = \{v \neq x : x \notin N^+(v) \cup N^{++}(v)\}.$$

The remaining part of T is precisely $R = N^{--}(x)$, so $T = P \dot{\cup} R$. Every arc leaving P ends in T . Also

$$T = N^+(x) \dot{\cup} \mathcal{M}(x). \tag{S4.1}$$

If $P = \emptyset$, then $d^{--}(x) = d^+(x) + \mu(x)$. Since x is deficient, $d^{++}(x) - d^{--}(x) \leq -\mu(x) - 1 \leq \mu(x) - 1$.

Assume henceforth $P \neq \emptyset$. Strong connectivity implies $C := \partial P \neq \emptyset$. Put

$$c = |C|, \quad X = \partial^2 P, \quad Z = R \setminus C, \quad j = |Z|, \quad s = |N^+(x) \cap Z|.$$

We have $C \subseteq R$, $c = d^{--}(x) - j$, and

$$|X| \leq c - 1 \tag{S4.2}$$

by Lemma S3.1.

Delete the s arcs from x to Z and denote the resulting graph by D' . When $s = 0$, let $D' = D$. The retained outneighbors of x lie in $P \cup C$. Every arc leaving $P \cup C$ has its head in X , while inside $P \cup C$ the vertices that are not retained outneighbors of x are exactly its original nonneighbors there. By (S4.1), the set Z contains $j - s$ original nonneighbors. Consequently

$$d_{2,D'}^+(x) \leq \mu(x) - (j - s) + |X| \leq \mu(x) + d^{--}(x) - 1 - 2j + s. \quad (\text{S4.3})$$

A deleted outneighbor in Z may become a second neighbor, but in that case it already belongs to X . It must not be counted a second time.

If $s = 0$, (S4.3) immediately proves (S2.1). If $s > 0$, arc-minimality makes x a Seymour vertex in D' : all other vertices retain their outneighbors and cannot gain second neighbors. Therefore

$$d^+(x) - s \leq d_{2,D'}^+(x) \leq \mu(x) + d^{--}(x) - 1 - 2j + s,$$

and hence

$$d^+(x) \leq \mu(x) + d^{--}(x) - 1 - 2(j - s).$$

Using $d^{++}(x) \leq d^+(x) - 1$ proves the slightly stronger inequality $d^{++}(x) - d^{--}(x) \leq \mu(x) - 2$ in this case.

This proves the local assertion of Theorem S2.1. Finally, exact directed distance-two pairs are counted equally by their sources and targets:

$$\sum_x d^{++}(x) = \sum_x d^{--}(x).$$

Summing (S2.1) gives $0 \leq 2M - n$, completing the proof.

S5 Residuals, whole vertices, and near-equality

Define the nonnegative integer residual

$$\eta(x) = \mu(x) - 1 + d^{--}(x) - d^{++}(x).$$

Equivalently, with $p(x) = |P|$,

$$\eta(x) = 2\mu(x) + g(x) - 1 - p(x).$$

We have the exact identity

$$\sum_x \eta(x) = 2M - n. \quad (\text{S5.1})$$

For $P \neq \emptyset$, the preceding proof supplies the refinements

$$s = 0 \implies \eta(x) \geq 2j, \quad s > 0 \implies \eta(x) \geq g(x) + 2(j - s).$$

Proposition S5.1. *If $\eta(x) = 0$, every nonneighbor of x belongs to $N^{++}(x)$. Thus if $2M = n$, every missing pair is reachable in both directions by paths of length two. More generally, at most $2M - n$ vertices have positive residual.*

Proof. For $P = \emptyset$, $\eta(x) = 2\mu(x) + g(x) - 1$, so zero residual forces $\mu(x) = 0$ and $g(x) = 1$. Otherwise $s > 0$ is impossible, and the $s = 0$ bound forces $j = 0$. Equality in (S4.3) and (S4.2) then gives $|N^{++}(x)| = \mu(x) + |X|$. Here $N^{++}(x) \subseteq \mathcal{M}(x) \dot{\cup} X$, since $T = P \dot{\cup} C$. Therefore all nonneighbors are second outneighbors. The final assertion follows from (S5.1). $\square$

Call x *whole* when $\mu(x) = 0$.

Proposition S5.2. *Every whole vertex of D is reachable from every other vertex in at most two steps. Consequently, if w_2 counts whole vertices with $g(x) = 2$, then $2M \geq n + w_2$.*

Proof. The local inequality and Lemma S3.2 give $p(x) \leq g(x) - 1 \leq 1$. Suppose $p(x) = 1$. Then $g(x) = 2$, $\eta(x) = 0$, and the preceding proof gives $P = \{u\}$, $j = s = 0$ and $N^+(x) = \{u\} \cup C$. By definition $C = N^+(u)$, so no vertex in C points to u . Thus the arc $x \rightarrow u$ has no alternative two-step path from x to u . Deleting it reduces the outdegree of x by one but does not increase its second outdegree. Vertex x remains deficient, as do all other vertices, contradicting arc-minimality. Hence $p(x) = 0$. A whole vertex now contributes $\eta(x) = g(x) - 1$ to (S5.1). $\square$

S6 Sink components and the global missing-pair bound

The global bound extends beyond strongly connected graphs, with a quantitative penalty for vertices outside a sink component. The local inequality (S2.1) is still asserted only in the strongly connected setting.

Theorem S6.1. *Let D be any arc-minimal all-deficient oriented graph on n vertices. Let S be a sink strongly connected component, put $k = |S|$, and put $o = n - k$. Then*

$$M(D) \geq \left\lceil \frac{k}{2} \right\rceil + \left\lceil \frac{o(k-2)}{2} \right\rceil \geq \left\lceil \frac{n}{2} \right\rceil + o. \quad (\text{S6.1})$$

In particular, every arc-minimal counterexample satisfies $M(D) \geq \lceil n/2 \rceil$, without a connectivity assumption. If equality holds, D is strongly connected.

Proof. The graph $D[S]$ is all-deficient and arc-minimal. Indeed, deleting arcs inside S leaves every outside outneighborhood unchanged and can only shrink its second neighborhood. A deletion preserving deficiency throughout S would therefore preserve it throughout D . Theorem S2.1 yields $M(D[S]) \geq \lceil k/2 \rceil$.

Write $O = V(D) \setminus S$, and let $\mu_S(v)$ count the missing pairs from $v \in O$ to S . As S is a sink, every adjacent pair from O to S is directed into S . A two-step path entering S cannot leave. Consequently

$$d_{1,D}^+(v) = d_{1,D[O]}^+(v) + k - \mu_S(v), \quad d_{2,D}^+(v) \leq d_{2,D[O]}^+(v) + \mu_S(v),$$

and hence

$$g_D(v) \geq g_{D[O]}(v) + k - 2\mu_S(v).$$

For any oriented graph H , $\sum_{v \in V(H)} g_H(v) \geq -2M(H)$: a present unordered pair supports at most one distance-two direction, and a missing pair at most two. Summing over O and using $g_D(v) \leq 2$ gives

$$2o \geq -2M(D[O]) + ok - 2M(O, S).$$

Thus $M(D[O]) + M(O, S) \geq \lceil o(k-2)/2 \rceil$, proving the first bound.

Every all-deficient oriented graph has minimum outdegree at least two. A sink is already a Seymour vertex; if v has unique outneighbor u , then every outneighbor of u is an exact second neighbor of v , so v is again a Seymour vertex. The arc count on $D[S]$ therefore gives $k \geq 5$. Now

$$\left\lceil \frac{k}{2} \right\rceil + \left\lceil \frac{o(k-2)}{2} \right\rceil \geq \left\lceil \frac{k}{2} \right\rceil + \left\lceil \frac{3o}{2} \right\rceil \geq \left\lceil \frac{n}{2} \right\rceil + o.$$

If D is not strongly connected then $o \geq 1$, proving the last assertion. $\square$

S7 Uniform external out-neighbourhoods

Lemma S7.1. *Suppose $\emptyset \neq B \subsetneq V(D)$ and there is a set C such that $N^+(v) \setminus B = C$ for every $v \in B$. Then $D[B]$ has no arcs.*

Proof. Strong connectivity gives $C = \partial B \neq \emptyset$. Every vertex of B points to every vertex of C , so no arc goes from C to B . Delete all arcs inside B . Each vertex of B then has first neighborhood C and second neighborhood exactly $\partial^2 B$. Lemma S3.1 makes it deficient. All other vertices retain their outneighbors and cannot gain second neighbors. Thus a nonempty internal arc set could be deleted without destroying the counterexample, contrary to arc-minimality. $\square$

S8 Zero-residual protection and good missing edges

Throughout the main argument, D is nonempty, strongly connected, all-deficient, and minimal under deletion of arbitrary nonempty sets of arcs. No assumption $n = 2\delta + 3$ is made. Write

$$g(x) = d_D^+(x) - d_D^{++}(x) > 0, \quad \eta(x) = \mu(x) - 1 + d_D^-(x) - d_D^{++}(x).$$

The established pointwise residual theorem gives

$$\eta(x) \geq 0, \quad \sum_x \eta(x) = 2M - n,$$

where M is the number of missing unordered pairs.

S8.1 A zero-residual source protects all of its far targets

Lemma S8.1. *If $\eta(x) = 0$ and $y \in U_D(x)$, then*

$$y \rightarrow x, \quad N_D^-(y) \subseteq N_D^-(x) \setminus \{y\}.$$

Proof. Use the target-bundle sets

$$W = N_D^-(x), \quad T = V(D) \setminus (\{x\} \cup W), \quad P = \{v : x \in U_D(v)\}, \quad R = T \setminus P = N_D^-(x).$$

If P is empty, $\eta(x) = 2\mu(x) + g(x) - 1 = 0$ forces $\mu(x) = 0$ and $g(x) = 1$. A far target y then points to x , since x is whole. No outneighbor of x points to y , so every predecessor of y belongs to $N_D^-(x) \setminus \{y\}$.

Now let $P \neq \emptyset$ and put $C = \partial P$, $X = \partial^2 P$, $j = |R \setminus C|$ and $s = |N_D^+(x) \cap (R \setminus C)|$. Strong connectivity gives $C \neq \emptyset$. The established sharpened target-bundle bounds are

$$s = 0 \implies \eta(x) \geq 2j, \quad s > 0 \implies \eta(x) \geq g(x) + 2(j - s).$$

Since $j \geq s$ and $g(x) > 0$, zero residual forces $j = s = 0$. Thus $C = R$, $T = P \dot{\cup} C$, and $|X| \leq |C| - 1$ by the external-boundary lemma. Every second outneighbor of x lies in $\mathcal{M}(x) \cup X$. Since $C \subseteq N_D^-(x)$, no vertex of C points directly to x . Thus $X \subseteq W$ and $\mathcal{M}(x) \subseteq T$, so these sets are disjoint. Zero residual gives

$$d_D^{++}(x) = \mu(x) + |C| - 1.$$

Equality of cardinalities therefore forces

$$N_D^{++}(x) = \mathcal{M}(x) \dot{\cup} X, \quad |X| = |C| - 1.$$

Since $T = N_D^+(x) \dot{\cup} \mathcal{M}(x)$, this yields $U_D(x) = W \setminus X$. For $y \in U_D(x)$, there is no arc from P to y because all out-arcs of P remain in T . There is no arc from C to y , because such a head outside $P \cup C$ would belong to X . Also $x \not\rightarrow y$. Hence $y \rightarrow x$ and all predecessors of y are in $W \setminus \{y\}$, as required. $\square$

S8.2 The published good-missing-edge transfer

A missing pair ab is *good* in the sense of Ghazal if either

$$\forall w \notin \{a, b\} : \quad w \rightarrow a \implies b \in N_D^+(w) \cup N_D^{++}(w),$$

or

$$\forall w \notin \{a, b\} : \quad w \rightarrow b \implies a \in N_D^+(w) \cup N_D^{++}(w).$$

The respective convenient orientations are $a \rightarrow b$ and $b \rightarrow a$. The following published result has no generalized-star assumption on the missing graph.

Theorem S8.2 (Ghazal [5, Theorem 2]). *If all missing pairs of a nonempty oriented graph are good, that graph has a Seymour vertex.*

Unweighted completion proof. Conveniently orient every missing pair to obtain a tournament T . Choose an order L maximizing the number of forward arcs and let f be its last vertex. Reverse the r added arcs directed out of f , obtaining T' . The order L gains r forward arcs; any other order gains at most r . Hence L remains a median order. The published feed-vertex theorem [7] gives the SNP for f in T' .

All added arcs incident to f now point inward, so $N_{T'}^+(f) = N_D^+(f)$. For a path $f \rightarrow y \rightarrow z$ in T' , the first arc is original. The second arc is original or an untouched convenient orientation: it cannot involve f , as $z = f$ would make a digon. Consequently z is reachable from f within two steps in D . If $z \in N_{T'}^{++}(f)$, it is not a direct outneighbor in D , so $z \in N_D^{++}(f)$. The tournament SNP therefore transfers to D . $\square$

S8.3 A bad missing edge has two positive-residual witnesses

Lemma S8.3. *If a missing pair ab is not good, there are distinct vertices u, v which are nonadjacent and satisfy*

$$\eta(u) > 0, \quad \eta(v) > 0.$$

Proof. Failure of both convenient orientations supplies $u, v \in V(D) \setminus \{a, b\}$ with

$$u \rightarrow a, \quad b \in U_D(u), \quad v \rightarrow b, \quad a \in U_D(v).$$

They are distinct, since $u = v$ would contradict $u \rightarrow a$ and $a \in U_D(v)$. An arc $u \rightarrow v$ would let u reach b in two steps, and an arc $v \rightarrow u$ would let v reach a in two steps. Thus uv is a missing pair.

If $\eta(u) = 0$, Lemma S8.1 at its far target b gives $N_D^-(b) \subseteq N_D^-(u) \setminus \{b\}$. But $v \rightarrow b$ would imply $v \rightarrow u$, a contradiction. Symmetrically, $\eta(v) = 0$ would imply $u \rightarrow v$. Nonnegativity and integrality of the residuals prove the claim. $\square$

Theorem S8.4 (Strict missing-pair margin at all orders). *The positive-residual set $Z = \{x : \eta(x) > 0\}$ induces a missing pair. In particular,*

$$\sum_x \eta(x) \geq 2, \quad M \geq \left\lceil \frac{n}{2} \right\rceil + 1.$$

Proof. Since D is all-deficient, Theorem S8.2 implies that some missing pair is not good. Lemma S8.3 places both endpoints of a missing witness pair inside Z . Each contributes at least one, so $2M - n = \sum_x \eta(x) \geq 2$. Integer rounding gives $M \geq \lceil (n+2)/2 \rceil = \lceil n/2 \rceil + 1$. $\square$

Corollary S8.5. *At order $n = 2\delta + 3$,*

$$M \geq \delta + 3$$

for every minimum outdegree δ . The one-unit residual layer $2M - n = 1$ is impossible at every odd order, and the zero-residual layer is impossible at every even order.

The strict margin above is an intermediate step; the parity-sensitive final bound is Theorem S31.1.

S8.4 Unprotected targets and a four-unit residual margin

Continue with a finite nonempty, strongly connected, arc-minimal all-deficient oriented graph. Here arc-minimal means minimal under deletion of arbitrary nonempty sets of arcs on the fixed vertex set, not merely under single-arc deletion. Write $\mathcal{F}(x)$ for the vertices $y \in U(x)$ which are not protected, and put $t(x) = |\mathcal{F}(x)|$. Protection means

$$y \rightarrow x, \quad N^-(y) \subseteq N^-(x) \setminus \{y\}.$$

All neighbourhoods, convenient orientations, and protected pairs below refer to the original graph.

Lemma S8.6 (The unprotected-target envelope). *Fix a vertex x with $P = P(x) \neq \emptyset$, and put*

$$\begin{aligned} Q &= N^{--}(x), \quad W = N^-(x), \quad C = \partial P, \quad H = Q \setminus C, \\ X &= \partial^2 P, \quad j = |H|, \quad s = |N^+(x) \cap H|. \end{aligned}$$

Then

$$j = s = 0 \implies t(x) \leq \eta(x), \quad j = s > 0 \implies t(x) \leq \eta(x) - g(x).$$

Proof. When $j = s$, every vertex of H is an original outneighbour of x . Delete all arcs $x \rightarrow H$, and call the resulting graph $\widehat{D}$; if $H = \emptyset$, set $\widehat{D} = D$. Its retained first neighbours lie in $P \cup C$. Also $\mathcal{M}(x) \subseteq P \cup C$, whereas X is disjoint from $P \cup C$. The target-deletion envelope, including deleted first neighbours that may become exact second neighbours, gives

$$N_{\widehat{D}}^{++}(x) \subseteq E := \mathcal{M}(x) \dot{\cup} X, \quad |E| \leq \mu(x) + |Q| - s - 1 = d_D^{++}(x) + \eta(x) - s.$$

For completeness, if a retained intermediate is in P , its outgoing arcs stay in $P \cup C$; if it is in C , an outgoing head outside that set is in X . Inside $P \cup C$, a head which is not a retained first neighbour is an original nonneighbour of x . A deleted head in H which becomes second belongs to X and is counted only once. The size bound uses $|X| \leq |C| - 1 = |Q| - s - 1$.

We claim

$$\mathcal{F}(x) \subseteq E \setminus N_{\widehat{D}}^{++}(x).$$

For a missing far target this follows from its membership in $\mathcal{M}(x)$. Otherwise a far target y lies in W and points to x . Being unprotected, it has a predecessor $z \notin W$. The vertex z cannot be x ; it cannot be in P , because no arc from P points into W ; and it cannot be in $H \subseteq N^+(x)$, because that would give $x \rightarrow z \rightarrow y$. Thus $z \in C$ and $y \in X$. Finally,

deleting arcs cannot create a path of length at most two to an originally unreachable target, so $y \notin N_D^{++}(x)$.

If $s = 0$, subtracting the original second neighbourhood from E gives $t(x) \leq \eta(x)$. If $s > 0$, every other vertex has unchanged first neighbourhood and cannot gain second neighbours. Arc-minimality therefore makes x a Seymour vertex in $\widehat{D}$, so $d_{\widehat{D}}^{++}(x) \geq d_D^+(x) - s$. Consequently

$$t(x) \leq |E| - d_{\widehat{D}}^{++}(x) \leq d_D^{++}(x) + \eta(x) - s - (d_D^+(x) - s) = \eta(x) - g(x).$$

No minimality or counterexample property is assumed for $\widehat{D}$. $\square$

Corollary S8.7 (One unit of residual). *If $\eta(x) = 1$, then $t(x) \leq 1$.*

Proof. If $P(x) = \emptyset$, the identity $1 = 2\mu(x) + g(x) - 1$ and $g(x) \in \{1, 2\}$ force $\mu(x) = 0$ and $g(x) = 2$. For a whole vertex every far target is protected: it points to x , and each of its predecessors must also point to x , because the opposite orientation would give a two-step path from x to that target. Thus $t(x) = 0$.

If $P(x) \neq \emptyset$, the existing local bounds are

$$s = 0 \implies \eta(x) \geq 2j, \quad s > 0 \implies \eta(x) \geq g(x) + 2(j - s).$$

For residual one, the first case forces $j = s = 0$. The second forces $g(x) = 1$ and $j = s > 0$. Apply Lemma S8.6; in the second case it even gives $t(x) = 0$. $\square$

Lemma S8.8 (A two-unit dichotomy). *Suppose $\eta(x) = 2$ and r is a missing partner of x with $N^+(r) \cap U(x) \neq \emptyset$. Then either $t(x) \leq 2$, or $r \rightarrow x$ is a convenient orientation of the missing edge rx .*

Proof. If $P(x) = \emptyset$, every other vertex reaches x within two steps, which proves convenience of $r \rightarrow x$ immediately. Otherwise use the notation of Lemma S8.6. If $s > 0$, the bound $2 \geq g(x) + 2(j - s)$ forces $j = s$, and that lemma gives $t(x) \leq 2 - g(x) \leq 1$. If $s = j = 0$, it gives $t(x) \leq 2$. The only remaining case is $s = 0, j = 1$. Write $H = \{q\}$. Since $q \in Q$ but $q \notin N^+(x)$, it is an original missing partner of x .

The retained original first neighbours lie in $P \cup C$, and the same envelope argument gives

$$N_D^{++}(x) \subseteq E_0 := (\mathcal{M}(x) \setminus \{q\}) \cup X.$$

The boundary inequality gives $|X| \leq |Q| - 2$, hence

$$|E_0| \leq \mu(x) - 1 + |Q| - 2 = d_D^{++}(x),$$

where the last equality uses $\eta(x) = 2$. Thus the envelope is saturated:

$$N_D^{++}(x) = E_0.$$

Every vertex in $P \cup C$ is reachable from x within at most two steps: it is either an original first neighbour, or a missing partner other than q and hence an exact second neighbour by saturation. This assertion is about reachability within two steps, not about membership of all these vertices in the exact second neighbourhood. Every arc leaving $P \cup C$ has its head in X , also an exact second neighbour of x . A missing partner $r \neq q$ lies in $P \cup C$, so none of its outneighbours can lie in $U(x)$. The hypothesis on r therefore forces $r = q$.

Finally $q \notin P$ and $q \notin C = \partial P$, so no vertex of $P(x)$ points to q . Every predecessor of q is thus outside $P(x)$ and reaches x within two steps. This is precisely convenience of $q \rightarrow x$. $\square$

S8.5 Exceptional partners and a shared failure-head budget

Let D be a finite nonempty, strongly connected, all-deficient oriented graph minimal under deletion of arbitrary nonempty arc sets. Fix x with $P(x) \neq \emptyset$, and write

$$Q = N^{--}(x), \quad W = N^-(x), \quad C = \partial P(x), \quad X = \partial^2 P(x), \quad H = Q \setminus C.$$

Let $B = H \cap \mathcal{M}(x)$, $k = |B|$, and $s = |H \cap N^+(x)|$, so $|H| = k + s$. Delete the s arcs from x into H , obtaining $\widehat{D}$; if $s = 0$, it is the original graph. Put

$$E = (\mathcal{M}(x) \setminus B) \dot{\cup} X, \quad L = E \setminus N_{\widehat{D}}^{++}(x).$$

Lemma S8.9 (Shared head budget outside exceptional partners). *Every $r \in B$ admits the convenient orientation $r \rightarrow x$. Moreover*

$$\bigcup_{r \in \mathcal{M}(x) \setminus B} (N^+(r) \cap U(x)) \subseteq L,$$

and

$$|L| \leq \begin{cases} \eta(x) - 2k, & s = 0, \\ \eta(x) - g(x) - 2k, & s > 0. \end{cases}$$

The bound concerns the union of distinct far heads, not the sum of the sizes of the individual partner contributions.

Proof. The partition $V(D) = \{x\} \sqcup W \sqcup P(x) \sqcup C \sqcup H$ implies $\mathcal{M}(x) \setminus B \subseteq P(x) \cup C$. All retained outneighbours of x lie in $P(x) \cup C$. Every arc leaving this union has its head in X , whereas a head inside the union that is not an original outneighbour of x is in $\mathcal{M}(x) \setminus B$. Therefore the target envelope gives $N_{\widehat{D}}^{++}(x) \subseteq E$, with the displayed union disjoint. For $r \in \mathcal{M}(x) \setminus B$, any head of r in $U(x)$ is likewise in E . An original far head cannot become reachable after deleting arcs, so it belongs to L .

The strict external-boundary inequality gives $|X| \leq |C| - 1 = |Q| - s - k - 1$. Hence

$$|E| \leq \mu(x) - k + |Q| - s - k - 1 = d_D^{++}(x) + \eta(x) - s - 2k.$$

If $s = 0$, subtract $d_D^{++}(x)$. If $s > 0$, the nonempty deletion can create a Seymour vertex only at x , since all other first neighbourhoods are unchanged and their second neighbourhoods cannot grow. Minimality therefore gives $d_{\widehat{D}}^{++}(x) \geq d_D^{++}(x) - s$; subtracting proves the second bound. Finally every $r \in B$ lies outside both $P(x)$ and $C = \partial P(x)$, so no member of $P(x)$ points to r . All predecessors of r thus reach x within two steps, which is exactly convenience of $r \rightarrow x$. $\square$

For instance, if $\eta(x) = 3$, $s = 0$, and $k = 1$, all nonexceptional missing partners together contribute at most one far head. If $\eta(x) = 3$, $s > 0$, and $k = 1$, the bound forces $g(x) = 1$ and L empty; every missing partner with an outneighbour in $U(x)$ must then be the unique exceptional partner, directed conveniently into x . Neither statement supplies a universal upper bound on $t(x)$.

S9 Canonical convenient orientations and small failure-source sets

Draft status: this argument has undergone internal adversarial review; it has not undergone external peer review.

In this section D is an arbitrary finite, nonempty oriented graph. No strong-connectivity, minimality, or residual hypothesis is imposed. All neighbourhoods and directed distances in the definitions below refer to the original D . Put

$$U_D(x) = V(D) \setminus (\{x\} \cup N_D^+(x) \cup N_D^{++}(x)), \quad g(x) = d_D^+(x) - d_D^{++}(x).$$

In particular, a target in $U_D(x)$ is neither a first nor an exact second neighbour of x .

Following Ghazal [5, Definition 1], a missing pair ab is good if at least one of its orientations is convenient. We write $G(a \rightarrow b)$ for the statement

$$\forall x \in V(D) \setminus \{a, b\} : \quad x \rightarrow a \implies b \in N_D^+(x) \cup N_D^{++}(x).$$

For any missing pair define its directional failure sets by

$$\begin{aligned} F_{ab} &= \{x \in V(D) \setminus \{a, b\} : x \rightarrow a, b \in U_D(x)\}, \\ F_{ba} &= \{x \in V(D) \setminus \{a, b\} : x \rightarrow b, a \in U_D(x)\}. \end{aligned}$$

It is bad precisely when both sets are nonempty.

Define the protection relation by

$$y \triangleleft p \iff y \rightarrow p \text{ and } N_D^-(y) \subseteq N_D^-(p) \setminus \{y\}.$$

It implies $y \in U_D(p)$: a path $p \rightarrow z \rightarrow y$ would require $z \rightarrow p$, contradicting $p \rightarrow z$. Let

$$\mathcal{F}(x) = \{y \in U_D(x) : y \not\triangleleft x\}, \quad t(x) = |\mathcal{F}(x)|.$$

The relation $\triangleleft$ is a strict partial order. Indeed, $y \triangleleft p$ gives $d_D^-(y) < d_D^-(p)$, and if $y \triangleleft p \triangleleft r$, then $y \in N_D^-(p) \subseteq N_D^-(r)$ gives $y \rightarrow r$, and the two containments compose to $N_D^-(y) \subseteq N_D^-(r) \setminus \{y\}$.

Choose a linear order $\prec_0$ extending $\triangleleft$. Orient each good missing pair by its unique convenient direction if only one exists; if both exist, orient it from earlier to later in $\prec_0$. We call these the canonical convenient orientations associated with $\prec_0$.

Lemma S9.1 (Compatibility of canonical convenient orientations). *Suppose $y \triangleleft p$ and both pq and qy are good missing pairs. If the canonical orientation of pq is $p \rightarrow q$, then the canonical orientation of qy is $y \rightarrow q$.*

Proof. There are two convenient-direction implications:

$$G(p \rightarrow q) \implies G(y \rightarrow q), \quad G(q \rightarrow y) \implies G(q \rightarrow p).$$

For the first, every predecessor of y is a predecessor of p . For the second, every vertex reaching y within two steps also reaches p within two steps: replace the last arc into y by an arc into p using the protected in-neighbourhood containment. The relevant sources are distinct from the missing-pair endpoints, as required by convenience.

If the chosen directions were $p \rightarrow q$ and $q \rightarrow y$, these implications would make both pairs convenient in both directions. The canonical tie-breaking rule would force $p \prec_0 q \prec_0 y$, contrary to $y \triangleleft p$ and hence $y \prec_0 p$. $\square$

Lemma S9.2 (At most three failure sources). *Let $Z \subseteq V(D)$ satisfy $|Z| \leq 3$ and*

$$F_{ab} \cup F_{ba} \subseteq Z \text{ for every bad missing pair } ab.$$

Choose canonical convenient orientations using some linear extension $\prec_0$ of protection. Let K be the oriented graph on Z whose arcs are only these added orientations of missing pairs inside Z , and put

$$S = \{z \in Z : d_K^+(z) = 0\}.$$

If

$$t(z) \leq g(z) + 1 \quad (z \in Z \setminus S), \quad (\text{SF1})$$

then D has a Seymour vertex.

Proof. First, every missing pair inside Z is good, so the definition of K accounts for every such pair. Otherwise its two endpoints and two opposite failure sources would be four distinct vertices of Z . Opposite sources are distinct, and are themselves non-adjacent: if $u \in F_{ab}$ and $v \in F_{ba}$, either arc between them would create one of the forbidden two-step paths.

No two vertices of S form a missing pair, since its orientation in K would give one of them positive out-degree. For every bad pair, at least one direction consequently has no failure source in S . Indeed, if both directions had an S -source, those two opposite sources would form a missing pair inside S . Orient each bad pair in a direction avoiding failures from S , and complete the graph to a tournament T using these orientations and the canonical orientations of all good pairs.

We record two path facts in the original graph. Every failure head of a bad edge is unprotected: if $x \in F_{ab}$, an opposite witness $v \in F_{ba}$ would, under $b \triangleleft x$, give $v \rightarrow x \rightarrow a$. Thus

$$x \in F_{ab}, ab \text{ bad} \implies b \in \mathcal{F}(x). \quad (\text{SF2})$$

Second, if the canonical convenient direction is $p \rightarrow q$ and $y \triangleleft p$, then $G(p \rightarrow q)$ directly implies $G(y \rightarrow q)$. Thus a missing pair qy is automatically good, regardless of the number of failure sources. Compatibility gives

$$y \triangleleft p, \quad qy \text{ missing} \implies y \rightarrow q \text{ in } T. \quad (\text{SF3})$$

Choose an order L maximizing the forward arcs of T , with feed vertex f . When any r added arcs leaving f are reversed towards f , all were backward in L . The forward-arc count of L increases by r , while that of every other order increases by at most r . Thus L remains a median order after either reversal used below. The feed-vertex theorem [7] gives the SNP for f in the resulting tournament.

Case A: $f \notin Z \setminus S$. Reverse every added arc leaving f , obtaining T' . Then $N_{T'}^+(f) = N_D^+(f)$. An exact two-step head is reached first through an original out-neighbour. An original second arc or a convenient added second arc reaches an original first or second neighbour. A bad added second arc producing a genuinely new head would be a failure from f . This is excluded if $f \notin Z$ by the source-cover hypothesis, and if $f \in S$ by the bad-edge choices. An added arc incident to f cannot be the second arc of a two-step path starting at f , because a tournament has no two-cycle. Therefore

$$N_{T'}^{++}(f) \subseteq N_D^{++}(f).$$

The SNP in T' transfers to D , proving the conclusion.

Case B: $f \in Z \setminus S$. Choose an arc $f \rightarrow q$ of K . Reverse all other added arcs leaving f , but retain this one. Then

$$N_{T'}^+(f) = N_D^+(f) \dot{\cup} \{q\}. \quad (\text{SF4})$$

We claim the following exact-neighbourhood containment:

$$N_{T'}^{++}(f) \subseteq (N_D^{++}(f) \setminus \{q\}) \cup (\mathcal{F}(f) \setminus \{q\}). \quad (\text{SF5})$$

Take an exact two-step head y and a path $f \rightarrow a \rightarrow y$ in T' . It is neither an original first neighbour nor q . If $a \in N_D^+(f)$, an original or convenient second arc makes y an original second neighbour. A bad added second arc with a new head makes y an unprotected original far target by (SF2). These added second arcs are not incident to f , so their chosen orientations have not been reversed.

It remains to check the new intermediate $a = q$. The only issue is a head $y \in U_D(f)$ protected from f . An original arc $q \rightarrow y$ would force $q \rightarrow f$ by protection, contrary to the original missing pair fq . If qy was missing, (SF3) forces its canonical direction to be $y \rightarrow q$. Neither endpoint is f , so that arc is unchanged in T' . If $y \rightarrow q$ was original, it too is unchanged. In all cases $q \rightarrow y$ is impossible. Thus every new far head through q is unprotected. Since q is now first, it must be removed from both parts of the envelope. This proves (SF5); it explicitly accounts for all paths through the retained added first neighbour.

The original missing partner q is either in $N_D^{++}(f)$ or in $U_D(f)$. In the latter case it is unprotected, since it does not point to f . Hence q belongs to exactly one of the disjoint sets $N_D^{++}(f)$ and $\mathcal{F}(f)$. Removing it in (SF5) gives

$$d_{T'}^{++}(f) \leq d_D^{++}(f) + t(f) - 1, \quad d_{T'}^+(f) = d_D^+(f) + 1.$$

The count is of distinct second-neighbourhood heads, not paths or added arcs. By (SF1),

$$d_{T'}^+(f) - d_{T'}^{++}(f) \geq g(f) + 2 - t(f) \geq 1,$$

contradicting the feed-vertex theorem for the same median feed f . Case B is impossible, while Case A gives the required Seymour vertex. $\square$

The cardinality bound ensures that every internal missing pair of Z is good. The protected-head compatibility itself has no source-cardinality restriction. All convenience and protection statements remain properties of the original graph; no such property is transferred to an augmented graph without the explicit path checks above.

Relation to earlier completion arguments. Retaining one added out-arc at the feed vertex already appears in Ghazal's proof of [6, Theorem 3]. That theorem imposes a specific hypothesis on the entire missing graph. It does not directly assert Lemma S9.2 for a small set of failure sources. The canonical-orientation compatibility and the protected-head path check above are supplied explicitly here. No comprehensive novelty or priority claim is made.

S10 Dangerous witnesses of a bad missing edge

For a vertex v , a far target $y \in U(v)$ is called protected from v if

$$y \rightarrow v, \quad N^-(y) \subseteq N^-(v) \setminus \{y\}.$$

Let $t(v)$ be the number of far targets not protected from v . We first prove a completion statement for an arbitrary oriented graph whose vertices all fail the second-neighbourhood inequality. In this statement $g(v) = d^+(v) - d^{++}(v)$ is any positive integer.

A missing edge ab is good in Ghazal's sense [5] if at least one of

$$\forall x \notin \{a, b\}, \quad x \rightarrow a \implies b \in N^+(x) \cup N^{++}(x),$$

$$\forall x \notin \{a, b\}, \quad x \rightarrow b \implies a \in N^+(x) \cup N^{++}(x)$$

holds. Such an implication specifies its convenient orientation. A failure for the orientation $a \rightarrow b$ is a source x with $x \rightarrow a$ and $b \in U(x)$.

Lemma S10.1 (A localized completion obstruction). *Let D be any nonempty oriented graph with $g(v) > 0$ for every vertex. Then there is a bad missing edge ab and distinct witnesses u, v such that*

$$u \rightarrow a, \quad b \in U(u), \quad v \rightarrow b, \quad a \in U(v), \quad t(u) \geq g(u), \quad t(v) \geq g(v).$$

The witnesses u, v are non-adjacent.

Proof. Put $B_t = \{x : t(x) \geq g(x)\}$. Suppose the conclusion fails. Every bad missing edge then has at least one orientation with no failure whose source belongs to B_t . Give it such an orientation. Give every good missing edge a *fully convenient* orientation, as in Ghazal's completion proof. For good edges we do not merely choose an orientation avoiding failures from B_t .

The completed graph is a tournament T . Choose a median order and let f be its last, or feed, vertex. Reverse every added arc leaving f so that all added arcs incident to f point towards it; call the result T' . The order remains a local median order: interval feedback conditions not involving f do not change, and feedback conditions involving its last position only improve. By the feed-vertex theorem [7], f has the second-neighbourhood property in T' . Its first out-neighbourhood is exactly the original $N_D^+(f)$.

Let $E = N_{T'}^{++}(f) \setminus N_D^{++}(f)$ be the set of new second *heads*. For $z \in E$, choose a two-step path $f \rightarrow a \rightarrow z$ in T' . The first arc is original. The second must be an added arc; otherwise z would already be an original second neighbour. Neither a nor z equals f , since a tournament has no two-cycle. Thus the second arc was not among those reversed at the feed vertex, and still has its chosen completion orientation. The equality of the first neighbourhoods ensures that $z \in U_D(f)$.

This added edge cannot be good, since its convenient orientation allows no such failure. It is therefore a bad edge az with a failure from f . Any failure head of a bad edge is unprotected: an opposite failure witness $w \rightarrow z$, $a \in U(w)$ would, if z were protected from f , imply $w \rightarrow f \rightarrow a$, a contradiction.

If $f \in B_t$, our choice of orientations excludes the failure, so E is empty. If $f \notin B_t$, we have

$$E \subseteq \{y \in U_D(f) : y \text{ is unprotected from } f\}, \quad |E| \leq t(f) < g(f).$$

The count is of distinct heads, not arcs or two-step paths. In either case,

$$|N_{T'}^{++}(f)| \leq d_D^{++}(f) + |E| < d_D^+(f) = |N_{T'}^+(f)|,$$

contradicting the feed-vertex theorem. Hence some bad edge has failures from B_t in both directions.

The two witnesses are distinct and outside $\{a, b\}$. An arc $u \rightarrow v$ would yield $u \rightarrow v \rightarrow b$, and an arc $v \rightarrow u$ would yield $v \rightarrow u \rightarrow a$. Both are forbidden. Thus u, v are non-adjacent. $\square$

Lemma S10.2 (The safe residual budget). *Let D be a strongly connected counterexample minimal under deletion of any nonempty set of arcs. If $\eta(v) < g(v)$, then*

$$t(v) \leq \eta(v) < g(v).$$

Proof. The established gap and residual theorems give $g(v) \in \{1, 2\}$ and $\eta(v) \geq 0$. Put $P = P(v)$, $R = N^{--}(v)$, and $W = N^-(v)$. If P is empty, the identity $\eta(v) = 2\mu(v) + g(v) - 1$ and $\eta(v) < g(v)$ force $\mu(v) = 0$. Every far target y then points to v , and every $x \rightarrow y$ must point to v , since otherwise $v \rightarrow x \rightarrow y$. Thus all far targets are protected and $t(v) = 0$.

Suppose $P \neq \emptyset$, and write $C = \partial P \subseteq R$, $j = |R \setminus C|$, and $s = |N^+(v) \cap (R \setminus C)|$. The previously proved outgoing-bundle refinement gives

$$s > 0 \implies \eta(v) \geq g(v) + 2(j - s), \quad s = 0 \implies \eta(v) \geq 2j.$$

Since $s \leq j$, the safe inequality excludes $s > 0$. Also $\eta(v) < g(v) \leq 2$ gives $\eta(v) \leq 1$, and hence $j = 0$. Thus $C = R$.

Let $X = \partial^2 P$. Then $X \subseteq W$, and the external-boundary lemma gives $|X| \leq |R| - 1$. Every vertex of $Y = W \setminus X$ is a protected far target of v : no vertex of P or R points to it, and its in-neighbours therefore lie in $W \setminus \{y\}$. Using $\eta(v) = \mu(v) - 1 + |R| - d^{++}(v)$, we obtain

$$|Y| \geq |W| - |R| + 1 = |U(v)| - \eta(v).$$

Consequently at most $\eta(v)$ far targets can be unprotected. □

Proposition S10.3 (Dangerous residual witnesses). *Every strongly connected counterexample minimal under deletion of any nonempty set of arcs has a bad missing edge with opposite failure witnesses u, v such that u, v are non-adjacent and*

$$\eta(u) \geq g(u), \quad \eta(v) \geq g(v).$$

In particular,

$$2M - n = \sum_x \eta(x) \geq g(u) + g(v).$$

Proof. Apply Lemma S10.1. The contrapositive of Lemma S10.2 says that $t(x) \geq g(x)$ implies $\eta(x) \geq g(x)$. Non-negativity of all residuals proves the sum bound. □

This strengthens the strict missing-pair theorem established earlier. If every vertex has gap two, then $2M - n \geq 4$, and parity yields

$$M \geq \lceil n/2 \rceil + 2.$$

If instead $2M - n = 2$, then the two dangerous witnesses are exactly the two positive-residual vertices; each has residual one and gap one, and they are non-adjacent.

S10.1 Using only cross-witness pairs as the source graph

Let D be an arbitrary finite nonempty oriented graph, with the failure sets, protection relation, unprotected-target count t , and canonical convenient orientations defined in Section S9. Let Z contain all directional failure sources of every bad missing pair. No bound on $|Z|$ is imposed. Define the undirected *cross-witness graph* $\mathcal{B}_Z$ on Z by putting an edge xy whenever some bad missing pair ab has $x \in F_{ab}$ and $y \in F_{ba}$, or the reverse assignment. Every such

edge is an original missing pair. Assume that all edges of $\mathcal{B}_Z$ are good. This is automatic when $|Z| \leq 3$, since an internal bad pair of Z would require four distinct vertices of Z .

Orient $\mathcal{B}_Z$ by the canonical convenient orientations of its edges, and put

$$S_{\mathcal{B}} = \{z \in Z : d_{\mathcal{B}_Z}^+(z) = 0\}.$$

This graph contains only cross-witness pairs, not every missing pair inside Z , and not the original arcs of $D[Z]$.

Lemma S10.4 (Cross-witness source transfer). *If the chosen protection-compatible reference order gives*

$$t(z) \leq g(z) + 1 \quad (z \in Z \setminus S_{\mathcal{B}}),$$

then D has a Seymour vertex.

Proof. We check the two uses of the source graph in the proof of Lemma S9.2. Although $S_{\mathcal{B}}$ may contain an original missing pair not belonging to $\mathcal{B}_Z$, no bad pair can have failures from $S_{\mathcal{B}}$ in both directions. Such opposite sources would form a cross-witness edge, and its orientation would give one of them positive out-degree in $\mathcal{B}_Z$. Thus every bad pair can still be oriented so as to avoid failure sources in $S_{\mathcal{B}}$.

Complete with these bad directions and all canonical good directions, and choose a median feed f . If $f \notin Z \setminus S_{\mathcal{B}}$, reverse all its added outgoing arcs. The original first neighbourhood is retained, and no added second arc produces a failure from f . The feed SNP therefore transfers back to D .

If $f \in Z \setminus S_{\mathcal{B}}$, retain one added cross-witness arc $f \rightarrow q$ and reverse its other added outgoing arcs. This retained arc is a canonical convenient orientation of a good missing pair, with $f, q \in Z$. For a protected far head $y \triangleleft f$, each predecessor of y is a predecessor of f . Hence convenience of $f \rightarrow q$ directly implies convenience of $y \rightarrow q$ whenever qy is missing. Thus qy is automatically good, independent of $|Z|$. Canonical compatibility then orients it $y \rightarrow q$. An original arc $q \rightarrow y$ would force $q \rightarrow f$ by protection, contrary to the missing pair fq . Consequently the same exact envelope holds:

$$N_{T'}^{++}(f) \subseteq (N_D^{++}(f) \setminus \{q\}) \cup (\mathcal{F}(f) \setminus \{q\}).$$

The first neighbourhood has gained exactly q , and q removes one element from the disjoint second/far envelope. Thus $d_{T'}^+(f) - d_{T'}^{++}(f) \geq g(f) + 2 - t(f) \geq 1$, contradicting the feed-vertex theorem. The same backward-arc reversal argument preserves the median order throughout. $\square$

In terms of the dependency digraph, the hypothesis says that every missing pair with positive out-degree has in-degree zero: each tail of a losing relation is good. It excludes every directed walk of two arcs, including a two-cycle. Merely excluding simple directed paths on three distinct vertices would not be the same condition.

Proposition S10.5 (One two-unit source among at most three sources). *Let D be a strongly connected all-deficient oriented graph minimal under deletion of arbitrary nonempty arc sets. Suppose its positive residual support has at most three vertices, at most one of which has residual two, with every other positive residual equal to one. Then D cannot exist. In particular, residual four cannot have distribution $(2, 1, 1)$.*

Proof. Put $Z = \{x : \eta(x) > 0\}$. Zero-residual protection guarantees that every bad-edge directional failure source belongs to Z . For every residual-one vertex, Corollary S8.7 gives $t(x) \leq 1 \leq g(x) + 1$.

If there is no residual-two vertex, any canonical reference order meets the cross-witness transfer threshold. If there is such a vertex u and $t(u) \leq 2$, the same is true, since $g(u) \geq 1$. In either case Lemma S10.4 gives a Seymour vertex, a contradiction.

It remains that $\eta(u) = 2$ and $t(u) > 2$. Each cross-witness neighbour r of u has $N_D^+(r) \cap U_D(u) \neq \emptyset$. The two-unit dichotomy, Lemma S8.8, therefore makes $r \rightarrow u$ a convenient direction. Since ru is missing, r, u are incomparable in the protection partial order. We may choose a linear extension placing every such r before u simultaneously. Indeed, adding these precedence constraints would create a cycle only if the original protection order already had a path from u to some r , which would make the missing pair comparable.

Every cross-witness edge incident to u is now canonically directed into u : a unique convenient direction is already inward, and a double-convenient edge is directed inward by the chosen order. Thus $u \in S_B$. Only residual-one vertices require a budget, and they have it. Applying the cross-witness transfer gives the required contradiction. This proof includes $P(u) = \emptyset$; no bound on $t(u)$ was inferred from that case. $\square$

The later support theorem excludes every two-positive allocation; no such allocation remains a frontier problem.

S10.2 A completion with only one possible failure source

Lemma S10.6 (One possible failing feed). *Let D be a finite nonempty oriented graph and let $v \in V(D)$. Suppose every missing pair incident to v is good, every bad missing pair admits a direction whose failure sources are contained in $\{v\}$, and $t(v) \leq g(v)$. Then D has a Seymour vertex.*

Proof. Suppose D is all-deficient. Complete good pairs by canonical convenient directions and each bad pair by a direction with failures only at v , obtaining T . If a global median order ends at $f \neq v$, reverse its added out-arcs inward. The order remains median, the first neighbourhood of f becomes the original one, and no genuinely new second head is possible: original second arcs and convenient added arcs reach original first or second neighbours, while no chosen bad direction fails at f . The tournament feed SNP transfers to D . Thus every global median order of this fixed T must end at v .

If v has any added outgoing arc $v \rightarrow q$, it is a canonical good direction. Retain it and reverse all other added outgoing arcs of v . The safe retention argument for good directions has no restriction on the number or budgets of other failure sources: for a protected far head y of v , convenience of $v \rightarrow q$ implies convenience of $y \rightarrow q$, and canonical compatibility excludes the added direction $q \rightarrow y$. Original $q \rightarrow y$ is excluded by protection. Consequently

$$d_{T'}^+(v) - d_{T'}^{++}(v) \geq g(v) + 2 - t(v) \geq 2,$$

contradicting the feed theorem for the unchanged median order. Therefore v has no added outgoing arc, and $N_T^+(v) = N_D^+(v)$.

Any new second head of v is an unprotected original far target. It follows that

$$d_T^{++}(v) \leq d_D^{++}(v) + t(v) \leq d_D^+(v) = d_T^+(v).$$

The feed SNP forces equality throughout, so the strong form of the Havet–Thomasse feed theorem gives $|N_T^+(v)| = |G_L| = |N_T^{++}(v)|$. Its global median sedimentation lemma [7, Theorem 1 and Lemma 1] produces another global median order of the same T , with v placed before $N_T^+(v) \cup G_L$. This last set is nonempty because D is all-deficient. The new last vertex is not v , contradicting the uniqueness of the possible feed. This proves the lemma. $\square$

Theorem S10.7 (Two failure sources, one with a unit budget). *Let D be a finite nonempty oriented graph. Suppose every directional failure source of every bad missing pair belongs to $\{u, v\}$, and $t(v) \leq 1$. Then D has a Seymour vertex.*

Proof. If all pairs are good, use the good-missing-edge theorem. Otherwise, each bad pair has opposite failure witnesses u, v . Neither directional failure set can contain both: the two sets are disjoint, and both are nonempty. Each such pair consequently has a direction whose only failure source is v . Every missing pair incident to v is good, since a bad one would need two distinct failure sources outside its endpoints, leaving only u . If $g(v) \leq 0$, v already has the SNP. Otherwise $t(v) \leq 1 \leq g(v)$, and Lemma S10.6 applies. $\square$

Corollary S10.8 (No two-source residual distribution with a unit). *In a strongly connected all-deficient graph minimal under deletion of arbitrary nonempty arc sets, a positive-residual support of two vertices cannot have residuals $(a, 1)$ for any positive integer a . In particular residual four cannot be distributed as $(3, 1)$, and residual five cannot be distributed as $(4, 1)$.*

Proof. Zero-residual source protection places every bad-edge directional failure source in the positive support. The residual-one estimate gives $t \leq 1$ at its unit vertex. Apply Theorem S10.7. $\square$

S11 A simultaneous completion with two failure sources

All convenience, protection, far sets and failure sets below are defined in the original oriented graph D . Write $y \triangleleft z$ when $y \rightarrow z$ and $N_D^-(y) \subseteq N_D^-(z) \setminus \{y\}$. This is a strict partial order; it implies $y \in U_D(z)$ and that reaching y within two steps reaches z within two steps. Write $\mathcal{F}(z)$ for the unprotected far targets and $t(z) = |\mathcal{F}(z)|$. Good missing pairs and convenient directions have Ghazal's meaning [5].

Fix a linear extension of $\triangleleft$. Canonically orient each good missing pair in its unique convenient direction, or according to that extension if both directions are convenient. The compatibility rule is

$$\begin{aligned} y \triangleleft z, \quad z \rightarrow q \text{ canonically added,} \quad qy \text{ missing} \\ \implies y \rightarrow q \text{ is the added direction.} \end{aligned} \tag{UT1}$$

Indeed $G(z \rightarrow q)$ implies $G(y \rightarrow q)$ by predecessor containment, so qy is good. If $G(q \rightarrow y)$ also holds, reachability through the protected target gives $G(q \rightarrow z)$. Choosing $z \rightarrow q \rightarrow y$ would then require $z \prec q \prec y$ in the reference extension, contrary to $y \triangleleft z$.

S11.1 The bad pairs form a uniformly orientable complete bipartite set

Lemma S11.1. *Suppose D has at least one bad missing pair and all directional failure sources of bad pairs lie in $\{u, v\}$. Then uv is missing, every missing pair incident to u or v is good, and, with*

$$A = N_D^+(u) \cap U_D(v), \quad B = N_D^+(v) \cap U_D(u),$$

the bad pairs are exactly $A \times B$. The sets A, B are nonempty, disjoint and avoid u, v . Orienting all bad pairs as $A \rightarrow B$ makes u their only failure source.

Proof. Opposite failure witnesses are distinct, avoid both endpoints, and are nonadjacent: either arc between them would create a forbidden two-step path. With only two possible witnesses, they must be u, v , and each directional failure set of a bad pair is a singleton.

This proves that uv is missing. A bad pair incident to one of u, v would have at most one available failure source, impossible.

Label the endpoints of a bad pair so that $u \rightarrow a$, $b \in U_D(u)$, $v \rightarrow b$ and $a \in U_D(v)$. Then $a \in A, b \in B$. Conversely, for $a \in A, b \in B$, an arc $a \rightarrow b$ would let u reach b within two steps, while $b \rightarrow a$ would let v reach a . Thus ab is missing and bad, with precisely these two witnesses. The remaining assertions follow from the definitions. $\square$

Lemma S11.2 (Exact simultaneous transfer at protected sources). *Under the hypotheses of Lemma S11.1, orient all bad pairs as $A \rightarrow B$ and all good pairs canonically, obtaining a tournament T . For every $z \notin \{u, v\}$ with $t(z) = 0$, put $K_z = N_T^+(z) \setminus N_D^+(z)$. Then*

$$N_T^{++}(z) = N_D^{++}(z) \setminus K_z, \quad g_T(z) = g_D(z) + 2|K_z|. \quad (\text{UT2})$$

This lemma does not assume that D is all-deficient or arc-minimal.

Proof. Every missing partner of z is an exact second outneighbor: if it were far, it could not satisfy the directed adjacency required for protection, contrary to $t(z) = 0$. Hence $K_z \subseteq N_D^{++}(z)$.

Let $y \in U_D(z)$. It is protected for z . We rule out every possible path $z \rightarrow a \rightarrow y$ in T . If a is an original outneighbor, an original or convenient second arc would make y originally reachable. A chosen bad second arc fails only at u , not at z , so it also cannot introduce this far head.

If $z \rightarrow a$ is an added good arc, an original $a \rightarrow y$ would force $a \rightarrow z$, contradicting that za was missing. If ay is missing, (UT1) orients it as $y \rightarrow a$.

Finally, if $z \rightarrow a$ is an added bad arc, $z \in A$ and $a \in B$. Thus z is far from v , and $v \rightarrow a$. Protection of y makes y far from v as well. An original $a \rightarrow y$ is again impossible by protection. If ay is good, the direction $a \rightarrow y$ fails at v , so its only convenient direction is $y \rightarrow a$. If ay is bad, $a \in B$ implies $y \in A$, and the uniform rule also chooses $y \rightarrow a$. This last case explicitly includes two consecutive added bad arcs.

No original far target of z becomes reachable within two steps. Original two-step paths are retained, and the only original second neighbors excluded from the exact second neighborhood are those in K_z , now first neighbors. This proves the equality and its degree consequence. $\square$

S12 At least three vertices with unprotected far targets

The following argument uses protection rather than the numerical residual of vertices outside the two candidate sources. In particular, it is stronger than excluding a two-vertex positive residual support. All far sets, convenience conditions, and protected pairs refer to the original oriented graph D .

Theorem S12.1. *Let D be a finite nonempty strongly connected all-deficient oriented graph minimal under deletion of arbitrary nonempty arc sets. Then at least three vertices have an unprotected far target:*

$$|\{x : t(x) > 0\}| \geq 3.$$

Proof. A directional failure witness of a bad missing pair always has an unprotected far target: a witness in the opposite direction would, under protection, create a forbidden two-step path. Thus, if at most one vertex had an unprotected far target, every missing pair would be good. Ghazal's good-pair theorem would give a Seymour vertex [5, Theorem 2].

Suppose therefore that the only vertices with $t(x) > 0$ are u, v . There is a bad pair, again by the good-pair theorem, and all its directional failure sources lie in $\{u, v\}$. Lemma S11.1 gives the missing pair uv , goodness of every missing pair incident to either vertex, and the nonempty sets

$$\mathcal{A} = N_D^+(u) \cap U_D(v), \quad \mathcal{B} = N_D^+(v) \cap U_D(u).$$

The bad pairs are exactly $\mathcal{A} \times \mathcal{B}$. Orient all of them as $\mathcal{A} \rightarrow \mathcal{B}$, so their only directional failure source is u .

We will canonically orient all good pairs using a reference linear extension in which every original missing partner of v precedes v . Such an extension exists. Missing partners are incomparable with v in the protected partial order: a protected chain would force an arc. Adding all precedence requirements $r \prec v$ for $r \in \mathcal{M}(v)$ cannot create a cycle, because such a cycle would require an original protected chain from v to one of these partners. Let T be the resulting tournament. By Lemma S11.2, every vertex outside $\{u, v\}$ remains deficient in this one tournament. It is enough to prove that v also remains deficient.

The empty far-in set. If $P(v) = \emptyset$, every other vertex reaches v within two steps. Thus $r \rightarrow v$ is convenient for every missing partner r . The chosen reference extension makes every added incident arc point into v , whether the reverse direction is also convenient or not. No first neighbour is added at v . Its original first neighbours cannot introduce a new far second head: good directions are convenient, and the selected bad directions fail only at u . Hence $g_T(v) = g_D(v) > 0$.

Assume henceforth that $P = P(v) \neq \emptyset$ and put

$$\begin{aligned} Q &= N_D^{--}(v), \quad C = \partial P, \quad X = \partial^2 P, \quad H = Q \setminus C, \\ B &= H \cap \mathcal{M}(v), \quad k = |B|, \quad s = |H \cap N_D^+(v)|. \end{aligned}$$

Delete the s outgoing arcs from v into H , obtaining $\widehat{D}$; when $s = 0$ leave D unchanged. Define

$$E = (\mathcal{M}(v) \setminus B) \dot{\cup} X, \quad L = E \setminus N_D^{++}(v), \quad p = |P|.$$

Lemma S8.9 gives

$$|L| \leq \lambda := \begin{cases} \eta(v) - 2k, & s = 0, \\ \eta(v) - g_D(v) - 2k, & s > 0. \end{cases} \quad (\text{S12.1})$$

It also gives convenience of $r \rightarrow v$ for every $r \in B$.

Which missing partners become first neighbours. Write $K = N_T^+(v) \setminus N_D^+(v)$ and $h = |K|$. Our reference order selects $r \rightarrow v$ whenever that direction is convenient. Consequently every $r \in K$ has an original predecessor $a \in P(v)$ with $a \rightarrow r$: otherwise $r \rightarrow v$ would be convenient. Every exceptional partner in B is directed into v , so $K \subseteq \mathcal{M}(v) \setminus B \subseteq E$.

For a missing partner $r \notin B \cup \{u\}$, one has $t(r) = 0$. It cannot belong to $P(v)$, because its far target v would then be protected, forcing the original arc $v \rightarrow r$ despite the missing pair. Thus $r \in C$, and some $a \in P$ points to it. The direction $r \rightarrow v$ is not convenient, so the good pair rv is canonically oriented as $v \rightarrow r$. The only nonexceptional partner which might instead be directed into v is u . Therefore

$$h \geq \mu(v) - k - 1. \quad (\text{S12.2})$$

A shared envelope for all added first intermediates. We claim

$$N_T^{++}(v) \subseteq (N_D^{++}(v) \cup L) \setminus K, \quad K \subseteq N_D^{++}(v) \cup L. \quad (\text{S12.3})$$

Original first intermediates do not introduce far heads, because the chosen bad directions fail only at u and good directions are convenient. Consider an added first intermediate $r \in K$ and an original far target $y \in U_D(v) \setminus L$. The original partner-budget lemma already rules out $r \rightarrow y$ in D .

If ry was missing, it is good: v originally points to neither r nor y , so it cannot witness a failure of either direction; the sole remaining possible source u cannot supply both opposite witnesses. Choose $a \in P$ with $a \rightarrow r$, as above. Every path of length at most two from P stays inside $P \cup C \cup X$. Moreover every vertex of this union which is originally far from v lies in L . Indeed, such a vertex is either a missing partner outside B or a member of X , hence is in E ; an originally far target cannot become reachable after deleting arcs and so is not in $N_{\hat{D}}^{++}(v)$. Thus $y \notin P \cup C \cup X$ and $y \in U_D(a)$. The direction $r \rightarrow y$ is not convenient, so the good pair ry is oriented uniquely as $y \rightarrow r$. This excludes a new path through r to y and proves the first inclusion.

For the second inclusion, $K \subseteq E$. If a member of K belongs to $N_{\hat{D}}^{++}(v)$, its two-step path already existed in D ; it was not an original first neighbour, since it was missing from v . It therefore belongs to $N_D^{++}(v)$. Otherwise it belongs to L . Notice that L may also contain original second neighbours lost in $\hat{D}$, and may contain deleted original first neighbours. No disjointness between L and $N_D^{++}(v)$ is assumed.

Removing the whole set K from the union in (S12.3) gives

$$d_T^{++}(v) \leq d_D^{++}(v) + \lambda - h, \quad g_T(v) \geq g_D(v) + 2h - \lambda. \quad (\text{S12.4})$$

If $h = 0$, its first neighbourhood is entirely original and has no chosen-direction failures, so $g_T(v) = g_D(v) > 0$ independently of this upper estimate. We can thus assume $h > 0$.

The nonempty deleted bundle. If $s > 0$, substitute (S12.2) and (S12.1) into (S12.4):

$$g_T(v) \geq 2g_D(v) + 2\mu(v) - 2 - \eta(v) = p + g_D(v) - 1 \geq 1.$$

The equality uses the original identity $p = 2\mu(v) + g_D(v) - 1 - \eta(v)$. Only the original graph's arc-set minimality is used to obtain the nonempty-bundle bound in (S12.1).

No deleted outgoing arc and the singleton exception. If $s = 0$, the same substitution gives $g_T(v) \geq p - 1$. This is positive when $p \geq 2$. For $p = 1$ it is positive unless $h = \mu(v) - k - 1$: any larger integer h improves the estimate by two. In this last case $u \notin B$ and the canonical direction on uv is $u \rightarrow v$. Write $P = \{a\}$. If $a \neq u$, then u is a nonexceptional missing partner outside P , hence belongs to C . Thus $a \rightarrow u$, while $v \in U_D(a)$. This makes $u \rightarrow v$ nonconvenient, contradicting its canonical choice. If $a = u$, then $C = N_D^+(u)$. Since $s = 0$, all of $H = B$ consists of missing partners of v , as does the singleton $P = \{u\}$. The partition into P, C, H therefore gives $N_D^+(v) \subseteq C = N_D^+(u)$. But then $\mathcal{B} = N_D^+(v) \cap U_D(u) = \emptyset$, contrary to the nonempty bad-pair bipartition. This excludes the only zero-gap estimate case.

Two Seymour vertices in one fixed tournament. We have proved $g_T(v) > 0$ for the selected completion. Every other vertex except possibly u has $t(z) = 0$ and remains deficient by Lemma S11.2. Thus T has at most one Seymour vertex. The original all-deficient graph has positive out-degree at every vertex, and adding arcs cannot create a sink. Havet–Thomasse's two-Seymour-vertex theorem for tournaments with no sink [7, Theorem 2] is a contradiction. The comparison is in this one tournament, not between completions reoriented separately for different candidate feed vertices. $\square$

Corollary S12.2. *Every finite nonempty strongly connected arc-minimal all-deficient oriented graph has at least three vertices with positive residual:*

$$|\{x : \eta(x) > 0\}| \geq 3.$$

In particular, a two-vertex positive residual support is impossible, without any upper bound on either residual.

Proof. Zero residual makes every far target protected, so $\eta(x) = 0$ implies $t(x) = 0$. The preceding theorem therefore applies. $\square$

Remark S12.3. The theorem bounds the number of vertices with an unprotected far target, not the number which actually occur as directional failure sources of bad missing pairs. A third vertex with $t > 0$ need not be a bad-pair witness; no claim that there must be three actual bad failure sources is made here.

S13 Unit failure heads and safe retained completion arcs

All reachability, convenience, protection, and failure sets in this section are defined in the original graph D . For a far target $y \in U_D(x)$, protection means

$$y \rightarrow x, \quad N_D^-(y) \subseteq N_D^-(x) \setminus \{y\}.$$

Write $\mathcal{F}(x)$ for the unprotected far targets of x , and $t(x) = |\mathcal{F}(x)|$. A protected target has the useful reachability implication

$$\begin{aligned} & y \text{ protected for } x \\ \implies & [z \rightsquigarrow y (\leq 2) \implies z \rightsquigarrow x (\leq 2)]. \end{aligned}$$

This follows directly from the incoming-neighborhood containment.

For a missing pair ab , let

$$F_{ab} = \{x : x \rightarrow a, b \in U_D(x)\}.$$

The direction $a \rightarrow b$ is convenient precisely when F_{ab} is empty. The pair is bad precisely when both F_{ab} and F_{ba} are nonempty. If it is bad and $x \in F_{ab}$, then $b \in \mathcal{F}(x)$: an opposite-direction witness would force an arc toward x if b were protected, contradicting the cross-witness nonadjacency.

Conveniently orient all good missing pairs using the canonical rule: choose the only convenient direction when unique; if both directions are convenient, follow a fixed linear extension of the protected-target strict partial order. This rule has the compatibility property

$$\begin{aligned} & y \text{ protected for } f, \quad f \rightarrow q \text{ canonically added good arc,} \\ & qy \text{ missing} \implies y \rightarrow q \text{ is chosen.} \end{aligned} \tag{UF1}$$

In particular, no cardinality bound on the failure-source set is needed here. Indeed $G(f \rightarrow q)$ implies $G(y \rightarrow q)$, since every predecessor of y is a predecessor of f ; hence qy is automatically good. If its opposite direction is also convenient, then $G(q \rightarrow y)$ implies $G(q \rightarrow f)$ by the reachability implication above. Both pairs would then be oriented by the reference order, and the choices $f \rightarrow q \rightarrow y$ would contradict y preceding f .

Lemma S13.1 (Safe retention with unit failure heads). *Suppose every directional failure source of a bad missing pair has $t(x) \leq 1$. Complete the good pairs canonically, and orient the bad pairs arbitrarily to form a tournament T . For any added arc $f \rightarrow q$, reverse all the other added out-arcs of f inward to form T' . Then*

$$N_{T'}^+(f) = N_D^+(f) \dot{\cup} \{q\},$$

$$N_{T'}^{++}(f) \subseteq (N_D^{++}(f) \setminus \{q\}) \cup (\mathcal{F}(f) \setminus \{q\}), \quad (\text{UF2})$$

and consequently

$$d_{T'}^{++}(f) \leq d_D^{++}(f) + t(f) - 1.$$

The retained pair fq may be good or bad.

Proof. The first-neighborhood assertion is immediate. Consider a second-neighborhood path $f \rightarrow a \rightarrow y$ in T' . If a is an original outneighbor of f , an original or convenient second arc makes y originally reachable within two steps. A nonconvenient bad second arc can only introduce an unprotected far head of f . Second arcs in such a path are untouched by the modification, since they do not involve f .

Now let $a = q$. A protected far target y of f cannot receive an original arc from q : that would force $q \rightarrow f$, whereas fq is missing. If qy is missing and fq is good, its chosen direction is $y \rightarrow q$ by (UF1).

If fq is bad, choose an opposite-direction failure source r : $r \rightarrow q$ and $f \in U_D(r)$. Protection of y forces $y \in U_D(r)$ as well. Thus the direction $q \rightarrow y$ fails at r . If qy were bad, its opposite witness would make $y \in \mathcal{F}(r)$, while the bad pair fq already makes $f \in \mathcal{F}(r)$. These two distinct heads contradict $t(r) \leq 1$. Therefore qy is good, with $y \rightarrow q$ its only convenient direction. The arc $q \rightarrow y$ again cannot occur.

All new second heads are therefore in $\mathcal{F}(f)$, and q itself is excluded because it is now a first neighbor. The two sets in (UF2) are disjoint. Originally missing fq puts q in exactly one of $N_D^{++}(f)$ and $U_D(f)$; in the latter case q is necessarily unprotected. Excluding it removes one unit from their combined cardinality, proving the final estimate. $\square$

S13.1 A completion whose sink sources avoid failure

Suppose all bad-edge directional failure sources lie in a set Z , and all have $t \leq 1$. Orient every missing pair inside Z first, using the canonical rule on good pairs. Let K consist of these oriented internal missing pairs and let $S = \{z \in Z : d_K^+(z) = 0\}$. The set S contains no missing pair. Assume the already chosen bad internal directions have no failure source in S . Every remaining bad pair can then be oriented to avoid failures in S : otherwise opposite witnesses in S would form a missing pair. The remaining pairs do not change K .

If $t(z) \leq g(z) + 1$ for $z \in Z \setminus S$, this completion gives a Seymour vertex in D . To see this, choose a global median order and its last vertex f . If f is outside $Z \setminus S$, reverse all its added out-arcs inward; all its remaining first arcs are original, and it is not a failure source of any chosen direction, so its tournament SNP transfers directly to D . Otherwise retain any added arc $f \rightarrow q$ in K , reversing the other added out-arcs inward. The order remains median. By Lemma S13.1, its new deficiency is at least $g(f) + 2 - t(f) \geq 1$, contradicting the tournament feed theorem. This reasoning does not infer SNP merely from acyclicity of a dependency digraph.

S13.2 Four unit failure sources reduce to one reciprocal cycle

Proposition S13.2. *Let D be a finite nonempty all-deficient oriented graph. Let Z be the union of the directional failure-source sets of all bad missing pairs. Suppose $|Z| \leq 4$ and $t(x) \leq 1$ for every $x \in Z$. Then the only case not excluded by the preceding completion argument has the following structure:*

1. *The four failure-source vertices induce a directed cycle of length four, with the two diagonals missing.*
2. *These diagonals are the only bad missing pairs of D , and lose to each other.*
3. *Every other missing pair is good.*
4. *Each failure-source vertex has exactly one unprotected far head, its incoming neighbor on the four-cycle.*

Proof. Every bad-edge directional failure source is in Z . An internal bad pair requires four distinct vertices of Z ; if $|Z| \leq 3$, the case of no internal bad pair below applies immediately. Otherwise $|Z| = 4$. For each active such source, write $h(x)$ for its unique unprotected far head.

Two bad missing pairs inside Z cannot share an endpoint. Suppose ab and ac were both bad, and let d be the fourth vertex. The witnesses for ab must be c, d . As ac is missing, c cannot point to a ; hence $c \rightarrow b$, $a \in U_D(c)$, and the opposite witness has $d \rightarrow a$ and $b \in \mathcal{F}(d)$. But d is also a directional failure source for ac , which would give it an unprotected head in $\{a, c\}$, different from b . This contradicts $t(d) \leq 1$. Thus internal bad pairs form a matching, with at most two members.

If there is no internal bad pair, use the canonical internal directions. If there is exactly one, say ab , its two witness sources c, d form a missing pair which is good. Its canonical direction makes one of c, d , say c , a nonsink of K . Orient ab in the direction failed by c . There are no other possible directional failure sources for that pair, because its endpoints are in Z and only c, d remain. This internal direction therefore avoids every sink of K . External bad pairs can be oriented to avoid those sinks as above. The threshold $t \leq 1 \leq g + 1$ applies to every nonsink source, so both cases contradict that D is all-deficient.

It remains that there are two internal bad pairs, say ab and cd . Label their witnesses so that $c \rightarrow a$, $d \rightarrow b$, $h(c) = b$ and $h(d) = a$. The witnesses for cd are necessarily a, b . Because opposite arcs are forbidden, they must satisfy $a \rightarrow d$, $b \rightarrow c$, $h(a) = c$ and $h(b) = d$. Thus the four cross pairs are precisely $c \rightarrow a \rightarrow d \rightarrow b \rightarrow c$; the two diagonals are missing and lose to each other.

Every bad pair has its two endpoints among the heads of its opposite failure witnesses. All four heads just identified belong to Z , so there can be no other bad pair outside Z . The internal matching classification leaves only ab and cd . The full dependency digraph consequently consists of this one two-cycle and isolated missing pairs. $\square$

Theorem S13.3 (At most four unit failure sources). *Let D be any finite nonempty oriented graph. If the union Z of the bad-edge directional failure sources has at most four vertices, and $t(x) \leq 1$ for every $x \in Z$, then D has a Seymour vertex.*

Proof. Suppose instead that D is all-deficient. By Proposition S13.2, only the reciprocal cycle remains. Label its vertices $0, 1, 2, 3$, with original arcs $0 \rightarrow 1 \rightarrow 2 \rightarrow 3 \rightarrow 0$, missing diagonals $02, 13$, and

$$h(0) = 3, \quad h(1) = 0, \quad h(2) = 1, \quad h(3) = 2.$$

These diagonals are the only bad pairs. Complete them by $0 \rightarrow 2$ and $1 \rightarrow 3$, and complete all good pairs canonically, obtaining T . The selected bad directions have failures only at 3 and 0, respectively. Take a global median order L of T and let f be its last vertex.

If $f \notin \{0, 3\}$, reverse all added outgoing arcs of f . The result still has median order L , has the original first neighbourhood of f , and has no new second head, because neither selected bad direction fails at f . Its feed SNP transfers to D , a contradiction. If $f = 0$, retain the added arc $0 \rightarrow 2$ and reverse the other added outgoing arcs. Lemma S13.1 gives deficiency at least $g(0) + 2 - t(0) > 0$, again contradicting the feed theorem. Consequently every global median order of this fixed T must end at 3.

If 3 has any added outgoing arc $3 \rightarrow q$, retain it and reverse the other added outgoing arcs. The same lemma gives positive deficiency at this median feed, since $t(3) = 1$. Thus 3 has no added outgoing arc at all, and $N_T^+(3) = N_D^+(3)$. The only possible new second head at 3 is its unique unprotected far head $h(3) = 2$. Therefore

$$N_T^{++}(3) \subseteq N_D^{++}(3) \cup \{2\}.$$

The feed SNP and $g(3) > 0$ imply

$$g(3) = 1, \quad d_T^+(3) = d_T^{++}(3).$$

Use the strong form of the Havet–Thomasse feed theorem [7, Theorem 1 and its proof]:

$$|N_T^+(3)| \leq |G_L| \leq |N_T^{++}(3)|,$$

where G_L consists of the incoming neighbours y of 3 for which some outgoing neighbour a occurs before y in L and satisfies $a \rightarrow y$. The equality just obtained forces $|N_T^+(3)| = |G_L|$. The sedimentation lemma [7, Lemma 1] then states that the order

$$\text{Sed}(L) = (\text{incoming vertices outside } G_L, 3, N_T^+(3) \cup G_L)$$

is still a global median order of the same tournament T , with each displayed group preserving its old internal order. Its last vertex is not 3, since $N_T^+(3)$ is nonempty in an all-deficient graph. This contradicts the fact that every global median order of T ends at 3. The reciprocal case is impossible as well. $\square$

Corollary S13.4 (No four-unit residual distribution). *In a strongly connected all-deficient graph minimal under arbitrary nonempty arc-set deletion, residual four cannot be distributed as $(1, 1, 1, 1)$. Combined with Proposition S10.5, this supplies an intermediate exclusion; all remaining residual-four allocations are excluded in the residual-exclusion part of this supplement.*

Proof. All bad-edge failure sources belong to the positive-residual support, and residual one gives $t \leq 1$. Apply Theorem S13.3. The support contains a missing pair, so a single positive vertex is impossible. The remaining integer partitions are as stated. $\square$

The sedimentation step uses a global median order and the equality $|N_T^+(f)| = |G_L|$. It does not infer the existence of two feed vertices from the existence of two Seymour vertices in a tournament, nor does it assert that an arbitrary tournament admits two median feeds.

S14 Bipartite witness graphs and nonnegative tournament gaps

Let $\mathcal{W}$ be the graph whose vertices are the directional failure sources of bad missing pairs. Join x, y when they are opposite failure witnesses for some bad pair. Its edges are original missing pairs. The two directional failure sets of each bad pair span a complete bipartite subgraph of $\mathcal{W}$.

Lemma S14.1 (Completion along a bipartition). *Suppose $\mathcal{W}$ is bipartite with parts L, R . Orient each bad missing pair in the direction whose failure sources belong to L , and all good pairs canonically relative to protection, obtaining T . For every vertex z , put $K_z = N_T^+(z) \setminus N_D^+(z)$. Then*

$$N_T^{++}(z) \subseteq (N_D^{++}(z) \setminus K_z) \cup (\mathcal{F}(z) \setminus K_z), \quad g_T(z) \geq g_D(z) + 2|K_z| - t(z).$$

If $z \notin L$ and $K_z = \emptyset$, its two neighbourhoods are unchanged. If $t(z) = 0$, the second-neighbourhood inclusion is the exact equality $N_T^{++}(z) = N_D^{++}(z) \setminus K_z$.

Proof. The direction choice is well-defined: the complete bipartite graph between the opposite failure sets makes each of these sets monochromatic, in opposite colours. Every chosen bad direction therefore fails only at vertices of L .

No added second arc through an original first neighbour reaches a protected far head: a good direction is convenient, and every failure head of a bad direction is unprotected. Through an added good first arc, the canonical protection compatibility rules out such a head. For an added bad first arc $z \rightarrow a$, choose an opposite witness $r \in R$, so $r \rightarrow a$ and $z \in U_D(r)$. If y is protected from z , then $y \in U_D(r)$ as well. An original $a \rightarrow y$ would force $a \rightarrow z$, contrary to the missing pair za . If ay is missing, its direction $a \rightarrow y$ fails at r . A good pair must therefore be oriented $y \rightarrow a$; for a bad pair, our choice avoiding R also forces $y \rightarrow a$. This checks all new first intermediates and all possible second arcs.

Thus every genuinely new far second head belongs to $\mathcal{F}(z)$. Every added first neighbour was missing from z , hence lies in exactly one of the disjoint sets $N_D^{++}(z)$ and $\mathcal{F}(z)$. Removing the entire K_z proves the count. For $z \notin L$ with no new first neighbours, no chosen bad direction can fail at z , so there are no new second heads. Original paths remain. The exact assertion for $t(z) = 0$ follows in the same way. $\square$

Lemma S14.2 (Only one selected failure source). *If every bad missing pair can be oriented so that its chosen directional failure sources lie in $\{c\}$, the preceding conclusion holds for canonical good directions and these bad directions. In particular, every $z \neq c$ with $t(z) \leq 1$ and $g_D(z) > 0$ remains deficient in the same completion, regardless of whether K_z is empty.*

Proof. All cross-witness edges are incident to c : the chosen failure set of every bad pair is precisely $\{c\}$, and the opposite set excludes c . Thus use $L = \{c\}$ in the previous lemma. If K_z is empty its gap is unchanged; otherwise $g_T(z) \geq g_D(z) + 2 - t(z) > 0$. $\square$

Lemma S14.3 (The zero-gap set has no sink). *Let T be a finite nonempty tournament with no sink, and suppose $g_T(z) \geq 0$ for every vertex. Then its zero-gap vertices induce a tournament with no sink. In particular there are at least three of them.*

Proof. The unweighted tournament SNP theorem ensures a zero-gap vertex exists. Suppose the zero-gap set S has a sink c in $T[S]$. The absence of a sink in T gives an outgoing neighbour h of c outside S . Assign weights $w(c) = 5/4$, $w(h) = 9/8$, and weight one to all

other vertices. The weighted gap at c is $1/8$. Every other zero-gap vertex points to c , so its weighted gap is at least $1/4 - 1/8 = 1/8$. Every originally positive-gap vertex has integer gap at least one, and its weighted gap is at least $1 - 1/4 - 1/8 > 0$. All weighted gaps are therefore strictly positive, a contradiction to the weighted tournament SNP theorem [3, Proposition 2.1].

For this rational-weight use, that theorem follows directly from the unweighted tournament theorem without assumptions about a weighted median optimization: multiply the weights by eight, replace each vertex by a transitive tournament of its integer weight, and orient between blocks as in T . Every vertex in a block has external gap equal to eight times the rational weighted original gap, and nonnegative additional internal gap. If all weighted gaps were positive, the expanded tournament would be all-deficient, contradicting the unweighted theorem. A tournament on at most two vertices always has a sink, proving the last assertion. $\square$

Part II

Residual exclusions and unbounded source control

Theorem S14.4 (A four-unit residual margin). *Every nonempty strongly connected arc-minimal all-deficient oriented graph on n vertices satisfies*

$$\sum_x \eta(x) = 2M - n \geq 4, \quad M \geq \left\lceil \frac{n}{2} \right\rceil + 2.$$

Equivalently, $|A(D)| \leq \lfloor n(n-2)/2 \rfloor - 2$.

Proof. The established bad-edge witness lemma implies that the set $Z = \{x : \eta(x) > 0\}$ contains a missing pair, and therefore the total residual is at least two. It also implies that every failure source of every bad missing edge lies in Z .

If the total residual is two, there are exactly two positive vertices, both of residual one. If it is three and has three positive vertices, all three likewise have residual one. In either case $|Z| \leq 3$, and Corollary S8.7 gives $t(z) \leq 1 \leq g(z) + 1$ for every $z \in Z$. The small-failure-source transfer lemma (Lemma S9.2) therefore gives a Seymour vertex, a contradiction. This application uses the transfer lemma's canonical convenient orientations and does not assume that an arbitrary convenient completion preserves all good missing edges.

It remains to exclude total residual three with $Z = \{u, v\}$, $\eta(u) = 2$, and $\eta(v) = 1$. The pair uv is missing. There is a bad missing edge, since otherwise the good-edge completion argument would give a Seymour vertex. Its two directional failure sources must be u, v . Consequently $N^+(v) \cap U(u) \neq \emptyset$. Apply Lemma S8.8.

If $t(u) \leq 2$, both positive vertices satisfy the transfer threshold $t(z) \leq g(z) + 1$, so Lemma S9.2 again gives a contradiction. Otherwise that dichotomy supplies convenience of $v \rightarrow u$. Choose the reference linear extension of the protected relation with v before u . This is possible because missing partners are incomparable in that relation; a protected chain would, by transitivity, force an arc between them. The canonical convenient orientation of uv is then $v \rightarrow u$: it is either the only convenient direction or the direction selected by the reference order. In the transfer lemma's internal missing-edge digraph on Z , the only vertex with positive out-degree is now v , and $t(v) \leq 1 \leq g(v) + 1$. The vertex u is a zero-out-degree vertex; the transfer construction directs bad missing edges to avoid its failures and requires no bound on $t(u)$. Thus the lemma applies once more, a contradiction.

The total residual is an integer, so it is at least four. Finally $2M - n$ has the parity of n : for odd n it is at least five. Both parity cases give the displayed missing-pair bound. $\square$

S15 Excluding residual four and the parity-sensitive margin

Theorem S15.1. *Let D be a finite nonempty strongly connected all-deficient oriented graph minimal under deletion of arbitrary nonempty arc sets. Then*

$$R = 2M - n = \sum_x \eta(x) \geq 5.$$

In particular,

$$M \geq \left\lceil \frac{n}{2} \right\rceil + 2 + \mathbf{1}_{\{n \text{ even}\}}.$$

For even n the residual is at least six, and for odd n it is at least five.

Proof. The earlier four-unit margin gives $R \geq 4$. If $R = 4$, at least three residuals are positive by Corollary S12.2. The only possible integer partitions are consequently $(2, 1, 1)$ and $(1, 1, 1, 1)$. The former is excluded by Proposition S10.5; the latter by Theorem S13.3 and the unit-residual bound $t \leq 1$. Thus $R \neq 4$, giving $R \geq 5$. Since $R = 2M - n$ has the parity of n , even order requires $R \geq 6$. Integer rounding gives the asserted missing-pair bound. $\square$

Corollary S15.2 (All arc-set-minimal counterexamples). *Let D be any finite nonempty all-deficient oriented graph minimal under deletion of arbitrary nonempty arc sets. Let S be a sink strongly connected component, with $k = |S|$ and $o = n - k$. Then*

$$M(D) \geq \left\lceil \frac{k}{2} \right\rceil + 2 + \mathbf{1}_{\{k \text{ even}\}} + \left\lceil \frac{o(k-2)}{2} \right\rceil \geq \left\lceil \frac{n}{2} \right\rceil + o + 2 + \mathbf{1}_{\{n \text{ even}\}}.$$

In particular the parity-sensitive uniform bound holds without strong connectivity. Equality in that uniform bound forces strong connectivity.

Proof. The sink component inherits all-deficiency and arc-set minimality, as established in the previous sink-component argument. Apply Theorem S15.1 to $D[S]$. The unchanged outside estimate is

$$M(D[V \setminus S]) + M(S, V \setminus S) \geq \left\lceil \frac{o(k-2)}{2} \right\rceil.$$

This proves the first inequality. Every all-deficient graph has minimum outdegree at least two, so $k \geq 5$. Thus the outside term is at least $\lceil 3o/2 \rceil$. For $n = k + o$, the elementary parity inequality

$$\left\lceil \frac{k}{2} \right\rceil + \left\lceil \frac{3o}{2} \right\rceil + \mathbf{1}_{\{k \text{ even}\}} \geq \left\lceil \frac{n}{2} \right\rceil + o + \mathbf{1}_{\{n \text{ even}\}}$$

follows from $\lceil 3o/2 \rceil = o + \lceil o/2 \rceil$: if n is even, either k is even or both k, o are odd, each giving the needed extra unit. If n is odd, the old rounding bound suffices. If $o > 0$, the result is strictly greater than the uniform bound, which proves the equality assertion. $\square$

At the odd boundary order $n = 2\delta + 3$, the missing-pair bound remains $M \geq \delta + 4$. At the first even order above it, $n = 2\delta + 4$, the bound improves to $M \geq \delta + 5$.

The next sections exclude residual five and improve the odd-order bound as well.

S16 Three positive residuals cannot include two units

All far sets, protection relations, convenience and failure sets are defined in the original graph. For a far target y of x , protection means $y \rightarrow x$ and $N^-(y) \subseteq N^-(x) \setminus \{y\}$. Write $\mathcal{F}(x)$ for the unprotected far targets and $t(x) = |\mathcal{F}(x)|$. As before, good missing pairs are oriented canonically: use the only convenient direction, or use a fixed linear extension of the protection partial order when both directions are convenient. Protected-target compatibility ensures that an added convenient $z \rightarrow a$ cannot be followed by an added arc from a to a protected far target of z .

S16.1 A simultaneous single-blame completion

Lemma S16.1. *Suppose every bad missing pair of an oriented graph D can be oriented so that all failure sources of its selected direction lie in $\{u\}$. Orient all bad pairs this way and all*

good pairs canonically, obtaining a tournament T . For every $z \neq u$, put $K_z = N_T^+(z) \setminus N_D^+(z)$. Then

$$N_T^{++}(z) \subseteq (N_D^{++}(z) \setminus K_z) \cup (\mathcal{F}(z) \setminus K_z), \quad (\text{TU1})$$

and

$$g_T(z) \geq g_D(z) + 2|K_z| - t(z).$$

If $K_z = \emptyset$, the second neighborhood is exactly unchanged.

Proof. All added first neighbors are missing partners, and therefore lie in the disjoint union $N_D^{++}(z) \dot{\cup} \mathcal{F}(z)$. We check that no protected far target y of z becomes a second neighbor in T .

If the first intermediate vertex is an original outneighbor, an original or convenient second arc cannot introduce an original far head. A selected bad second arc fails only at u , not at z . If the first arc $z \rightarrow a$ is added and good, protection excludes an original $a \rightarrow y$, and canonical compatibility chooses $y \rightarrow a$ whenever ay is missing.

Now let $z \rightarrow a$ be an added bad arc. Its opposite direction has a failure source $r \neq u$: $r \rightarrow a$ and $z \in U_D(r)$. The source cannot be u , since it already fails the selected direction. Protection makes $y \in U_D(r)$. An original $a \rightarrow y$ would force $a \rightarrow z$, contrary to missing za . If ay is missing, the direction $a \rightarrow y$ fails at r . It cannot be a selected convenient direction when ay is good, or a selected bad direction when ay is bad, because $r \neq u$. Thus the completion points from y to a . This explicitly rules out paths using two added bad arcs.

Every genuinely new far second head is therefore unprotected. Exclude all of K_z , now first neighbors, to obtain (TU1). The degree inequality follows by removing $|K_z|$ from the combined cardinality and adding $|K_z|$ to the first degree. If K_z is empty, every first intermediate is original and the first case above prevents every new far head, whether protected or not. Original second-neighborhood paths remain, so the second neighborhood is unchanged. $\square$

Corollary S16.2. *Under the preceding orientation hypothesis, if $t(z) \leq 1$ for every $z \neq u$, then D has a Seymour vertex.*

Proof. Suppose all vertices are deficient. For $K_z = \emptyset$, the completed degree gap is $g_D(z) > 0$. For $|K_z| \geq 1$, the bound gives $g_T(z) \geq g_D(z) + 2|K_z| - 1 > 0$. Thus every vertex except possibly u is deficient in the same T . All original outdegrees are positive; hence T has no sink. The published theorem giving at least two Seymour vertices in a sink-free tournament [7, Theorem 2] contradicts this simultaneous conclusion. $\square$

S16.2 A unit–unit witness pair is itself bad

The remainder assumes that D is strongly connected, all-deficient and minimal under deletion of arbitrary nonempty sets of arcs. No restriction on the order or minimum outdegree is imposed. The established residual and unit-target estimates are

$$\eta(x) = 2\mu(x) + g(x) - 1 - |P(x)| \geq 0, \quad \eta(x) = 1 \implies t(x) \leq 1.$$

Lemma S16.3 (The packet of an active unit source). *If $\eta(x) = 1$ and $t(x) = 1$, then $P(x) \neq \emptyset$ and*

$$\partial P(x) = N_D^-(x).$$

Proof. The sole unprotected far target prevents x from being whole: all far targets of a whole source are protected. Hence $\mu(x) \geq 1$, and $P(x) = \emptyset$ would give $1 = 2\mu(x) + g(x) - 1 \geq 2$.

So $P(x)$ is nonempty. Set $C = \partial P(x)$, $Q = N_D^-(x)$, $j = |Q \setminus C|$ and $s = |N_D^+(x) \cap (Q \setminus C)|$. The established target-bundle bounds give

$$s = 0 \Rightarrow \eta(x) \geq 2j, \quad s > 0 \Rightarrow \eta(x) \geq g(x) + 2(j - s).$$

If $s > 0$, unit residual forces $g(x) = 1$ and $j = s$. The equal-defect unprotected-target bound would then give $t(x) \leq \eta(x) - g(x) = 0$, a contradiction. Thus $s = 0$, and $1 \geq 2j$ forces $j = 0$. $\square$

Lemma S16.4. *Suppose opposite failure witnesses of a bad missing pair are vertices v, w with $\eta(v) = \eta(w) = 1$. Then vw is itself a bad missing pair.*

Proof. Label the original bad pair ab so that $v \rightarrow a$, $b \in U_D(v)$, $w \rightarrow b$ and $a \in U_D(w)$. The witness pair vw is missing. Cross-witness protection shows $\mathcal{F}(v) = \{b\}$ and $\mathcal{F}(w) = \{a\}$. The heads b, a are different from w, v , respectively, since $w \rightarrow b$ and $v \rightarrow a$ are not loops.

A missing partner which is far is necessarily unprotected. Thus the missing pair vw is reachable in both directions in exactly two steps: it uses neither source's sole unprotected head. Lemma S16.3 gives $w \in N_D^-(v) = \partial P(v)$ and $v \in N_D^-(w) = \partial P(w)$. Choose $p \in P(v)$ with $p \rightarrow w$, and $q \in P(w)$ with $q \rightarrow v$. Then v is far from p and w is far from q . The direction $w \rightarrow v$ fails at p , and $v \rightarrow w$ fails at q . Both directions fail, so vw is bad. $\square$

Theorem S16.5. *An arc-set-minimal strongly connected counterexample cannot have positive residual support $\{u, v, w\}$ with $\eta(v) = \eta(w) = 1$, regardless of the positive value $\eta(u)$. Equivalently, no allocation $(m, 1, 1)$ with $m \geq 1$ is possible. In particular the residual-five allocation $(3, 1, 1)$ is excluded.*

Proof. Every directional failure source of a bad missing pair has positive residual, by the zero-residual protection theorem. Since D is all-deficient, Ghazal's theorem [5, Theorem 2] guarantees at least one bad pair.

The vertices v, w cannot be opposite witnesses of any bad pair. Otherwise Lemma S16.4 would make vw bad as well. Its two opposite failure witnesses would have to be distinct positive-residual vertices outside $\{v, w\}$. Only u is available, a contradiction.

Consequently, for each bad pair, its two nonempty directional failure sets cannot have a unit source on each side. They are disjoint subsets of $\{u, v, w\}$. One direction therefore has no source in $\{v, w\}$ and has precisely u as its failure source. Choose this direction for every bad pair. In cross-witness graph language, the graph has no edge vw and all its nonempty edges lie in the star centered at u . This is an explicit choice of a single-blame orientation; no assertion about arbitrary acyclic dependency digraphs is used.

The unit estimate gives $t(v), t(w) \leq 1$. All other vertices except u have residual zero, hence $t = 0$. Corollary S16.2 now produces a Seymour vertex of D , the required contradiction. $\square$

S17 Five unit residuals and functional losing cycles

Throughout, D is a nonempty strongly connected all-deficient oriented graph, minimal under deletion of arbitrary nonempty sets of arcs. Let Z be the positive residual support, and let Z_{act} be the set of actual directional failure sources of bad missing pairs. Zero-residual protection gives $Z_{\text{act}} \subseteq Z$. For every unit-residual $x \in Z_{\text{act}}$, the unit budget and an actual bad failure give

$$t(x) = 1, \quad \mathcal{F}(x) = \{h(x)\}.$$

Thus every such source has a unique unprotected failure head $h(x)$, which is a far target distinct from x .

Define the missing-edge dependency digraph $\mathcal{L}$ as follows. Its vertices are all missing unordered pairs of D , and there is one arc

$$\{x, y\} \longrightarrow \{a, b\}$$

whenever $x \rightarrow a$, $b \in U(x)$, $y \rightarrow b$, and $a \in U(y)$, possibly after interchanging the endpoints. The witnesses and the two target endpoints are four distinct vertices. The source pair is missing. The digraph has no multiplicity: an arc records existence of witnesses, not their number. A missing pair is bad precisely when its dependency vertex has positive in-degree.

Lemma S17.1 (A unit cross-witness pair is itself bad). *If $\{x, y\}$ is the tail of an arc of $\mathcal{L}$ and $\eta(x) = \eta(y) = 1$, then the missing pair xy is bad. No restriction on the residuals of other vertices is required for this lemma. Moreover x and y reach one another at exact distance two.*

Proof. Both witnesses belong to Z_{act} and have residual one and $t = 1$. Fix x . Since xy is missing, $\mu(x) \geq 1$, and the identity $p(x) = 2\mu(x) + g(x) - 2$ gives $P(x) \neq \emptyset$. For the target-bundle quantities s, j , a positive s would force $j = s$ and $g(x) = 1$, and the unprotected-target envelope would give $t(x) \leq \eta(x) - g(x) = 0$. Hence $s = 0$, and the bound $\eta(x) \geq 2j$ gives $j = 0$. Consequently

$$\partial P(x) = N_D^-(x).$$

The same holds for y .

The unique unprotected head of x is the opposite target b , not y , since $y \rightarrow b$ and the witnesses are outside the target pair. If y were far from x , their missing pair would make y another unprotected far target. This contradicts $t(x) = 1$. Thus $y \in N_D^{++}(x)$, and symmetrically $x \in N_D^{++}(y)$.

It follows that $y \in N_D^-(x) = \partial P(x)$, so some $p \in P(x)$ points to y . Since $x \in U(p)$, this is a failure of the convenient direction $y \rightarrow x$. Likewise a predecessor of x in $P(y)$ supplies a failure of the direction $x \rightarrow y$. Both directions fail, so xy is bad. $\square$

Lemma S17.2 (All nontrivial unit dependency components are cycles). *Suppose every actual directional failure source has residual one. Then every nonisolated vertex of $\mathcal{L}$ has in- and out-degree one. In particular all its nontrivial components are directed cycles, and every bad missing pair has both endpoints in Z_{act} .*

Proof. A tail $\{x, y\}$ has only one possible target, namely $\{h(x), h(y)\}$, since every bad failure head is unprotected. Hence all out-degrees are at most one. Let E be the set of positive out-degree vertices. There are exactly $|E|$ arcs. By Lemma S17.1, every member of E is bad and has in-degree at least one. The total in-degree over E is therefore at least $|E|$, which already equals the total over all dependency vertices. Equality forces every arc head to lie in E , and every member of E to have in-degree one. All remaining vertices are isolated.

Every bad pair has positive in-degree and is thus in E . Every tail pair consists of actual failure sources, proving the last assertion. This is a degree-count argument; it is not a replacement of directed walks by simple paths, and it allows two-cycles. $\square$

Theorem S17.3 (No five-unit residual distribution). *Residual five cannot be distributed as five unit residuals:*

$$\sum_x \eta(x) = 5, \quad \eta(x) \in \{0, 1\} \text{ for every } x$$

is impossible in this minimal class.

Proof. Here $|Z| = 5$. If $|Z_{\text{act}}| \leq 4$, the general four-unit failure-source theorem, Theorem S13.3, already gives a Seymour vertex. Thus all five vertices are actual sources and $Z_{\text{act}} = Z$.

Let B be the undirected cross-witness graph on Z , whose edges are the tails E of dependency arcs. It has no isolated vertices: every actual failure has an opposite witness and hence participates in a tail pair. By Lemma S17.2, every bad pair is internal to Z , so every failure head $h(x)$ belongs to Z . The map

$$\{x, y\} \mapsto \{h(x), h(y)\}$$

permutes E , since it is exactly the successor map on the dependency cycles. As B has no isolated vertices, this edge surjectivity makes $h : Z \rightarrow Z$ surjective: each vertex appears in a target edge and therefore in the image of h . Thus h is a permutation. It has no fixed point, because $h(x) \in U(x)$.

The edge set of B is invariant under applying h to both endpoints. For every edge $\{x, y\} \in E$, its image is disjoint from it, and the original graph has the cross-arcs

$$x \rightarrow h(y), \quad y \rightarrow h(x). \quad (\text{U5})$$

Every edge of B , in contrast, is an original missing pair.

A derangement on five vertices has cycle type either five or three plus two. Suppose first that h is a five-cycle. The unordered pair orbits are its five consecutive pairs and its five distance-two pairs. A consecutive pair intersects its image, so none can belong to E . The other orbit is therefore the only possible nonempty invariant edge set, and E must consist of all five distance-two pairs. In particular $\{x, h^2(x)\} \in E$. Its required cross-arc in (U5) is $x \rightarrow h^3(x)$. But $\{x, h^3(x)\}$ is also a distance-two pair and belongs to E , so it is missing. This is a contradiction.

Now suppose h has a three-cycle on a, b, c and a transposition on d, e . Every pair within either cycle intersects its image and is forbidden in E . The six cross pairs between the two cycles form one orbit under h . Consequently the nonempty invariant edge set is all of $K_{3,2}$. The edge $\{a, d\}$ requires the original arc $a \rightarrow h(d) = e$, while $\{a, e\}$ belongs to the same edge set and is missing. This is again a contradiction. Both derangement types are excluded. $\square$

S18 Excluding the residual allocation (2,2,1)

Continue with a finite nonempty strongly connected all-deficient graph minimal under deletion of arbitrary nonempty arc sets. Let $\mathcal{B}$ be its cross-witness graph: two vertices are adjacent when they occur as opposite directional failure witnesses of a bad missing pair. Every edge of $\mathcal{B}$ is an original missing pair.

Lemma S18.1 (Large two-unit vertices have one witness partner). *If $\eta(x) = 2$ and $t(x) > 2$, then $d_{\mathcal{B}}(x) \leq 1$. For its sole witness partner c , when one exists, $c \rightarrow x$ is a convenient orientation.*

Proof. If $P(x) = \emptyset$, the identity $2 = 2\mu(x) + g(x) - 1$ and $g(x) \in \{1, 2\}$ force $\mu(x) = 1$, so there is only one missing partner. That partner is conveniently directed into x because every vertex reaches x within two steps. Otherwise the local residual and unprotected-target bounds force $s = 0, j = 1$ when $t(x) > 2$. The exceptional missing-partner set is $B_x = \{q\}$, and Lemma S8.9 has zero remaining head budget. Thus every missing partner $r \neq q$ satisfies $N_D^+(r) \cap U_D(x) = \emptyset$. A cross-witness partner necessarily has a nonempty such intersection, so it must be q . That lemma also gives convenience of $q \rightarrow x$. $\square$

Lemma S18.2 (A two-unit leaf in a source star). *Suppose the cross-witness graph is a star with center c , and v is a leaf with $\eta(v) = 2$. Orient every bad missing pair in the direction whose only failure source is c . Canonically orient good pairs using a reference extension which places all good missing partners of v before v . The resulting tournament T keeps v deficient. If $t(v) > 2$, the completion also adds the first arc $c \rightarrow v$.*

Proof. Each bad pair has one directional failure set equal to $\{c\}$: otherwise opposite noncentral witnesses would create an edge between leaves of the star. This justifies the prescribed orientation. Since cv is missing, a selected bad direction cannot be an outgoing arc from v : failure at c would require the original arc $c \rightarrow v$. Thus all added first neighbours of v come from good pairs.

The reference extension exists, because every missing partner of v is incomparable with v in the protected order. If $t(v) \leq 2$, the nonselected-source budget from Lemma S14.2 keeps v deficient: with no added first neighbour its gap is unchanged, and with at least one it is at least $g_D(v) + 2 - t(v) \geq 1$.

For $t(v) > 2$ and $P(v) = \emptyset$, one has $\mu(v) = 1$ and $g(v) = 1$, so the sole missing partner is c . The direction $c \rightarrow v$ is convenient and is selected by the reference extension. There are no added first neighbours at v , and its original first intermediates cannot create a far head because the selected bad directions fail only at c .

For $t(v) > 2$ and $P(v) \neq \emptyset$, the preceding lemma identifies the unique exceptional partner as c . Thus $s = 0, j = 1$ and the local envelope saturates:

$$E = (\mathcal{M}(v) \setminus \{c\}) \dot{\cup} \partial^2 P(v) = N_D^{++}(v).$$

Every vertex in $P(v) \cup \partial P(v) \cup \partial^2 P(v)$ is originally reachable from v within two steps. The convenient direction $c \rightarrow v$ is selected, so any added first neighbour r lies in $\mathcal{M}(v) \setminus \{c\} \subseteq E$. Because all good partners were placed before v , an added good direction $v \rightarrow r$ means that $r \rightarrow v$ is not convenient. Therefore there is an original $p \in P(v)$ with $p \rightarrow r$. In particular $p \neq c$, since c is outside $P(v)$. This argument applies to the actual added-first set; it does not assume that every nonexceptional partner has residual zero or belongs to $\partial P(v)$.

Let $y \in U_D(v)$. An original $r \rightarrow y$ is impossible by saturation. Moreover p cannot reach y within two steps: all such paths from $P(v)$ stay in the displayed reachable union. If ry is good, $r \rightarrow y$ is nonconvenient because it fails at p . If ry is bad, a selected direction $r \rightarrow y$ would likewise fail at $p \neq c$, contradicting the choice of c as its only failure source. Thus the completion never directs r toward y . Original first intermediates of v are also safe because bad directions fail only at c . Writing $K = N_T^+(v) \setminus N_D^+(v)$, we obtain

$$K \subseteq N_D^{++}(v), \quad N_T^{++}(v) = N_D^{++}(v) \setminus K, \quad g_T(v) = g_D(v) + 2|K| > 0.$$

The convenience and selection of $c \rightarrow v$ also prove the final assertion in both large-budget cases. $\square$

Theorem S18.3. *The positive residual allocation $(2, 2, 1)$ is impossible.*

Proof. Let the positive vertices be u, v, w , with residuals 2, 2, 1. Every other vertex has $t = 0$, while Corollary S8.7 gives $t(w) \leq 1$. All directional failure sources lie in these three vertices. The cross-witness graph $\mathcal{B}$ is nonempty. Every missing pair inside this three-vertex set is good: a bad pair would require two further distinct failure witnesses within the set.

If $\mathcal{B}$ is a triangle, the two residual-two vertices each have two witness partners, and Lemma S18.1 forces $t(u), t(v) \leq 2$. All source thresholds $t(x) \leq g(x) + 1$ now hold, so the generic cross-witness transfer Lemma S10.4 gives a Seymour vertex.

Suppose $\mathcal{B}$ has one edge incident to the unit vertex w . The actual directional failure sources are just the endpoints of that edge; Theorem S10.7 applies. Thus the remaining one-edge possibility joins u and v . Orient all bad pairs with u as their sole selected failure source, and apply Lemma S18.2 to the leaf v . The generic single-selected-source completion budget keeps the nonselected unit w deficient, and keeps all zero-residual vertices deficient. Only u can be a Seymour vertex of this fixed tournament.

Next let $\mathcal{B}$ be a two-edge path centered at a residual-two vertex, say u . Select u as the sole failure source of every bad direction, and protect the other residual-two leaf v by Lemma S18.2. The nonselected unit leaf w remains deficient by its one-unit budget. Again all vertices except possibly u remain deficient in the same tournament.

Finally let the path be centered at the unit vertex w . Select w as the sole failure source. If both heavy leaves satisfy $t \leq 2$, the generic nonselected-source budget keeps both deficient for any canonical completion. Only w can then be Seymour. Otherwise choose a heavy leaf with $t > 2$, say v , and use the reference extension of Lemma S18.2. This keeps v deficient and adds the good first arc $w \rightarrow v$. The center w has no incident bad pair, since a selected direction cannot fail at its own endpoint. Its new first neighbours are therefore good ones. The center's general completion budget gives

$$g_T(w) \geq g_D(w) + 2|K_w| - t(w) \geq g_D(w) + 1 > 0,$$

because $|K_w| \geq 1$ and $t(w) \leq 1$. Thus only the other heavy leaf u can be Seymour.

These cases exhaust all nonempty graphs on three vertices. In every non-transfer case we produced one fixed completed tournament with at most one Seymour vertex. It has no sink, because every original vertex had positive out-degree. Havet–Thomasse's theorem provides at least two distinct Seymour vertices in such a tournament [7, Theorem 2], a contradiction. $\square$

S19 Excluding the residual allocation (2,1,1,1)

Continue in a finite nonempty strongly connected all-deficient graph minimal under deletion of arbitrary nonempty arc sets. Suppose its positive residual vertices are a heavy vertex u with $\eta(u) = 2$ and three unit vertices with residual one. Every other vertex has residual zero and hence $t = 0$; every unit vertex has $t \leq 1$ by Corollary S8.7. Let $\mathcal{B}$ be the actual cross-witness graph on these four vertices. It is nonempty, and its edges are original missing pairs.

Theorem S19.1. *The positive residual allocation (2,1,1,1) is impossible.*

Proof. No unit–unit witness edge. If no edge of $\mathcal{B}$ joins two units, it is a star centered at u . Orient each bad pair so that its only selected failure source is u , and good pairs canonically. All other vertices remain strictly deficient in the same completed tournament by Lemma S14.2. There is at most one Seymour vertex. The tournament has no sink, contradicting Havet–Thomasse's two-Seymour-vertex theorem [7, Theorem 2].

Classification after a unit–unit edge. Let ab be a unit–unit edge of $\mathcal{B}$, and let c be the third unit. Lemma S16.4 shows that the original missing pair ab is itself bad. Its opposite witnesses must be the two remaining positive vertices u, c . After exchanging a, b if necessary, their original relations are

$$u \rightarrow a, \quad c \rightarrow b, \quad a \in U_D(c), \quad b \in U_D(u), \quad uc \in E(\mathcal{B}). \quad (\text{S19.1})$$

The unique unprotected far target of c is thus $h(c) = a$. The original arcs prohibit witness edges ua and cb . There is no witness edge ca either: a cross witness opposite c would have to point to its unique unprotected head a , impossible when that witness is a itself. Consequently $\mathcal{B}$ is either the matching $\{ab, uc\}$ or the path $a - b - u - c$. In particular this classification leaves no additional unit-unit edge or alternative four-source graph.

The path. If $ub \in E(\mathcal{B})$, the heavy vertex u has witness degree two, so Lemma S18.1 gives $t(u) \leq 2$. Use the bipartition $L = \{b, c\}$, $R = \{a, u\}$ and complete according to Lemma S14.1. Zero-residual vertices remain strictly deficient. The nonselected unit a remains strictly deficient: with no new first neighbour its gap is unchanged, and otherwise its gap is at least $g_D(a) + 2 - t(a) > 0$. The nonselected heavy u likewise remains strictly deficient, using $t(u) \leq 2$. The selected units b, c have gap at least zero, and strictly positive gap if they gain a first neighbour. Thus all tournament gaps are nonnegative, and its zero-gap set has at most two vertices. The tournament has no sink. This contradicts Lemma S14.3.

The matching and its two families of bad pairs. Suppose now $\mathcal{B} = \{ab, uc\}$. Each bad pair has singleton opposite failure sets corresponding to one of these two edges; otherwise the complete cross-witness graph between its two failure sets would create another edge in $\mathcal{B}$. Since $h(c) = a$, the bad pairs witnessed by u, c are all of the form ax , with selected direction $x \rightarrow a$ failing at c . Put $r = h(b)$ and $y = h(a)$. The units a, b witness precisely the single bad pair ry , with original arcs $a \rightarrow r$, $b \rightarrow y$, $y \in U_D(a)$ and $r \in U_D(b)$. Select $r \rightarrow y$, which fails at a . These two families are disjoint: the second pair avoids both a, b , whereas every pair in the first family contains a . This is the completion with bipartition $L = \{a, c\}$, $R = \{b, u\}$.

Observe that $r \neq u$: otherwise $a \rightarrow r = u$ would contradict the original arc $u \rightarrow a$ in (S19.1). Every selected bad arc incident to u therefore points into u . The star family cannot contain u as an endpoint, since its witnesses include u ; the sole other pair ry can contain u only as y , and is then directed $r \rightarrow u$.

Choose the reference extension for good directions so that all good missing partners of u precede u . If $P(u) = \emptyset$, every direction into u is convenient, so no first neighbour is added there. Its original first intermediates have no selected-direction failures, and u stays deficient. Otherwise let the target-bundle sets be

$$\begin{aligned} P &= P(u), \quad C = \partial P, \quad X = \partial^2 P, \quad H = N_D^{--}(u) \setminus C, \\ B &= H \cap \mathcal{M}(u), \quad k = |B|, \quad s = |H \cap N_D^+(u)|. \end{aligned}$$

Use the envelope E and remainder L_0 from Lemma S8.9. Its size satisfies

$$|L_0| \leq \lambda = \begin{cases} 2 - 2k, & s = 0, \\ 2 - g_D(u) - 2k, & s > 0. \end{cases}$$

Write $K = N_T^+(u) \setminus N_D^+(u)$. Every member of K comes from a good missing pair. By the reference choice its reverse direction is nonconvenient, so it has an original predecessor $p \in P$ pointing to it. Every exceptional partner in B is conveniently directed into u , and every bad incident direction is already incoming. Thus $K \subseteq \mathcal{M}(u) \setminus B \subseteq E$.

We will rule out a new far head outside L_0 through each added first intermediate. An original second arc is excluded by the partner-head budget. A good second arc is also excluded: a far target outside L_0 is outside $P \cup C \cup X$, so it is far from the chosen predecessor $p \in P$, making that outgoing direction nonconvenient. This is the same cone argument as in the shared-envelope proof, and does not suppose that p has residual zero. For a bad second arc, the star family has head a , already an original first neighbour of u . It cannot be a far

second head. The only other possibility is $r \rightarrow y$ through the added keeper r . If $y = u$, it is a self endpoint and is not a second neighbour. Assume $y \neq u$ and that this is a genuinely new far head outside L_0 . Its predecessor $p \in P$ is far from y , so p is a directional failure witness for $r \rightarrow y$. That direction's failure set is exactly $\{a\}$, forcing $p = a$.

A short protected-reachability obstruction. The preceding putative bad path gives $a \in P(u)$, so $u \in U_D(a)$. Since $t(a) = 1$, its only unprotected head is y , and $y \neq u$, the far target u must be protected for a :

$$u \rightarrow a, \quad N_D^-(u) \subseteq N_D^-(a) \setminus \{u\}.$$

The missing pair cu and $h(c) = a \neq u$ imply that c reaches u within two steps: otherwise u would be another unprotected far target of the unit c . Reaching the protected predecessor u within two steps implies reaching a within two steps. This contradicts $a \in U_D(c)$ in (S19.1). This excludes the sole bad second arc which could create a far head outside the packet remainder.

We have now shown

$$N_T^{++}(u) \subseteq (N_D^{++}(u) \cup L_0) \setminus K, \quad K \subseteq N_D^{++}(u) \cup L_0.$$

No disjointness of L_0 and the original second neighbourhood is needed. Removing K from the union gives $g_T(u) \geq g_D(u) + 2|K| - \lambda$. If $K \neq \emptyset$, this is positive because $\lambda \leq 2$. If $K = \emptyset$, its first neighbourhood is entirely original, and no selected bad direction fails at the nonselected vertex u ; its gap is unchanged and positive.

The matching contradiction in the same tournament. The heavy u , the nonselected unit b , and all zero-residual vertices are strictly deficient in T . The selected units a, c have nonnegative gaps by their one-unit budgets. Thus every tournament gap is nonnegative, and the zero-gap set is contained in $\{a, c\}$. The no-sink tournament has at most two zero-gap vertices, contrary to Lemma S14.3. The three cases prove the theorem. $\square$

S20 Excluding residual five and the uniform three-unit margin

Theorem S20.1. *Let D be a finite nonempty strongly connected all-deficient oriented graph minimal under deletion of arbitrary nonempty arc sets. Then*

$$R = 2M - n = \sum_x \eta(x) \geq 6, \quad M \geq \left\lceil \frac{n}{2} \right\rceil + 3.$$

At odd order $R \geq 7$ by parity; at even order $R \geq 6$. Equivalently, $|A(D)| \leq \lfloor n(n-2)/2 \rfloor - 3$.

Proof. The earlier parity-sensitive margin gives $R \geq 5$. Every positive residual support has at least three vertices by Corollary S12.2. If $R = 5$, its only possible positive integer partitions are

$$(3, 1, 1), \quad (2, 2, 1), \quad (2, 1, 1, 1), \quad (1, 1, 1, 1, 1).$$

They are excluded, respectively, by Theorem S16.5, Theorem S18.3, Theorem S19.1, and Theorem S17.3. Hence $R \neq 5$ and $R \geq 6$. Its parity agrees with that of n . Therefore integer rounding gives $M \geq \lceil (n+6)/2 \rceil = \lceil n/2 \rceil + 3$ in both cases. $\square$

Corollary S20.2 (Extension to all arc-set-minimal counterexamples). *Let D be any finite nonempty all-deficient oriented graph minimal under deletion of arbitrary nonempty arc sets. If a sink strongly connected component has k vertices and $o = n - k$ vertices lie outside it, then*

$$M(D) \geq \left\lceil \frac{k}{2} \right\rceil + 3 + \left\lceil \frac{o(k-2)}{2} \right\rceil \geq \left\lceil \frac{n}{2} \right\rceil + o + 3.$$

In particular the uniform three-unit margin holds without strong connectivity, and equality forces strong connectivity.

Proof. The sink component inherits deficiency and arc-set minimality. Apply Theorem S20.1 internally, and add the established outside-component estimate $M(D[V \setminus S]) + M(S, V \setminus S) \geq \lceil o(k-2)/2 \rceil$. Since every all-deficient graph has minimum outdegree at least two, $k \geq 5$. Thus the outside term is at least $\lceil 3o/2 \rceil$. The old rounding inequality

$$\left\lceil \frac{k}{2} \right\rceil + \left\lceil \frac{3o}{2} \right\rceil \geq \left\lceil \frac{k+o}{2} \right\rceil + o$$

proves the second bound. If $o > 0$ it is strictly stronger than the uniform bound, giving the equality conclusion. $\square$

At both $n = 2\delta + 3$ and $n = 2\delta + 4$, the new missing-pair bound gives $M \geq \delta + 5$. The next unresolved residual values are six at even order and seven at odd order. The full conjecture is not settled by this result. The bound concerns the stated deletion-minimal class; it is not transferred backwards to an arbitrary denser counterexample.

S21 A protection-preserving completion at arbitrary residual

All graphs are finite, nonempty and oriented. Put $g_D(x) = d_D^+(x) - d_D^{++}(x)$, let $U_D(x)$ be the targets not reached within two steps, and let $\mathcal{F}(x)$ be its unprotected far targets, with $t(x) = |\mathcal{F}(x)|$. Protection $y \triangleleft x$ means $y \rightarrow x$ and $N_D^-(y) \subseteq N_D^-(x) \setminus \{y\}$. This is a strict partial order. A missing direction $a \rightarrow b$ is convenient when every original predecessor of a reaches b within two steps. A bad pair has nonempty failure sets in both directions.

Lemma S21.1 (A universal reference completion). *Choose any linear extension $\prec$ of protection. Orient good pairs canonically, using their unique convenient direction or $\prec$ if both directions are convenient. Orient every bad pair forward in $\prec$. In the resulting tournament T , for every vertex x ,*

$$N_T^{++}(x) \subseteq (N_D^{++}(x) \setminus K_x) \cup (\mathcal{F}(x) \setminus K_x), \quad g_T(x) \geq g_D(x) + 2|K_x| - t(x),$$

where $K_x = N_T^+(x) \setminus N_D^+(x)$. Every original protected far target stays protected and far in T . No bound on t , residual, source support, or the cross-witness graph is required.

Proof. Fix $y \triangleleft x$. A two-step path through an original first neighbor cannot newly reach y : original or convenient second arcs already reach an original first or second target, while a bad directional failure head is unprotected. A path through a canonically added good first arc is excluded by canonical protection compatibility.

For an added bad first arc $x \rightarrow q$, choose an opposite failure witness r , so $r \rightarrow q$ and $x \in U_D(r)$. Protection gives $y \in U_D(r)$. An original $q \rightarrow y$ would force $q \rightarrow x$, contrary to missing xq . If qy is good, $q \rightarrow y$ fails at r and cannot be conveniently chosen. If qy is bad,

$x \prec q$ and $y \prec x$ force its reference direction $y \rightarrow q$. Thus no protected far head becomes a second neighbor.

All added first neighbors were missing partners. Each lies in exactly one of $N_D^{++}(x)$ and $\mathcal{F}(x)$, and all of K_x is removed from their union because distance two is exact. The count follows. In a tournament a far target is automatically protected, so every original protected pair remains protected in T too. $\square$

Lemma S21.2 (Avoiding all nonunit failure sources). *Alternatively, orient good pairs canonically and choose bad directions whose actual failures all have $t \leq 1$. The same simultaneous envelope holds at every vertex, including vertices with arbitrarily large t . Such a choice exists exactly when the cross-witness graph induced by $\{x : t(x) > 1\}$ has no edge.*

Proof. Only a path using a bad added first arc $x \rightarrow q$ needs a new argument. Use the opposite witness r as above. If $t(r) > 1$, an added bad second arc $q \rightarrow y$ failing at r is excluded by the orientation hypothesis. If $t(r) \leq 1$, badness of qy would give r both unprotected heads x, y , impossible. A good second pair again points $y \rightarrow q$. The feasibility equivalence follows because the two directional failure sets of each bad pair span a complete bipartite cross-witness graph. They cannot both meet a cross-witness-independent set, and a witness edge inside that set obstructs both directions of its defining bad pair. $\square$

S22 A maximal completion and its rigid median orders

Call an added pair *flexible* if it is bad or good in both directions. All remaining added pairs have a fixed, uniquely convenient direction.

Lemma S22.1 (A globally maximal reference completion). *Among the reference completions in the first lemma, there is one T_* such that every global median order of T_* puts every flexible added arc forward. Every such order extends original protection.*

Proof. A protected pair $y \triangleleft x$ in a tournament has the usual domination property: $y \rightarrow x$, every predecessor of y points to x , and every outneighbor of x receives an arc from y . If a median order put x before y , swapping these two positions would gain their mutual arc and lose none of the intervening pair contributions. Thus every median order of a protection-preserving completion extends original protection.

Over all pairs consisting of an admissible reference completion and an order, maximize the number of forward arcs. Choose a maximizing completion T_* and a median order L . If a flexible added edge were backward in L , re-complete the graph using L itself as reference. This is admissible because L extends protection. All changed edges are precisely flexible backward edges made forward; uniquely convenient directions are unchanged. The forward score strictly increases, a contradiction. The same argument applies to every median order of the fixed T_* , since all have the maximizing score. $\square$

Theorem S22.2 (Unit failure budgets with flexible witness closure). *Let D be any finite nonempty oriented graph. Suppose every actual directional failure source of a bad pair has $t \leq 1$, and every cross-witness missing pair is either bad or convenient in both directions. Then D has a Seymour vertex. There is no bound on source count and no budget condition on vertices which never witness a bad pair.*

Proof. Suppose D is all-deficient. Use T_* above and take its median feed f . If f is not an actual bad failure source, reverse all its added outgoing arcs. The same order stays median;

its original first neighborhood and the absence of bad failures make its SNP transfer directly to D . Thus f is an actual source and has $t(f) \leq 1$. An added outgoing arc at f cannot be flexible, so is a canonical good direction. Retaining one and reversing the other added out-arcs leaves the order median and gives gap at least $g_D(f) + 2 - t(f) > 0$, contradicting the feed theorem. Hence $K_f = \emptyset$. The unit envelope and feed SNP force $g_D(f) = t(f) = 1$ and $g_T(f) = 0$. There is a new far head y on a path $f \rightarrow a \rightarrow y$ with the second arc added bad. Choose its opposite witness r . The cross-witness pair fr is missing and flexible by hypothesis. The source r is not y and cannot be a second unprotected far target of f , so $r \in N_D^{++}(f) \subseteq N_T^{++}(f)$. The strong feed theorem and equality give $G_L = N_T^{++}(f)$. Thus $r \in G_L$. Global sedimentation moves f before r and preserves the median score. The flexible added arc $r \rightarrow f$, incoming because $K_f = \emptyset$, is now backward in another median order of the same T_* . This contradicts the preceding lemma. $\square$

Corollary S22.3 (A nonunit actual failure source is unavoidable). *Every strongly connected all-deficient oriented graph minimal under arbitrary nonempty arc-set deletion has an actual bad directional failure source x with $\eta(x) \geq 2$. Inactive positive-residual vertices alone cannot supply all the nonunit residuals.*

Proof. If every actual source had residual one, the unit budget would give $t \leq 1$ at those sources, regardless of the residuals elsewhere. Every cross-witness pair would be bad by the unit-unit tail lemma, which imposes no residual condition on other vertices. The theorem would give a Seymour vertex. Zero-residual vertices cannot be actual bad failure sources because all their far targets are protected. Thus some actual source has residual at least two. $\square$

In particular this gives an alternative short proof of the entire pure-unit residual exclusion. It does not yet handle a nonunit actual feed with multiple new heads or nonflexible cross-witness pairs.

S23 Packets with a nonempty missing far-in overlap

Now D is strongly connected, all-deficient and minimal under deletion of arbitrary nonempty arc sets. Fix u , put $p = |P(u)|$, and let $\ell = |\mathcal{M}(u) \cap P(u)|$. Assume the chosen bad directions avoid all failures in $P(u) \cup \{u\}$, and use a reference putting all good missing partners of u before u .

Lemma S23.1 (The keeper-loss form of the packet). *With the established packet B_u, k, s, E, L , let K be the actual added first set and put $\beta = \mu(u) - k - |K|$. Then*

$$0 \leq \beta \leq \ell, \quad N_T^{++}(u) \subseteq (N_D^{++}(u) \cup L) \setminus K, \quad K \subseteq N_D^{++}(u) \cup L.$$

For $P(u) \neq \emptyset$,

$$g_T(u) \geq \begin{cases} p + 1 - 2\beta, & s = 0, \\ p + g_D(u) + 1 - 2\beta, & s > 0. \end{cases}$$

In particular $p \geq 2\ell$ suffices to keep u strictly deficient, without any bound on $\eta(u)$, $t(u)$, or the number of sources. The exact all-keeper criterion is the case $\ell = 0$.

Proof. An incoming bad incident direction would fail at a predecessor in $P(u)$, so cannot be chosen. Every outgoing good incident direction has a nonconvenient reverse direction by the reference rule, and every outgoing bad direction also has a nonconvenient reverse.

Each actual keeper therefore has an original predecessor in $P(u)$ pointing to it. Exceptional partners have a convenient incoming direction and are good, so none is a keeper. Every nonexceptional missing partner outside $P(u)$ belongs to $\partial P(u)$ and is uniquely directed outward. Only the ℓ missing partners inside $P(u)$ can be lost keepers.

The packet predecessor blocks every good or chosen bad second arc to a far head outside L , exactly as in the all-keeper proof. Original first intermediates have no selected failure at u . This proves the envelope. Then $|K| = \mu - k - \beta$ and the packet bound $|L| \leq \eta - 2k$ for $s = 0$, or $\eta - g_D - 2k$ for $s > 0$, give the displayed inequalities using $p = 2\mu + g_D - 1 - \eta$. No assumption is made that the deletion graph remains all-deficient. $\square$

Why simultaneous use is still an open step. For a collection H of heavy targets, avoiding all their far-in failure sources requires a compatible choice on every bad pair, and their inward reference priorities must be acyclic. Neither follows from single-target feasibility. The explicit six-vertex control below shows that even three originally adjacent targets can have incompatible priority families. These issues are not removed by cancelling the numerical residual inside each separate packet.

S24 A precise unbounded certificate and its unresolved existence

Theorem S24.1 (A conditional global heavy-cover certificate). *Suppose, in the minimal all-deficient class, $H = \{x : t(x) > 1\}$ and $I = H \cup \bigcup_{u \in H} P(u)$ satisfy the following conditions. The cross-witness graph induced by I is independent; the protection order plus inward priorities for all good partners of all targets in H is acyclic; and $|P(u)| \geq 2|\mathcal{M}(u) \cap P(u)|$ for every $u \in H$. Choose the resulting canonical good completion and bad directions avoiding I . Define*

$$Q = \{x \notin H : g_D(x) = t(x) = 1, K_x = \emptyset, x \text{ fails a selected bad direction}\}.$$

If Q is empty or its original induced tournament has a sink, then D cannot be all-deficient. There is no support-size or residual bound in this sufficient criterion.

Proof. Independence permits the bad choices. Every chosen failure source has $t \leq 1$, so the second global envelope lemma applies. Every target in H has positive completed gap by the overlap packet lemma (the empty P case has unchanged neighborhoods). All other gaps are nonnegative. The zero-gap set is exactly Q : when K_x is empty, original second paths persist, and a selected bad failure introduces its sole unprotected head; conversely, with no such failure the gap is unchanged. An added first neighbor or original gap at least two makes the gap positive. Two vertices of Q cannot be missing partners, because their supplemental arc would give one a first neighbor. Thus $D[Q] = T[Q]$ is a tournament. The completed tournament has no sink. The nonnegative-gap zero-set theorem forbids both an empty zero set and a zero set with a sink. This yields the contradiction. $\square$

This is a sufficient certificate, not a proof that every minimal counterexample admits such a certificate. Its independence, acyclic priority and final zero-core requirements are the remaining global selection problem. The maximal reference completion provides a different always-feasible construction, but does not by itself supply the necessary heavy packet budgets.

S25 Rejected global shortcuts, with exact controls

Three inward target priorities. The six-vertex out masks $[26, 52, 41, 0, 0, 0]$ use targets $0, 1, 2$. Protection contains $0 \triangleleft 3, 1 \triangleleft 4, 2 \triangleleft 5$. The missing partner priorities require $3 \prec 1, 4 \prec 2, 5 \prec 0$, creating a directed cycle. All target gaps equal one, but the other vertices are sinks, so the control is not a Seymour counterexample. The strong nine-vertex control $[26, 52, 41, 64, 128, 256, 18, 36, 9]$ preserves the same priority conflict and has gap minus one at every vertex.

Skew-equilibrium weights. For out masks $[2, 4, 24, 17, 3]$ and positive weights $(1, 2, 2, 1, 1)$, weighted incoming and outgoing neighborhoods balance at every vertex. The weighted first-minus-second gaps are nevertheless $(0, 0, -1, 0, 1)$. This control is an incomplete oriented graph, not a tournament. An equilibrium of the skew adjacency game of an arbitrary oriented graph is therefore not automatically an expansion weighting for the second-neighborhood matrix. For tournaments Fisher’s losing-density lemma does provide reverse second-neighborhood expansion; the two normalized feasible density sets actually coincide, as proved in the follow-up fragment. The present control does not contradict that lemma. Seacrest’s Farkas alternative remains a legitimate equivalent formulation, not a proof of the required expansion weighting.

Arc-weighted median feeds. The existential arc-weighted tournament theorem must not be upgraded to “every weighted median feed has arc-SNP.” The independent exact control `audit_canonical_arc_weighted_feed.results.json` already refutes this even for canonical completions with original and uniquely convenient arcs weighted 37, flexible arcs weighted 1, and all added outgoing feed arcs reversed inward. Its six-vertex original graph is $[0, 36, 17, 34, 2, 4]$, its median feed is 2, and its first and second arc-weights are 74 and 38. Other vertices have SNP; neither the ordinary conjecture nor the published existential theorem is refuted.

Verification boundary. The global graph-level program checked all 59049 five-vertex oriented graphs and 30000 fixed-seed samples of orders six through twelve. It checked 1130480 priority-completion vertex envelopes, including 53430 multiple bad-first cases; 400791 unit-selected envelopes; and 464674 overlap-packet completions, 249930 with positive overlap. These finite checks attack local identities, not the missing global certificate-existence theorem or the full SNC.

S25.1 An all-keeper packet criterion

Let D be a finite nonempty strongly connected all-deficient oriented graph minimal under deletion of arbitrary nonempty arc sets. All failure, convenience, protection, and reachability predicates refer to D . Fix u and suppose

$$\mathcal{M}(u) \cap P(u) = \emptyset. \quad (\text{AK1})$$

Choose one canonical reference placing all good missing partners of u before u . This is possible because original missing pairs are incomparable in the protection partial order.

Lemma S25.1 (All-keeper packet completion). *If every chosen bad direction has all its failure sources outside $P(u) \cup \{u\}$, then u is strictly deficient in the completed tournament T . No bound on $\eta(u)$ or $t(u)$ is needed.*

If $P(u) = \emptyset$, its neighborhoods are unchanged. Otherwise, with the usual packet notation B_u, k, s, E, L , one has exactly

$$J = N_T^+(u) \setminus N_D^+(u) = \mathcal{M}(u) \setminus B_u \quad (\text{AK2})$$

and the simultaneous envelope

$$N_T^{++}(u) \subseteq (N_D^{++}(u) \cup L) \setminus J, \quad J \subseteq N_D^{++}(u) \cup L. \quad (\text{AK3})$$

Consequently

$$g_T(u) \geq \begin{cases} |P(u)| + 1, & s = 0, \\ |P(u)| + g_D(u) + 1, & s > 0. \end{cases} \quad (\text{AK4})$$

Proof. If a bad incident pair ur were completed by $r \rightarrow u$, its nonempty failure set would contain some q with $q \rightarrow r$ and $u \in U_D(q)$. Thus $q \in P(u)$, forbidden by the direction selection. Every incident bad pair is therefore directed $u \rightarrow r$. Its opposite-direction witness q satisfies $q \in P(u)$ and $q \rightarrow r$. If $P(u)$ is empty there are no bad incident pairs. Every incoming good direction into u is then convenient and is selected by the reference, so $J = \emptyset$. No chosen direction fails at u , hence its exact second neighborhood is unchanged.

Assume $P = P(u) \neq \emptyset$. Write

$$\begin{aligned} Q &= N_D^{--}(u), \quad C = \partial P, \quad X = \partial^2 P, \quad H = Q \setminus C, \\ B_u &= H \cap \mathcal{M}(u), \quad k = |B_u|, \quad s = |H \cap N_D^+(u)|. \end{aligned}$$

The partner packet gives

$$E = (\mathcal{M}(u) \setminus B_u) \dot{\cup} X, \quad L = E \setminus N_D^{++}(u), \quad |L| \leq \begin{cases} \eta(u) - 2k, & s = 0, \\ \eta(u) - g_D(u) - 2k, & s > 0. \end{cases}$$

All nonexceptional missing partners lie in $P \cup C$, and (AK1) puts them in C . Every such good partner r has a predecessor $q \in P$ pointing to it, so $r \rightarrow u$ is nonconvenient and the chosen good direction is $u \rightarrow r$. Every good exceptional partner admits the convenient incoming direction, selected by the reference. A bad exceptional partner is impossible: every incident bad pair has the original predecessor $q \in P$ identified above; (AK1) then makes its missing endpoint a member of C , not H . This proves (AK2). Each $r \in J$ has an original predecessor $q_r \in P$ pointing to r , whether its chosen first arc is good or bad.

Through an original first intermediate of u , an originally far second head would require a chosen direction failing at u , which is excluded. Through an added first intermediate $r \in J$, original far second heads lie in L by the partner-head budget. Suppose an originally far head $y \notin L$ receives an added direction $r \rightarrow y$. Every vertex reachable from P within two steps is in $P \cup C \cup X$, and every vertex of this union far from u belongs to L . Thus $y \in U_D(q_r)$. The direction $r \rightarrow y$ fails at $q_r \in P$. It cannot be a chosen good direction, because it is nonconvenient; and it cannot be a chosen bad direction, because such failures were excluded. The first inclusion in (AK3) follows, with J removed since distance two is exact. The second follows because $J \subseteq E$, and a missing endpoint reached in $\widehat{D}$ was already an original exact second neighbor.

Now $h = |J| = \mu(u) - k$, so the envelope gives $g_T(u) \geq g_D(u) + 2h - |L|$. Substituting the packet bound and the identity $|P(u)| = 2\mu(u) + g_D(u) - 1 - \eta(u)$ gives (AK4). No comparison between different completions is involved. $\square$

Corollary S25.2 (A sufficient condition for the bad directions). *If (AK1) holds and the cross-witness graph induced by $P(u)$ has no edge, then bad pairs can all be directed as required by the lemma.*

Proof. Every cross-witness edge is an original missing pair. By (AK1) there is no cross-witness edge from u to $P(u)$; by hypothesis there is none inside $P(u)$. Thus $P(u) \cup \{u\}$ is independent in the cross-witness graph. If both directional failure sets of a bad pair intersected this set, their opposite sources would form an edge inside it. Therefore at least one direction avoids the whole set. Select that direction independently for each bad pair. $\square$

S26 A fixed-completion criterion for an independent heavy cover

All graph predicates in this section refer to the original finite nonempty oriented graph D . Let A be its actual bad-direction failure sources, and let B be the cross-witness graph on A : the two directional failure sets of each bad pair span their complete bipartite cross-witness graph. Write $t(x)$ for the number of unprotected original far heads of x .

Theorem S26.1 (An independent heavy cover without protection successors). *Suppose there is a nonempty set H of vertices such that:*

1. *Every source $f \in A \setminus H$ satisfies $t(f) \leq 1$.*
2. *$B[H \cap A]$ is independent.*
3. *No $h \in H$ is a protected far target of any source $f \in A \setminus H$; equivalently, no original relation $h \triangleleft f$ occurs between those sets.*

Then D has a Seymour vertex. No source-support, order, residual, or far-budget bound is imposed on H or on inactive vertices.

Proof. Suppose all original gaps are positive. Independence allows each bad pair to be oriented so that all its selected directional failures avoid H : its two failure sets cannot both meet H , since such opposite witnesses would give an edge of $B[H \cap A]$. Complete good pairs canonically by any fixed reference extension of protection. Call the resulting fixed tournament T . Every selected bad failure source has $t \leq 1$, so the simultaneous safe envelope applies at every vertex:

$$N_T^{++}(x) \subseteq (N_D^{++}(x) \setminus K_x) \cup (\mathcal{F}_D(x) \setminus K_x), \quad g_T(x) \geq g_D(x) + 2|K_x| - t(x).$$

Among all global median orders of this one fixed tournament, choose L maximizing $\sum_{h \in H} \text{pos}_L(h)$, and let f be its feed.

If $f \notin A$, or if $f \in H$, reverse every added outgoing feed arc inward. The order remains global median, its first neighborhood is original, and it has no new far second head: good directions are convenient, and no selected bad direction fails at f . Feed SNP would transfer to the original deficient vertex, a contradiction.

Thus $f \in A \setminus H$ and $t(f) \leq 1$. The envelope and feed SNP force $K_f = \emptyset$, $g_D(f) = t(f) = 1$ and $g_T(f) = 0$. Writing y for its unique unprotected original far head, the strong feed theorem then gives

$$G_L = N_T^{++}(f) = N_D^{++}(f) \cup \{y\}, \quad |N_T^+(f)| = |G_L|.$$

For every $h \in H$, condition 3 excludes original protected farness from f . Since $h \neq f$, it is therefore an original first neighbor, an original exact second neighbor, or the sole unprotected

original far head y . In all three cases

$$H \subseteq N_T^+(f) \cup G_L.$$

Global sedimentation remains a median order and strictly increases the position of each vertex of this union, by the exact positional identity

$$\text{pos}_{\text{Sed}(L)}(h) - \text{pos}_L(h) = 1 + |\{b \in B_L : \text{pos}_L(b) > \text{pos}_L(h)\}| > 0,$$

where $B_L = V(T) \setminus (N_T^+(f) \cup G_L \cup \{f\})$ denotes the *bad vertices of the median order*, not the cross-witness graph. It therefore increases the chosen sum by at least $|H| > 0$, a contradiction. This proves the criterion. $\square$

The strong feed inequality and the global sedimentation assertion are the unweighted results of Havet–Thomasse [7, Theorem 1 and Lemma 1]; the exact positional identity follows by counting the moved bad vertices.

Corollary S26.2 (Simplifying the second phase of the one-exception proof). *Once the unrestricted maximum argument has proved that a unique exceptional actual source u has no original protection successor, the second phase needs only an arbitrary completion avoiding all failures at u , and a median order of that fixed completion maximizing the position of u . No second family maximum or fixed-tail case analysis is necessary.*

Proof. Apply the theorem with $H = \{u\}$. Independence is automatic; the other sources have $t \leq 1$, and the first phase supplied condition 3. The proof uses only the simultaneous envelope in the fixed completion and its global sedimentation operation. $\square$

Corollary S26.3 (A global alternative for minimal counterexamples). *In the strong arbitrary-arc-set-deletion-minimal all-deficient class, put $H = \{h \in A : \eta(h) \geq 2\}$. The unit closure makes H nonempty. A counterexample must have either:*

1. *a cross-witness edge joining two vertices of H ; or*
2. *an actual unit source f and an $h \in H$ with $h \triangleleft f$ in original protection.*

Proof. If neither occurs, all hypotheses of the fixed-completion criterion hold. Actual sources outside H have residual one and hence $t \leq 1$. The criterion contradicts all-deficiency. $\square$

Audit boundary. The theorem does not show that every heavy cover is independent or that its protection-successor condition always holds. It exhibits the specific two global obstructions that remain after unit budgets are controlled. Its proof uses one fixed tournament, not a comparison of numerical gaps from different completions. Ordinary two-Seymour vertex theorems are not used, and global median orders, not merely local median orders, are essential.

A local protection-obstruction control. The ten-vertex out masks

$$[540, 541, 96, 96, 96, 1, 3, 28, 128, 256]$$

have an actual unit source $f = 0$ with $g(f) = \eta(f) = t(f) = 1$ and $\partial P(f) = N_D^-(f) = \{2, 3, 4\}$. The actual source $h = 1$ has $g(h) = 2$, $\eta(h) = 2$, $t(h) = 1$, and $h \triangleleft f$, because $N_D^-(h) = \{6\} \subseteq N_D^-(f) \setminus \{h\} = \{5, 6\}$. The bad missing pair $\{7, 9\}$ has failure sources $\{0, 1\}$ for $9 \rightarrow 7$ and $\{8\}$ for $7 \rightarrow 9$. Several other vertices have nonpositive gap, so this

is not an SNC counterexample and does not meet the complete minimal-counterexample hypotheses. It does show that a heavy-protecting-unit relation is not contradicted by the unit's local residual, positive gap and full packet alone. The independent program `audit_heavy_protects_unit_control.js` verifies all these claims.

S27 An arbitrary-order one-exception completion theorem

All predicates refer to the original finite nonempty oriented graph D . Let A be its actual bad-direction failure sources, $\mathcal{F}(x)$ its unprotected far targets and $t(x) = |\mathcal{F}(x)|$. Protection is $y \triangleleft x$ if $y \rightarrow x$ and $N_D^-(y) \subseteq N_D^-(x) \setminus \{y\}$. A missing pair is flexible if it is bad or is good with both directions convenient. An opposite pair of bad-direction witnesses is an original missing cross-witness pair.

Theorem S27.1 (At most one uncontrolled actual source). *Suppose a vertex u satisfies:*

1. *every source $x \in A \setminus \{u\}$ has $t(x) \leq 1$;*
2. *every cross-witness pair between two sources in $A \setminus \{u\}$ is flexible.*

Then D has a Seymour vertex. There is no condition on the budgets of inactive vertices, and no bound on the order or source support.

Proof. Assume $g_D(x) > 0$ for every vertex. For *every* linear extension σ of protection, let T_σ complete every flexible pair forward in σ and keep every uniquely convenient good direction fixed. The universal reference-priority envelope preserves every original protected far pair and gives

$$g_{T_\sigma}(x) \geq g_D(x) + 2|K_x| - t(x), \quad K_x = N_{T_\sigma}^+(x) \setminus N_D^+(x). \quad (\text{SX1})$$

Any global median order of such a tournament extends original protection: if $y \triangleleft x$ were reversed in the order, swapping their positions gains their mutual arc and loses none of the intervening pair contributions, by tournament domination.

Among all pairs (T_σ, L) , with L any vertex order, first maximize the unweighted forward-arc score Φ and then maximize $\text{pos}_L(u)$. The selected L is a global median order of the selected tournament T . Every flexible added arc is forward in every global median order of T : otherwise use that median order as a new reference, which is allowed because the family contains *every* protection extension. All flexible backward pairs become forward, while fixed directions and original arcs are unchanged, strictly improving Φ . Let f be the chosen feed.

If $f \notin A$, reverse all added outgoing feed arcs inward. This keeps the same order global median: its score increases by the number of reversals and every competing score increases by at most that number. Its first neighborhood becomes original. No new far second head can occur through an original first step: a good second direction is convenient and a bad second direction would fail at $f \notin A$. Its tournament feed SNP therefore transfers to the originally deficient vertex, a contradiction.

Suppose $f \in A \setminus \{u\}$. The envelope and $t(f) \leq 1$ force $K_f = \emptyset$, since any added first neighbor gives a strictly positive completed gap. Original exact second neighbors then persist. Feed SNP, positive original gap and the envelope force

$$g_D(f) = t(f) = 1, \quad g_T(f) = 0, \quad N_T^{++}(f) = N_D^{++}(f) \cup \{y\},$$

where y is its sole unprotected original far head. The strong feed theorem gives $|N_T^+(f)| \leq |G_L| \leq |N_T^{++}(f)|$; equality of the two neighborhoods therefore yields $G_L = N_T^{++}(f)$.

The new head y has a path $f \rightarrow a \rightarrow y$ with original first arc and added bad second arc. Choose an opposite witness r . Then $r \rightarrow y$ is original, so $r \neq y$, and fr is originally missing. Since a missing far target is unprotected and y is the only such head at f , $r \in N_D^{++}(f) \subseteq G_L$.

Global sedimentation remains a median order and puts f before every vertex of G_L . If $r \neq u$, condition 2 makes fr flexible; its direction was $r \rightarrow f$ because $K_f = \emptyset$. Sedimentation makes it backward in another median order of T , contradicting the maximum-score argument. If $r = u$, sedimentation strictly increases the position of u while preserving the score, because for every $z \in N_T^+(f) \cup G_L$ its position increases by

$$1 + |\{b \in B_L : \text{pos}_L(b) > \text{pos}_L(z)\}| > 0,$$

where B_L is the set of bad vertices of the median order. This contradicts the secondary maximum. Both cases are impossible.

Thus the *selected* median feed is u . If $u \notin A$, the outside-source transfer already gives a contradiction. Otherwise the last position at u shows that it has no original protection successor: there is no q with $u \triangleleft q$, since every median order extends protection.

Apply the fixed-completion heavy-cover criterion (Theorem S26.1) with $H = \{u\}$. Its independence condition is automatic; all other actual sources have $t \leq 1$, and the absence of a protection successor at u supplies its final condition. It contradicts all-deficiency. The second phase uses a different fixed completion solely to apply that criterion; no numerical gap or optimal score is transferred between the completions. $\square$

Corollary S27.2 (Two nonunit actual failure sources). *In every finite nonempty strongly connected all-deficient oriented graph minimal under arbitrary nonempty arc-set deletion, at least two distinct actual bad-direction failure sources have $\eta \geq 2$. Consequently every positive residual allocation $(m, 1^k)$ is excluded for arbitrary $m \geq 2$ and arbitrary finite $k \geq 0$.*

Proof. Actual sources have positive residual, and a source of residual one has $t \leq 1$. The unit–unit cross-witness-tail lemma says that opposite unit sources have a bad, hence flexible, original missing pair. If at most one actual source were nonunit, choose it as u , or choose any vertex when none exists. The general theorem applies, a contradiction. Residuals of inactive vertices are unrestricted. $\square$

Proof status. The first phase claims the feed property only for its selected score/position optimum, not for every median order. Closure under all protection extensions is explicit and essential. The fixed-cover criterion has a separate proof, so the theorem and corollary have no circular dependency. The remaining multiple-exception classes are not excluded, and no proof of the full SNC is asserted.

S28 Multiple retained cross-witness arcs and allocations $(m, 2, 1)$

All neighborhoods, far sets, protection, and convenience refer to the original oriented graph D . Write $g(x) = d_D^+(x) - d_D^{++}(x)$, $P(x) = \{p : x \in U_D(p)\}$, and $\mathcal{F}(x) = \{y \in U_D(x) : y \text{ is not protected from } x\}$. Let $t(x) = |\mathcal{F}(x)|$ and let $\mathcal{B}$ be the cross-witness graph. Opposite failure sources of a bad missing pair are joined in $\mathcal{B}$; every such edge is an original missing pair.

Lemma S28.1 (A multiple-retention source criterion). *Let Z cover the directional failure sources of all bad missing pairs, and suppose every edge of $\mathcal{B}_Z$ is good. Orient its edges by canonical convenient orientations, and put $S = \{z : d_{\mathcal{B}_Z}^+(z) = 0\}$. If*

$$t(z) \leq g(z) + 2d_{\mathcal{B}_Z}^+(z) - 1 \quad (z \in Z \setminus S),$$

then D has a Seymour vertex.

Proof. Orient bad pairs to avoid failure sources in S . This is possible: opposite sources in S would form a cross-witness edge with an outgoing endpoint, contradicting its definition. Complete all good pairs canonically and choose a median feed f in the resulting tournament. If $f \notin Z \setminus S$, reverse every added arc leaving f ; its feed SNP transfers to D , by the usual no-failure argument.

Otherwise retain every outgoing cross-witness arc at f and reverse its other added outgoing arcs. Put $J = N_{\mathcal{B}_Z}^+(f)$ and $r = |J| \geq 1$. These reversals are all of backward arcs in the chosen median order, which remains median. Each retained arc is a canonical good direction. Canonical protected-target compatibility, applied separately to every $q \in J$, prevents an original or added arc $q \rightarrow y$ when y is protected from f . Original first intermediates can introduce only unprotected far heads. Hence

$$N_{T'}^{++}(f) \subseteq (N_D^{++}(f) \setminus J) \cup (\mathcal{F}(f) \setminus J).$$

Every missing partner in J belongs to exactly one of the disjoint sets $N_D^{++}(f)$ and $\mathcal{F}(f)$. Thus

$$d_{T'}^+(f) - d_{T'}^{++}(f) \geq g(f) + 2r - t(f) \geq 1,$$

contradicting the median-feed SNP. Paths between retained first neighbors cause no difficulty: all their heads are first neighbors and are explicitly removed from the second-neighborhood envelope. $\square$

S28.1 A unit source in a three-source triangle

For the remainder, assume that D is strongly connected, all-deficient, and minimal under deletion of arbitrary nonempty arc sets on its fixed vertex set. Its positive residual support is $Z = \{u, v, w\}$ with

$$\eta(u) = m \geq 1, \quad \eta(v) = 2, \quad \eta(w) = 1.$$

All directional failure sources belong to Z , and all missing pairs inside Z are good. In particular, every edge of $\mathcal{B}$ is good. A unit source satisfies $t(w) \leq 1$; an active unit source has $\mathcal{F}(w) = \{a\}$ for some a .

Lemma S28.2 (The unit head in a witness triangle). *If $\mathcal{B}$ is a triangle, then $t(v) \leq 2$. Its active unit vertex has a unique unprotected head a , and*

$$u \rightarrow a, \quad v \rightarrow a, \quad w \notin P(u) \cup P(v).$$

Every bad missing pair having w as a directional failure source has one directional failure set exactly $\{w\}$.

Proof. The residual-two witness-degree lemma gives $t(v) \leq 2$, since v has two cross-witness partners. An opposite witness to w points originally to its sole unprotected head a ; both u and v are such opposite witnesses. Consequently $a \neq u, v$. As wu, wv are missing, neither u

nor v can be a protected far target of w , and neither equals its sole unprotected far target. Thus w reaches each of them in exactly two steps, proving the displayed exclusion from their far-in sets.

If w fails one direction of a bad pair, its failure head is a . Neither u nor v can fail this same direction, since both point originally to a . These are all other possible sources, so the directional failure set containing w is precisely $\{w\}$. $\square$

Lemma S28.3 (Nonexceptional triangle partners). *Assume the triangle situation and $P(u) \neq \emptyset$. In the target-packet notation, let B_u be the exceptional missing-partner set. If $w \notin B_u$, then $u \rightarrow w$ is the unique convenient orientation. If $v \notin B_u$, then $u \rightarrow v$ is convenient; moreover, if $v \rightarrow u$ is not convenient, some $p \in P(u)$ points originally to v .*

Proof. A nonexceptional missing partner belongs to $P(u) \cup \partial P(u)$. The preceding lemma excludes w from $P(u)$, so $w \in \partial P(u)$. A predecessor in $P(u)$ makes $w \rightarrow u$ nonconvenient. Since uw is good, its only convenient direction is $u \rightarrow w$.

If $v \notin P(u)$, the identical boundary argument makes $u \rightarrow v$ the unique convenient direction. Suppose instead $v \in P(u)$, so u is far from v . To prove convenience of $u \rightarrow v$, suppose $p \in P(v)$ points to u . The source p cannot be u or v , and it cannot be w because $w \notin P(v)$. Therefore $\eta(p) = 0$. Protection of the far target v from p then gives $v \rightarrow p$, which together with $p \rightarrow u$ contradicts $u \in U_D(v)$. There is no such predecessor p , exactly the convenience condition for $u \rightarrow v$. Finally, nonconvenience of $v \rightarrow u$ is equivalent to the asserted predecessor in $P(u)$. $\square$

S28.2 The non-triangular witness cases

Lemma S28.4 (A unit-centered star with one residual-two leaf). *More generally, let the three positive residuals be $(m, 2, 1)$ with any $m \geq 1$. If $\mathcal{B}$ is the two-edge star centered at the unit vertex, then D cannot exist.*

Proof. Call the heavy leaf u , the residual-two leaf v , and the unit center w . Select every bad direction with only w as its failure source. If $t(v) > 2$, the residual-two leaf-completion lemma keeps v deficient and adds the good first arc $w \rightarrow v$. The unit budget then keeps w deficient as well; all zero-residual vertices remain deficient. Only u could be Seymour, contrary to the two-Seymour vertex theorem for sink-free tournaments.

Assume $t(v) \leq 2$. Choose a canonical reference extension placing all good missing partners of u before u . The active unit has unique unprotected head a , with $u \rightarrow a$ and $a \neq u$. Thus $w \notin P(u)$, because wu is missing and u is not its unique far head. Every missing pair incident to u is good: a bad such pair would require both other sources v, w , whereas w can witness neither direction (it neither points to u nor has u as a far head). If $P(u) = \emptyset$, all its missing partners are directed inward and its gap is unchanged.

Otherwise use the packet E, L, B_u, k, s for u , with

$$|L| \leq \lambda = \begin{cases} \eta(u) - 2k, & s = 0, \\ \eta(u) - g(u) - 2k, & s > 0. \end{cases}$$

Every added first neighbor r has a predecessor in $P(u)$ and is nonexceptional. That predecessor is not w . For a far head outside L , an original $r \rightarrow y$ is excluded by the partner budget; an added good direction is excluded by that predecessor; and an added bad direction would fail at a source other than the selected w . Therefore, with $J = N_T^+(u) \setminus N_D^+(u)$ and $h = |J|$,

$$N_T^{++}(u) \subseteq (N_D^{++}(u) \cup L) \setminus J, \quad J \subseteq N_D^{++}(u) \cup L.$$

All zero-residual nonexceptional partners belong to $\partial P(u)$, as does w when nonexceptional. These are directed outward from u . Only v may be an additional inward nonexceptional partner, so $h \geq \mu(u) - k - 1$. For $h > 0$ this gives

$$g_T(u) \geq g(u) + 2h - \lambda \geq \begin{cases} |P(u)| - 1, & s = 0, \\ |P(u)| + g(u) - 1, & s > 0. \end{cases}$$

For $h = 0$ its original neighborhoods are unchanged. Thus $g_T(u) \geq 0$. The nonselected v remains strictly deficient because $t(v) \leq 2$; w has $g_T(w) \geq g(w) - 1 \geq 0$; and every other vertex has positive gap. The zero-gap set has at most two vertices, contradicting the zero-set theorem for a sink-free tournament with all gaps nonnegative. $\square$

Lemma S28.5 (A residual-two-centered star with an arbitrary heavy leaf). *The two-edge star centered at v is impossible for the allocation $(m, 2, 1)$, regardless of the positive value m .*

Proof. Write its center as $c = v$, its arbitrary-residual leaf as u , and its unit leaf as w . The witness-degree lemma gives $t(c) \leq 2$. Orient all bad pairs with c as their sole selected failure source, and use a reference extension placing all good missing partners of u before u . All zero-residual vertices and the nonselected unit w remain deficient. If c has an added first neighbor, then $g_T(c) \geq g(c) + 2 - t(c) > 0$, leaving only u as a possible Seymour vertex. Assume, therefore, $K_c = \emptyset$.

The good missing pair cu must be directed $u \rightarrow c$. The reference choice then means $c \rightarrow u$ is nonconvenient; in particular some $p_0 \in P(u)$ points to c , and c is nonexceptional. If $P(u) = \emptyset$ this is already impossible. Apply the packet notation to u .

An added first neighbor of u cannot be bad, since its chosen direction would need the nonexistent original arc $c \rightarrow u$. Every added first neighbor has a predecessor $p \in P(u)$ by the reference choice. Outside L , original second arcs from such neighbors and added good second arcs are excluded as before. An added bad second arc is also excluded unless its predecessor is $p = c$. In that exception $c \in P(u)$, and the new far head is a member of $\mathcal{F}(c) \setminus \{u\}$: the missing target u itself is already unprotected from c . Thus the union of additional heads outside L has size at most one.

There is at most one bad missing pair incident to u . Indeed its two witnesses must be c, w ; the selected inward direction must fail at c , and the opposite direction at w . If a is the unit's unique unprotected head, that pair is ua , requires $c \in P(u)$, and has a nonexceptional. Moreover $a \notin \{c, u, w\}$. Every other nonexceptional missing partner is directed outward: zero-residual partners cannot lie in $P(u)$; a missing partner w cannot lie there either, since $c \rightarrow a$ and cu is missing imply $a \neq u$; and c is outward by $K_c = \emptyset$. Consequently $h \geq \mu(u) - k$ if ua is not bad, and $h \geq \mu(u) - k - 1$ otherwise.

In the former case the packet bound, including the possible one extra head, gives $g_T(u) \geq |P(u)|$ for $s = 0$, and a stronger bound for $s > 0$. Thus u is deficient. In the latter case $c \in P(u)$, and $p_0 \neq c$ makes $p = |P(u)| \geq 2$. In fact there is no extra far head outside L in this latter case. To see this, a selected bad second arc $r \rightarrow y$ through an added first neighbor $r \in K_u$ has sole selected failure source c . Its opposite witness cannot be u : both ur is originally missing and y is originally far from u , so u points originally to neither endpoint. Hence its opposite witness must be w , whose unique unprotected head is a . This forces $r = a$. But the bad incident pair ua was directed $a \rightarrow u$, so $a \notin K_u$, a contradiction. We therefore have

$$g_T(u) \geq \begin{cases} p - 1, & s = 0, \\ p + g(u) - 1, & s > 0, \end{cases}$$

and both are positive. No numerical upper bound on m has been used in this argument.

Thus, when $K_c = \emptyset$, every vertex except possibly c is deficient. When $K_c \neq \emptyset$, every vertex except possibly u is deficient. In each case this is one fixed sink-free completed tournament with at most one Seymour vertex, a contradiction. $\square$

S28.3 The triangle at arbitrary heavy residual

Lemma S28.6 (Good incident pairs and good keeper–far pairs). *In the triangle situation every missing pair incident to u, v, w is good. Moreover, if r is a missing partner of u and $y \in U_D(u)$, then any missing pair ry is good.*

Proof. Let a be the unit's sole unprotected head. We have $u \rightarrow a$, $v \rightarrow a$, and uw, vw missing. A bad missing pair incident to u would need the two failure sources v, w . But w can fail neither direction: it does not originally point to u , and u is not far from w . The latter follows because $u \neq a$ and uw is missing. The same argument works for v . A bad pair incident to w would need u, v ; the direction with tail w has no possible failure source, since neither u nor v points originally to w . All three incident-pair assertions follow.

Now $r, y \neq a$, since $u \rightarrow a$ is original whereas r is missing from u and y is far from u . The vertex u points originally to neither endpoint of ry , so cannot witness either direction. If ry were bad, its two opposite sources would have to be v, w . Any failure from w on a bad pair has its head equal to a , whereas neither endpoint is a . This is impossible. $\square$

Theorem S28.7. *A strongly connected all-deficient oriented graph minimal under arbitrary nonempty arc-set deletion cannot have precisely three positive residuals $(m, 2, 1)$, for any positive integer m . In particular this excludes $(3, 2, 1)$ in residual six and $(4, 2, 1)$ in residual seven in all orders.*

Proof. The cross-witness graph is nonempty. If it has a single edge incident to the unit vertex, there are only two actual failure sources, one unit, and the existing two-source theorem applies. If its single edge joins u, v , select all bad directions with failure source u and apply the residual-two leaf-completion lemma to v ; the nonselected unit and all other vertices remain deficient. The same argument works for the two-edge star centered at u . A star centered at w is excluded by Lemma S28.4, and a star centered at v by Lemma S28.5.

It remains that $\mathcal{B}$ is a triangle. By Lemma S28.2, $t(v) \leq 2$ and $t(w) \leq 1$. Use one canonical reference extension in which every original missing partner of u precedes u ; every such pair is good by Lemma S28.6. This simultaneous extension exists because every missing partner is incomparable with u in the protected partial order: any protected chain would force an original arc. Adding all precedence constraints cannot create a cycle.

If $P(u) = \emptyset$, all incident directions into u are convenient, and the reference choice directs both cross-witness edges into u . It is the sole cross-witness sink. The ordinary cross-witness transfer uses only the budgets at v, w and excludes this case. Assume henceforth $P = P(u) \neq \emptyset$.

Take the packet notation for u :

$$B_u = H \cap \mathcal{M}(u), \quad k = |B_u|, \quad E = (\mathcal{M}(u) \setminus B_u) \dot{\cup} \partial^2 P, \quad L = E \setminus N_D^{++}(u).$$

Its general head budget is

$$|L| \leq \lambda = \begin{cases} m - 2k, & s = 0, \\ m - g(u) - 2k, & s > 0. \end{cases}$$

There is no bound on m , k , or $t(u)$ beyond these established packet facts.

We first choose bad directions so that every selected failure head at u belongs to L , while no direction fails at a cross-witness sink. Only membership of v, w in B_u matters, not its total size.

If neither v nor w is exceptional, orient bad pairs to avoid the canonical cross-witness sinks. Every opposite witness of a selected failure at u is a nonexceptional missing partner, so its head lies in L by the partner budget.

If v is exceptional but w is not, the canonical directions are $v \rightarrow u$ and $u \rightarrow w$. The latter is unique convenient because $w \notin P(u)$, so w is a nonexceptional boundary partner. The only possible cross-witness sink is w . For every bad pair having w as a source choose the direction avoiding w ; the directional failure set containing w is exactly $\{w\}$. For every other bad pair its two failure sets are $\{u\}, \{v\}$; choose the one failing at v . These choices avoid w , and every selected failure at u has the nonexceptional opposite witness w , with its head in L .

If w is exceptional but v is not, the canonical direction is $w \rightarrow u$. If $v \rightarrow u$ is convenient, the reference choice also selects it. Then u is the sole cross-witness sink, and the ordinary cross-witness transfer already excludes the graph. Otherwise the canonical direction is $u \rightarrow v$, and the only possible cross-witness sink is v . For every bad pair having w as a source choose the direction whose failure set is $\{w\}$. For every other bad pair choose the direction failing at u rather than v . These choices avoid v , and every selected failure at u has the nonexceptional opposite witness v , with its head in L .

If both v, w are exceptional, both canonical cross-witness arcs point into u . Once again u is the sole sink and the ordinary cross-witness transfer excludes the graph. Hence it suffices to consider the stated constructions in the other cases. Let T be their one fixed completed tournament.

Put $J = N_T^+(u) \setminus N_D^+(u)$ and $h = |J|$. Every member r of J is good and nonexceptional. Since all partners precede u in the reference extension, the incoming direction $r \rightarrow u$ must be nonconvenient; therefore some $p \in P$ points originally to r . We claim the simultaneous envelope

$$N_T^{++}(u) \subseteq (N_D^{++}(u) \cup L) \setminus J, \quad J \subseteq N_D^{++}(u) \cup L. \quad (\text{T21})$$

An original first intermediate introduces a far head only through a chosen bad direction failing at u , and all such heads lie in L . For an added first intermediate $r \in J$, an original outgoing far head likewise lies in L by the partner budget. Consider an original far head $y \notin L$ with ry missing. This pair is good by Lemma S28.6. Every vertex reachable from P within two steps belongs to $P \cup \partial P \cup \partial^2 P$, and any member of this union originally far from u lies in L : it is in E and cannot gain a path after deleting arcs. Thus y is far from the predecessor p , making $r \rightarrow y$ nonconvenient. Its unique convenient direction is $y \rightarrow r$. This prevents all new far heads outside L . Finally $J \subseteq E$, and any member of E reached in $\hat{D}$ was already an exact second neighbor of u in D , because it was originally missing. This proves the second inclusion.

A zero-residual missing partner cannot lie in P : protection of its far target u would force the forbidden original arc $u \rightarrow r$. Every nonexceptional such partner lies in ∂P and is directed outward from u . The same is true of the unit w when nonexceptional, since $w \notin P(u)$. Only v can be an extra inward nonexceptional partner. Consequently

$$h \geq \mu(u) - k - 1.$$

Let $p = |P|$. Count the envelope in (T21), noting that all h vertices are removed from the union even if L overlaps the original second neighborhood. Using $p = 2\mu(u) + g(u) - 1 - m$,

we obtain

$$g_T(u) \geq g(u) + 2h - \lambda \geq \begin{cases} p - 1, & s = 0, \\ p + g(u) - 1, & s > 0, \end{cases} \quad g_T(u) \geq 0. \quad (\text{T22})$$

This bound also applies when $h = 0$; we do not assume that an empty added-first set alone prevents chosen failures at u .

We now show that every global median order of this same T must end at u . A feed outside the nonsink source set transfers its SNP to D after reversing all added outgoing arcs, since the bad directions avoid cross-witness sinks. This contradicts all-deficiency. A nonsink feed v or w is contradicted by retaining one outgoing canonical good cross-witness arc and using $t(v) \leq 2, t(w) \leq 1$. The backward-arc reversal preserves the global median order; the retained-arc argument accounts for all protected targets. Thus no vertex other than u is a global median feed of T .

Choose a global median order L_0 of T , ending at u . The feed SNP and (T22) force $d_T^+(u) = d_T^{++}(u)$. The strong Havet–Thomasse feed theorem [7, Theorem 1 and its proof] gives

$$|N_T^+(u)| \leq |G_{L_0}| \leq |N_T^{++}(u)|,$$

so these quantities are equal. Their global sedimentation lemma [7, Lemma 1] now yields a global median order

$$\text{Sed}(L_0) = (\text{bad incoming vertices}, u, N_T^+(u) \cup G_{L_0}),$$

with the displayed groups preserving their internal order. Because $d_D^+(u) > 0$, the last group is nonempty and its last vertex is not u . This contradicts the conclusion that every global median order of the fixed T ends at u .

All three-vertex witness graphs are exhausted. Hence no allocation $(m, 2, 1)$ is possible for any positive m . $\square$

Scope and audit notes. The multiple-retention lemma is a graph-level transfer criterion, not restricted to residual six. Every witness case, including the triangle, now holds for arbitrary $(m, 2, 1)$. No assertion $\eta = m \Rightarrow t \leq m$ is made. The proof allows large m , large exceptional packets, and arbitrarily large $t(u)$; it controls the heads actually introduced by one chosen completion instead. All reference orders and bad directions are chosen once per case before choosing a median feed. There is no comparison between independently reoriented feed completions. The final triangle step uses the global median sedimentation lemma, not the false general claim that every sink-free tournament has two distinct median feeds. That primary prerequisite was checked directly in Havet–Thomasse’s author PDF, Section 2, Theorem 1 and Lemma 1.

S29 A reduction of the residual allocation $(2, 2, 2)$

Assume D is strongly connected, all-deficient, and minimal under arbitrary nonempty arc-set deletion. Suppose its positive residual support is $\{c, v, w\}$, each of residual two. Let $\mathcal{B}$ be the actual cross-witness graph. Its edges are original missing pairs, and every actual directional failure source is a nonisolated vertex of this graph.

Lemma S29.1 (Retaining a good added first arc). *Complete the good pairs canonically and the bad pairs arbitrarily. If a global median feed f has a good added out-arc $f \rightarrow q$ and $t(f) \leq g_D(f) + 1$, then D has a Seymour vertex.*

Proof. Suppose D is all-deficient. Retain $f \rightarrow q$ and reverse every other added out-arc of f . The same order is still global median. Its first neighborhood is $N_D^+(f) \dot{\cup} \{q\}$. Protected-target compatibility excludes every path through q to a protected far target. An original first intermediate followed by a chosen bad arc can introduce only an unprotected far head; original or convenient second arcs introduce no far head. Hence

$$N_{T'}^{++}(f) \subseteq (N_D^{++}(f) \setminus \{q\}) \cup (\mathcal{F}(f) \setminus \{q\}).$$

Because fq was missing, q belongs to the disjoint union on the right before removal. Therefore $g_{T'}(f) \geq g_D(f) + 2 - t(f) \geq 1$, contradicting the tournament feed theorem. No restriction on t at other vertices is used here. $\square$

Lemma S29.2 (Two large star leaves are adjacent). *Suppose $\mathcal{B}$ is the two-edge path centered at c , and $t(v), t(w) > 2$. Then vw is an original adjacent pair.*

Proof. If $P(v)$ or $P(w)$ is empty, its vertex has exactly one missing partner, namely c , so vw is adjacent. Otherwise both large-unit packets have $s = 0, j = 1$, and their unique exceptional missing partner is c . Saturation gives

$$\mathcal{M}(v) \setminus \{c\} \subseteq N_D^{++}(v), \quad \mathcal{M}(w) \setminus \{c\} \subseteq N_D^{++}(w).$$

Suppose vw were missing. They then reach each other at exact distance two. Thus $w \notin P(v)$ and $w \in N_D^{--}(v) \setminus \{c\} = \partial P(v)$; the corresponding assertion holds with v, w interchanged. A predecessor $p \in P(v)$ with $p \rightarrow w$ makes $w \rightarrow v$ nonconvenient. A predecessor in $P(w)$ similarly makes $v \rightarrow w$ nonconvenient. Hence vw would be bad. A bad pair requires two distinct actual directional failure sources outside its endpoints. Only c is available, a contradiction. $\square$

Lemma S29.3 (Two adjacent vertices admit inward reference priorities). *If vw is an original adjacent pair, there is a reference linear extension of the protected-target partial order placing every missing partner of v before v , and every missing partner of w before w .*

Proof. Adding either family of priorities alone cannot create a cycle: an original protected chain between a vertex and its missing partner would force their adjacency. Suppose adding both creates a cycle, and take a shortest directed cycle in the relation generated by the old order and the two new priority families. After contracting old chains, it must use each new head v, w once. Thus there are missing partners r of v and q of w with old protected chains $v \triangleleft q$ and $w \triangleleft r$. Neither chain has length zero, since vw is adjacent. If $w \rightarrow v$, predecessor containment along the first chain forces $w \rightarrow q$, contrary to missing wq . If $v \rightarrow w$, the second chain forces $v \rightarrow r$, contrary to missing vr . These are the two possible orientations of the original adjacent pair, so the cycle is impossible. $\square$

Proposition S29.4 (The path case of $(2, 2, 2)$ is impossible). *Under the displayed residual allocation, $\mathcal{B}$ cannot be a two-edge path.*

Proof. Label its center c . The large-two-unit witness-degree lemma gives $t(c) \leq 2$. Orient all bad pairs so that their selected direction fails only at c . Every added out-arc of either leaf is good: failure of a selected bad out-arc at c would require the original arc $c \rightarrow v$ or $c \rightarrow w$, whereas both pairs are missing.

If no leaf has $t > 2$, all three actual sources satisfy $t \leq 2 \leq g_D + 1$, and the generic source transfer applies; all internal missing pairs among the three actual sources are good.

Otherwise choose the reference priorities of the large-leaf completion lemma for every large leaf. With one large leaf this is already available. With two, Lemmas S29.2 and S29.3 permit both families simultaneously. The proof of the leaf-completion lemma then keeps every large leaf deficient in this fixed tournament, and adds $c \rightarrow v$ for a chosen large leaf v .

Take any global median feed f . It cannot be a large leaf, whose tournament gap is positive. If $f = c$, retain its good added arc $c \rightarrow v$ and use Lemma S29.1, since $t(c) \leq 2$. If f is a small leaf and has an added out-arc, that arc is good and the same retention lemma applies. If it has no added out-arc, its original first arcs acquire no new far head, since the selected bad directions fail only at c ; its SNP transfers directly to D . Finally, if f is outside the actual source set, reverse all its added out-arcs. The original first arcs have no chosen-direction failures, so once more its SNP transfers directly to D . This last argument does not require $t(f) = 0$ and avoids improperly claiming that every outside vertex stayed deficient in the original completed tournament. All possible feeds lead to a contradiction. $\square$

Lemma S29.5 (A large nonselected vertex with a safe packet). *Suppose selected bad directions fail only at c , and $x \neq c$ has $\eta(x) = 2$ and $t(x) > 2$. Choose canonical good directions using a reference extension placing all good missing partners of x before x . If either $c \notin P(x)$, or every head of every selected bad arc is originally reachable from x within two steps, then the resulting tournament keeps x deficient.*

Proof. If $P(x)$ is empty, every missing direction into x is convenient; thus no incident pair is bad, every added incident arc is directed into x , and its second neighborhood is unchanged. Otherwise the large-two-unit packet has $s = 0, j = 1$ and a unique exceptional partner q . Its direction $q \rightarrow x$ is convenient, and saturation gives

$$E = (\mathcal{M}(x) \setminus \{q\}) \dot{\cup} \partial^2 P(x) = N_D^{++}(x).$$

In particular xq is good and is selected inward by the reference rule. Every added first neighbor r of x is nonexceptional. Moreover $r \rightarrow x$ is nonconvenient: for a good pair this follows from the reference priority, and for a bad pair it follows from badness itself. Consequently some $p \in P(x)$ points to r .

For $y \in U_D(x)$, the saturated partner budget excludes an original $r \rightarrow y$. Every path of length at most two from $P(x)$ stays in $P(x) \cup \partial P(x) \cup \partial^2 P(x)$, all of whose vertices are originally reachable from x . Thus y is far from p . An added good direction $r \rightarrow y$ is impossible because it fails at p . An added bad direction would also fail at p , impossible when $c \notin P(x)$. Under the alternative hypothesis it is impossible because y is an originally far head of x . Original first intermediates create no far head since selected directions do not fail at x . Writing $K = N_T^+(x) \setminus N_D^+(x)$, saturation also gives $K \subseteq N_D^{++}(x)$. Hence

$$N_T^{++}(x) = N_D^{++}(x) \setminus K, \quad g_T(x) = g_D(x) + 2|K| > 0.$$

$\square$

Lemma S29.6 (Adjacency of an inactive large vertex). *Suppose the actual source graph is the single edge uv , both endpoints have residual two and $t > 2$, and w has residual two and $t(w) > 2$ but is not an actual source. Then w is adjacent to at least one of u, v . If wc is missing for one endpoint c , then $c \notin P(w)$.*

Proof. The exceptional partner of each large active endpoint is its sole witness partner, namely the other endpoint. Thus if wc is missing, saturation at c puts $w \in N_D^{++}(c)$ and gives $c \notin P(w)$. If $P(w)$ is empty, w has only one missing partner and the adjacency assertion

is immediate. Otherwise let q be its unique exceptional partner. If wc is missing and $c \neq q$, saturation at w also gives $c \in N_D^{++}(w)$. The two vertices reach each other at exact distance two and are nonexceptional in each other's packets. They therefore belong to each other's full boundaries ∂P ; the two predecessor witnesses make both directions of wc nonconvenient. This would make wc bad, but its two actual witnesses must avoid its endpoints and there is only the other active source available. This is impossible. Thus any missing endpoint must equal q , and w cannot be missing from both distinct endpoints. $\square$

Theorem S29.7 (Exclusion of $(2, 2, 2)$). *The positive residual allocation $(2, 2, 2)$ is impossible.*

Proof. The source graph is nonempty and is supported on three vertices. A triangle is excluded by the witness-degree bound and generic source transfer. A two-edge path is excluded by Proposition S29.4. It remains that the actual sources are the endpoints u, v of one edge, with the third positive vertex w inactive. Internal missing pairs among these three vertices are all good.

If both active endpoints have $t \leq 2$, use the generic source transfer. If exactly one active endpoint is large, select the other as the sole failure source c , and use the large-leaf reference priorities to keep the large endpoint deficient. Its completion adds a good arc out of c . A median feed outside the actual source set transfers after reversing all added first arcs; the large leaf cannot be feed; and a feed at c is excluded by retaining its added good arc, since $t(c) \leq 2$. Thus both endpoints must be large.

Orient all bad pairs as a single-blame completion. They are exactly $A \times B$, where

$$A = N_D^+(u) \cap U_D(v), \quad B = N_D^+(v) \cap U_D(u).$$

Either endpoint can be chosen as the sole selected source. If $t(w) \leq 2$, choose either endpoint c , use the leaf lemma to keep the other endpoint deficient, and use the nonselected-source budget to keep w deficient. All zero-residual vertices likewise stay deficient. This fixed sink-free tournament would have at most one Seymour vertex, contradicting the two-vertex tournament theorem.

Now assume $t(w) > 2$. If w is missing from an active endpoint c , choose c as the selected source. By Lemma S29.6, the other endpoint x is adjacent to w , and $c \notin P(w)$. Use Lemma S29.3 to put all good missing partners of both x, w before them. The leaf completion keeps x deficient, and Lemma S29.5 keeps w deficient.

Finally suppose w is adjacent to both active endpoints. If some endpoint c is not in $P(w)$, select it and use the same simultaneous priorities for the other endpoint and w . If both endpoints belong to $P(w)$, their original adjacency to w must point as $w \rightarrow u$ and $w \rightarrow v$. Every bad endpoint in $A \cup B$ is then originally reachable from w within two steps. Choose either endpoint as c , and the alternative condition of Lemma S29.5 keeps w deficient; the same simultaneous priorities keep the opposite large leaf deficient.

In every remaining case every vertex except possibly c stays deficient in one fixed completed tournament. Zero-residual vertices use the single-blame protected-target envelope. The completion is sink-free because original outdegrees are positive. The two-Seymour-vertex theorem again gives the contradiction. $\square$

S30 Excluding the positive residual allocation $(2, 2, 1, 1)$

Let D be a finite nonempty strongly connected all-deficient oriented graph minimal under deletion of arbitrary nonempty arc sets on its fixed vertex set. Suppose the positive residual

support is

$$Z = \{u, v, a, b\}, \quad \eta(u) = \eta(v) = 2, \quad \eta(a) = \eta(b) = 1.$$

All definitions and all reachability assertions refer to the original graph D . Let $\mathcal{B}$ be its actual cross-witness graph, with isolated positive vertices included when convenient. Every directional failure source belongs to Z , every edge of $\mathcal{B}$ is an original missing pair, and an actual source is a nonisolated vertex of $\mathcal{B}$. Every other vertex has $t = 0$, and $t(a), t(b) \leq 1$.

Call a residual-two vertex *large* when $t > 2$; this does not mean that it is necessarily an actual directional failure source. The large-two-unit witness-degree lemma gives, for a large actual source x , a unique cross-witness partner c , and a convenient orientation $c \rightarrow x$. Consequently a cross-witness edge incident to a large residual-two source is good. If $P(x) \neq \emptyset$, its unique exceptional missing partner is precisely c , and the saturated packet has

$$s = 0, \quad B_x = \{c\}, \quad E_x = (\mathcal{M}(x) \setminus \{c\}) \dot{\cup} \partial^2 P(x) = N_D^{++}(x). \quad (2211-1)$$

In particular $c \in N_D^{--}(x)$ and $c \notin P(x)$. When $P(x) = \emptyset$, the residual identity instead gives $\mu(x) = 1$, so c is its only missing partner.

Lemma S30.1 (Large sources with distinct unit partners are adjacent). *Suppose $ab \notin E(\mathcal{B})$ and both u, v are large actual sources, but $uv \notin E(\mathcal{B})$. Then uv is originally adjacent.*

Proof. Each large source has exactly one cross-witness partner, necessarily a unit vertex. If either far-in set is empty, that vertex has only one missing partner, so uv is adjacent. Otherwise its exceptional partner is a unit, not the other residual-two vertex. If uv were missing, saturation puts $v \in N_D^{++}(u)$ and $u \in N_D^{++}(v)$. Thus $v \notin P(u)$ and $v \in N_D^{--}(u) \setminus B_u = \partial P(u)$, and likewise $u \in \partial P(v)$. A predecessor in $P(u)$ pointing to v makes $v \rightarrow u$ nonconvenient, and a predecessor in $P(v)$ pointing to u makes $u \rightarrow v$ nonconvenient. The pair uv would be bad. Its opposite directional witnesses must be the two remaining positive vertices a, b , which would put $ab \in E(\mathcal{B})$, a contradiction. $\square$

Lemma S30.2 (A unit–unit witness edge is impossible). *Under the displayed allocation, $ab \notin E(\mathcal{B})$.*

Proof. Suppose $ab \in E(\mathcal{B})$. The unit–unit witness-tail lemma makes the missing pair ab itself bad. Its two opposite sources must be u, v . Relabel the units, if necessary, so that

$$u \rightarrow a, \quad v \rightarrow b, \quad b \in \mathcal{F}(u), \quad a \in \mathcal{F}(v). \quad (2211-2)$$

In particular $uv \in E(\mathcal{B})$. The original arcs exclude ua and vb from $\mathcal{B}$; therefore

$$E(\mathcal{B}) \subseteq \{ab, uv, ub, va\}.$$

This graph is bipartite with $L = \{u, a\}$ and $R = \{v, b\}$. Complete every bad pair with its directional failure sources in L , and the good pairs canonically. The direction selected on the bad pair ab is $a \rightarrow b$, failing at u . Hence the selected unit a gains a first neighbor and is strictly deficient by $g_T(a) \geq g_D(a) + 2|K_a| - t(a) > 0$. The nonselected unit b is also strictly deficient: if K_b is empty its neighborhoods are unchanged, and otherwise the same budget is positive. Every zero-residual vertex is strictly deficient.

It remains to keep the nonselected residual-two vertex v deficient; the selected vertex u need not be controlled. If $t(v) \leq 2$, the nonselected-source budget does this, with the exact assertion when $K_v = \emptyset$.

Assume $t(v) > 2$. Its sole cross-witness partner is u . Choose the canonical reference extension putting every good missing partner of v before v . The convenience of $u \rightarrow v$ then selects this direction inward, so $u \notin K_v$. If $P(v) = \emptyset$, all missing directions into v are convenient, no bad pair is incident to v , and every missing incident pair is directed inward. Its gap is unchanged and positive.

Otherwise use the saturated packet (2211-1). Every $r \in K_v$ is nonexceptional, and its incoming direction $r \rightarrow v$ is nonconvenient. For a good pair this follows from the reference priority; for a bad pair it follows from badness. Thus there is an original predecessor $p \in P(v)$ with $p \rightarrow r$. This argument deliberately allows bad added first arcs at v .

Fix $y \in U_D(v)$. An original $r \rightarrow y$ is excluded by the zero remaining partner-head budget. The saturated cone $P(v) \cup \partial P(v) \cup \partial^2 P(v)$ consists entirely of vertices originally reachable from v within two steps; therefore y is far from every such predecessor p . An added good direction $r \rightarrow y$ is nonconvenient at p and cannot be selected. An added bad direction would also fail at p , forcing $p \in L = \{u, a\}$.

But $u \notin P(v)$, because $u \in N_D^-(v)$. If $p = a$, then $v \in U_D(a)$. By (2211-2), also $a \in U_D(v)$, so $v \rightarrow a$ is not an original arc and v cannot be protected from a . The unit's unique unprotected head must therefore be v . A bad direction failing at a can have only the head v , whereas the proposed second head y is distinct from v . This is impossible too.

Original first intermediates of v create no new far heads, since selected bad directions avoid v . All added first neighbors are in the saturated original second set. Consequently

$$N_T^{++}(v) = N_D^{++}(v) \setminus K_v, \quad g_T(v) = g_D(v) + 2|K_v| > 0.$$

In this one fixed completed tournament every vertex except possibly u is strictly deficient. It has no sink because the original outdegrees were positive. Havet–Thomasse's two-Seymour-vertex theorem for sink-free tournaments gives a contradiction. $\square$

Lemma S30.3 (No cross-witness edge can itself be bad). *Under the displayed allocation, every edge of $\mathcal{B}$ is good.*

Proof. By Lemma S30.2, ab is not a cross-witness edge. If the heavy-heavy edge uv were bad, its two remaining opposite sources a, b would put ab into $\mathcal{B}$. Thus a bad cross-witness edge, if any, must join a residual-two vertex and a unit; label it ua . Its opposite directional sources are the remaining heavy v and unit b , so $vb \in E(\mathcal{B})$. Because ua is itself bad, u is not large: a large residual-two source has a convenient incoming direction from every cross-witness partner. Hence $t(u) \leq 2$.

There are two possible original witness configurations. If $v \rightarrow u$ and $b \rightarrow a$, then $a \in \mathcal{F}(v)$ and the unit's unique head is $h(b) = u$. The original arcs exclude uv, ab as cross-witness edges, and ub cannot be a cross-witness edge because a partner of the unit b would have to point to $h(b) = u$, impossible for u itself. Thus $E(\mathcal{B}) \subseteq \{ua, vb, va\}$. Use the bipartition $L = \{u, v\}$, $R = \{a, b\}$. The selected direction on the bad pair ua is $u \rightarrow a$, failing at v , so $K_u \neq \emptyset$ and $g_T(u) \geq g_D(u) + 2 - t(u) > 0$. Both nonselected units stay strictly deficient, as do all zero-residual vertices. Only v could be Seymour.

In the other configuration, $v \rightarrow a$ and $b \rightarrow u$, we have $u \in \mathcal{F}(v)$ and $h(b) = a$. The original arcs exclude va, ub from $\mathcal{B}$, and $ab \notin E(\mathcal{B})$ by the preceding lemma. Thus $E(\mathcal{B}) \subseteq \{ua, vb, uv\}$. Use $L = \{v, a\}$, $R = \{u, b\}$. The selected direction on ua is $a \rightarrow u$, failing at v , so the selected unit a has a new first neighbor and stays strictly deficient. The nonselected residual-two vertex u , with $t(u) \leq 2$, stays strictly deficient, as does the nonselected unit b and every zero-residual vertex. Once more only v could be Seymour.

Each construction contradicts the two-Seymour-vertex theorem for its one fixed sink-free completed tournament. These two configurations exhaust the possible witness orientations of the bad pair ua . $\square$

Theorem S30.4 (Exclusion of $(2, 2, 1, 1)$). *A strongly connected all-deficient oriented graph minimal under arbitrary nonempty arc-set deletion cannot have positive residual allocation $(2, 2, 1, 1)$.*

Proof. The cross-witness graph is nonempty, and all of its edges are good by Lemma S30.3. We may take its source set to be the actual directional failure sources, not all positive-residual vertices. An inactive positive vertex has no threshold requirement in cross-witness source transfer.

If no residual-two actual source is large, the thresholds $t \leq 2 \leq g_D + 1$ hold at both residual-two actual sources, and $t \leq 1 \leq g_D + 1$ at both units. Apply cross-witness source transfer.

If exactly one residual-two actual source is large, put its unique cross-witness partner before it in the canonical reference extension. Its incident cross-witness edge is conveniently directed inward, making it a sink in the oriented cross-witness graph. All remaining actual sources satisfy the transfer thresholds. The same criterion gives a contradiction. This case includes a second large positive vertex which is not an actual failure source.

Suppose both u, v are large actual sources. If $uv \in E(\mathcal{B})$, their degree-one property forces $\mathcal{B}$ to have only the edge uv , because ab is absent. Select every bad direction with its sole failure source u . The residual-two leaf-completion lemma, with reference priorities at v , keeps v strictly deficient. Both nonselected units and every zero-residual vertex also stay strictly deficient in this same tournament. Only u could be Seymour, contradicting the two-Seymour-vertex theorem.

Finally $uv \notin E(\mathcal{B})$. By Lemma S30.1, uv is originally adjacent. The two-inward-priority lemma therefore gives one protected-order extension putting every missing partner of u before u , and every missing partner of v before v , simultaneously. Each large source's sole cross-witness edge is conveniently oriented inward under this extension. Both large vertices are consequently cross-witness sinks, and only unit vertices need the transfer thresholds. They satisfy those thresholds, so cross-witness source transfer again gives the contradiction. All cases are excluded. $\square$

S31 The final residual margin and its dependency chain

Theorem S31.1. *Every MIN graph satisfies $R = 2M - n \geq 7$, with $R \geq 8$ at even order. Every arc-set-minimal all-deficient graph satisfies*

$$M \geq \lceil n/2 \rceil + 3 + \mathbf{1}_{\{n \text{ even}\}}.$$

If o vertices are outside a sink strong component, add o to the right side.

Proof. Corollary S12.2 requires three positive residuals. Corollary S27.2 requires two actual sources of residual at least two. Thus total residual at most six leaves only $(2, 2, 1), (3, 2, 1), (2, 2, 2), (2, 2, 1, 1)$. The first two are excluded by Theorem S28.7, valid for all $(m, 2, 1)$; the last two by Theorems S29.7 and S30.4. Hence $R \geq 7$, and parity gives eight at even order. The outside-component estimate in Theorem S6.1 adds $\lceil o(k-2)/2 \rceil$ for a sink component of order $k \geq 5$. Using $k-2 \geq 3$ and the two parities gives the asserted outside penalty. $\square$

This proof uses four mixed partitions after the unbounded source theorem. Earlier pure-unit and one-heavy finite classifications remain valid but are no longer needed for the final margin. The five allocations still unresolved at $R = 7$ are $(3, 3, 1)$, $(3, 2, 2)$, $(3, 2, 1, 1)$, $(2, 2, 2, 1)$ and $(2, 2, 1, 1, 1)$. No assertion that these allocations are realizable is made.

Part III

Closure refinements, partner cycles, and negative-gap paths

S32 Fixing only the directions actually forced by protection

For an oriented graph D , write $a \triangleleft w$ when $a \rightarrow w$ and $N_D^-(a) \subseteq N_D^-(w) \setminus \{a\}$. As in the preceding draft this is a strict partial order, and a is an originally protected far head of w . A direction on a missing pair is called *protection-forced* if

$$a \rightarrow b \text{ is forced} \iff \exists w : a \triangleleft w, \quad w \rightarrow b \in A(D), \quad ab \text{ is missing.} \quad (\text{S32.1})$$

Here “forced” is relative to preserving all original protection relations, not to preserving every original convenient direction.

Lemma S32.1 (Forced directions are consistent and convenient). *Each forced direction is uniquely convenient. Every tournament completion which preserves all original protected pairs must orient that missing pair in its forced direction. In particular two opposite directions on the same pair cannot both be forced.*

Proof. Let $a \triangleleft w$ and $w \rightarrow b$ witness $a \rightarrow b$. Each original predecessor p of a also points to w , hence reaches b via $p \rightarrow w \rightarrow b$. Thus $a \rightarrow b$ is convenient. Conversely w is an original predecessor of b , but cannot reach its protected far head a within two steps; it witnesses failure of $b \rightarrow a$. The direction is uniquely convenient. In a completed tournament, preservation of $a \triangleleft w$ forces every outneighbor of w to be an outneighbor of a , so $a \rightarrow b$ is mandatory. Opposite forced directions would both be convenient while each has a witnessed nonconvenient reverse, which is impossible. $\square$

Fix these directions, but release *all other* missing pairs: this includes bad pairs, pairs convenient in both directions, and uniquely convenient pairs not forced by protection. For any linear extension $\prec$ of original protection, orient every released pair forward in $\prec$. Call the resulting tournament $T_\prec$. This fixes fewer convenient directions than the preceding canonical family, but is an alternative reference family, not a guaranteed set-theoretic superset: the remaining released directions must now all arise from the same order. Its median-score optimization nevertheless dominates the old canonical-family optimization, as proved below.

Theorem S32.2 (The fewer-fixed-arcs family preserves original protection). *Every $T_\prec$ preserves every original relation $y \triangleleft x$. Consequently, with $K_x = N_{T_\prec}^+(x) \setminus N_D^+(x)$,*

$$N_{T_\prec}^{++}(x) \subseteq (N_D^{++}(x) \setminus K_x) \cup (\mathcal{F}_D(x) \setminus K_x),$$

and the exact identity $g_{T_\prec}(x) - g_D(x) = 2|K_x| - c_x$ holds with $c_x \leq t_D(x)$ as before. There is no residual or support bound.

Proof. Fix $y \triangleleft x$. Only a newly incoming arc $q \rightarrow y$ needs checking for the required incoming-neighborhood nesting.

If $q \rightarrow y$ is forced, choose w with $q \triangleleft w$ and $w \rightarrow y$ original. Since $y \triangleleft x$, $w \rightarrow x$ is original. If qx is missing it is forced $q \rightarrow x$, using the same w . If $x \rightarrow q$ were original, $q \triangleleft w$ would imply $x \rightarrow w$, contrary to $w \rightarrow x$. Thus an original qx also has the required direction $q \rightarrow x$.

Otherwise qy is released, so $q \prec y \prec x$. An original $x \rightarrow q$ would make $y \rightarrow q$ forced, using $y \triangleleft x$; that contradicts release of qy . Similarly, a missing pair qx forced $x \rightarrow q$ has a witness w with $x \triangleleft w$ and $w \rightarrow q$ original. Transitivity gives $y \triangleleft w$, again forcing $y \rightarrow q$, a contradiction. In all remaining cases $q \rightarrow x$ is original, forced, or released forward. Thus every predecessor of y also points to x in $T_{\prec}$.

This proves preservation. A protected far head remains far: if $x \rightarrow z \rightarrow y$ existed in the completed tournament, $z \rightarrow y$ would force $z \rightarrow x$, a digon. Such a head cannot be newly first either, because $y \rightarrow x$ was original. Every original exact second path survives unless its endpoint becomes first; the claimed envelope and the disjoint-cost identity follow directly. $\square$

Why simply releasing every direction is false. The three-vertex original graph $1 \rightarrow 0 \rightarrow 2$ has protection $1 \triangleleft 0$. The protection extension $2 \prec 1 \prec 0$ would orient the missing pair as $2 \rightarrow 1$ if every pair were released, creating $0 \rightarrow 2 \rightarrow 1$ and destroying protection. Here $1 \rightarrow 2$ is exactly a forced direction under the definition. The theorem does *not* use the false statement that original incoming protection already supplies full outgoing domination in an incomplete graph.

S33 A maximal fewer-fixed-arcs completion has no added feed out-arcs

Theorem S33.1 (All released arcs are forward in every median order). *There is a fewer-fixed-arcs completion T_* for which every global median order extends original protection and puts every released added arc forward. Every median feed f of this fixed T_* satisfies*

$$K_f = \emptyset, \quad N_{T_*}^+(f) = N_D^+(f).$$

Proof. In a tournament an originally protected pair now has both incoming and outgoing domination. If $y \triangleleft x$, the usual two-position swap shows that no median order can put x before y ; the mutual arc is gained and no intermediate contribution is lost. Hence every median order of a fewer-fixed-arcs completion extends original protection.

Maximize the forward score over all pairs consisting of a fewer-fixed-arcs completion and an order, and choose T_* attaining it. If any released added arc were backward in one of its median orders L , re-complete using L as reference. It is admissible by protection preservation. Forced arcs stay fixed, and precisely the backward released arcs become forward, strictly increasing the score. This is impossible. The argument applies to every median order of the same maximizing completion because all have the same median score.

A feed has no outgoing released arc, since every such arc would be backward. Nor can it be the tail a of a forced arc: its witness $a \triangleleft w$ puts a before w in every median order. Thus it has no outgoing added arc of either type. $\square$

Lemma S33.2 (Optimization over all protection-preserving completions). *The maximum median score over the fewer-fixed-arcs reference family equals the maximum over all tournament completions preserving original protection. In particular it is at least the maximum score over the preceding canonical family. No corresponding dominance of the weighted correction-budget objective is asserted.*

Proof. Take any protection-preserving completion T and one of its median orders L . The order extends original protection. Every forced pair has the same direction in T by its

mandatory-direction lemma. Normalize every remaining missing pair forward in L , obtaining a valid fewer-fixed-arcs completion. The order's forward score increases by exactly the number of previously backward released arcs, and none of its original or forced arcs changes. Thus its median score is at least that of T . The reverse inequality follows because the reference family consists of protection-preserving completions. $\square$

Lemma S33.3 (A single fixed partial graph controls the optimization). *Let Q be obtained from D by adding precisely its protection-forced directions, and let r be the number of released missing pairs. Then*

$$\max_{\prec} \text{medscore}(T_{\prec}) = \text{medscore}(Q) + r.$$

Every median order of a maximizing T_ is a median order of Q . Conversely every median order of Q , completed by orienting all released pairs forward in that order, attains the maximum above.*

Proof. In Q each original relation $y \triangleleft x$ already has both incoming and outgoing neighborhood nesting. Incoming nesting follows from the forced-arc case in the previous preservation proof. For outgoing nesting, an original $x \rightarrow q$ makes $y \rightarrow q$ original or forced. A forced $x \rightarrow q$ has a witness $x \triangleleft w$ and $w \rightarrow q$ original; then $y \triangleleft w$ forces $y \rightarrow q$ if missing. An opposite original $q \rightarrow y$ would give $q \rightarrow x$ original, contrary to missing xq .

The two-position swap for two-sided domination does not require a complete graph: absent arcs count zero and nesting ensures no lost intermediate contribution. Thus every median order of Q extends original protection. Completing such an order yields its maximum Q score plus all r released arcs. Conversely every order in any reference completion has score at most this number, since its Q part scores at most $\text{medscore}(Q)$ and its released part at most r . Equality forces both component maxima. $\square$

Corollary S33.4 (Both original neighborhoods survive at a partial median feed). *Every median feed f of the fixed partial graph Q satisfies*

$$N_Q^+(f) = N_D^+(f), \quad N_Q^{++}(f) = N_D^{++}(f).$$

Proof. Every median order of Q respects original protection. A tail of a forced arc protects an original vertex and so cannot be a feed. Thus f has no added first arc. Any new two-step path at f would use an original first arc and a forced second arc, but that second direction is convenient, so its original predecessor f already reaches the head within two steps in D . The exact first set is unchanged, which proves equality of the exact second sets as well. $\square$

This identifies the remaining task with obtaining an appropriate Seymour median feed in this fixed *incomplete* graph Q under the extremal hypotheses. The tournament feed theorem must not be applied directly to Q .

Corollary S33.5 (An unbounded exact feed obstruction). *If D were all-deficient, every median feed f of this T_* would have at least $g_D(f)$ newly reached unprotected far heads, with no added first neighbors. Each new head is reached via an original first arc followed by a released, nonconvenient added arc. None of these second arcs can be protection-forced.*

Proof. The feed theorem gives $g_{T_*}(f) \leq 0$. The exact first set and survival of all original second paths give $g_{T_*}(f) = g_D(f) - |A_f|$, so $|A_f| \geq g_D(f)$. There is no added first intermediate. Every new-head second arc fails convenience at f , whereas every forced arc is convenient. Protection preservation excludes every originally protected far head. $\square$

This is stronger first-neighborhood control than the preceding canonical family, where a feed could still have an outgoing uniquely convenient keeper. It is not a full SNC proof: a released uniquely convenient pair may now be chosen *against* its convenient direction, so the feed may acquire far heads even without witnessing a genuinely bad missing pair. The old actual-bad-source conclusions and direct good-completion transfer cannot be carried over unchanged.

An exact control against automatic feed transfer. Take the directed four-cycle blow-up with independent class sizes $(3, 2, 4, 1)$ and full arcs $C_a \rightarrow C_{a+1}$ cyclically. There are no original protection relations, so $Q = D$. Its out masks, with the classes consecutive in the displayed order, are

$$(24, 24, 24, 480, 480, 512, 512, 512, 512, 7).$$

The graph has 21 arcs and 24 directed four-cycles. Each arc lies in at most eight of those cycles; exactly the three arcs from the last class to the first attain eight. Every feedback arc set has size at least three. Equality forces all three selected arcs to be these last-to-first arcs. Deleting them makes the graph acyclic. Thus every global median order places the four classes consecutively in their cyclic order. Its forward score is 18, and there are $3!2!4! = 288$ such orders. Every possible maximizing-family feed is the singleton last-class vertex $f = 9$.

Here $d_D^+(f) = 3$ but $d_D^{++}(f) = 2$. Completing all released pairs forward creates four extra second heads in the third class; $K_f = \emptyset$ as proved, but $g_T(f) = -3$. There are Seymour vertices elsewhere, so this is not an SNC counterexample. It is the unweighted clone realization of the weighted four-cycle warning at the end of Section 2 of Havet–Thomasse’s median-order paper. Its formal residual is $\eta(f) = 1$ but $t(f) = 4$, so it violates the *minimal-class* unit budget. The new framework must therefore still use the extremal counterexample hypotheses rather than treating no added first neighbors as a complete transfer proof.

A unit-feed control that violates the full-boundary hypothesis. Another graph has out masks

$$(30, 224, 96, 96, 2, 320, 1, 1, 1).$$

Name its vertices $(f, a, b, c, d, r, w, z, y)$ in that order. Its original arcs are $f \rightarrow a, b, c, d$, $a \rightarrow r, w, z$, $b, c \rightarrow r, w$, $d \rightarrow a$, $r \rightarrow w, y$, and $w, z, y \rightarrow f$. The original protected pairs include $d \triangleleft a$ and $r \triangleleft w$. The forced additions are $d \rightarrow r, w, z$ and $r \rightarrow f$, so Q has 21 arcs.

Four arc-disjoint Q cycles are $fazf$, $fbwf$, $ferf$, and $fdaryf$. Thus every median order has at least four backward arcs. The order $(d, a, b, c, r, w, z, y, f)$ has exactly four, so it is globally median with forward score 17. Completing its released pairs forward gives $K_f = \emptyset$ and introduces just the head y , changing its gap from one to zero. Numerically $\mu(f) = 1$, $P(f) = \{d\}$, $\eta(f) = 1$, and $t(f) = 1$. The missing opposite witness $r \rightarrow f$ is protection-forced, so sedimentation does not contradict released-arc rigidity by reversing that arc.

This graph is *not* a minimal all-deficient graph: it has many Seymour vertices. More specifically its external packet has $\partial P(f) = \{a\}$ but $N_D^-(f) = \{a, b, c, r\}$. Therefore the minimal-class full-boundary lemma at an active unit source fails. In bundle notation $j = 3$, $s = 2$, and the required minimal inequality $1 \geq g(f) + 2(j - s) = 3$ is violated. The control does not obstruct a counterexample-specific proof which uses that full-boundary lemma to rule out forced incoming missing witnesses.

Relation to the canonical-only budget obstruction. For the six-vertex graph $(0, 49, 51, 6, 9, 8)$ from the companion file, there are no forced missing directions. The

protection extension with ranks $(5, 2, 1, 0, 3, 4)$ gives completion $(0, 49, 51, 7, 41, 9)$. Vertex 0 remains a sink, so $\lambda = e_0$ is an expansion density and its correction budget is zero. This defeats the previously proved $-1/3$ obstruction by admitting an alternative reference direction, not by changing or disputing the canonical-only theorem. In its cloned and amplified variants, this same orientation pattern can likewise be used internally and mixed equally across copies to obtain zero budget.

Independent finite audit. `audit_protection_forced_completion.js` tested all 59049 five-vertex oriented graphs and 20000 fixed-seed samples of orders six through twelve. It checked 158098 fewer-fixed-arcs completions, 529572 protected pairs, and 950484 vertex envelopes and exact cost identities. It encountered 195408 genuinely released reversals of uniquely convenient directions, so the expansion was exercised nonvacuously. It additionally checked every median order of every score-maximizing reference completion on 300 sampled six-vertex graphs, checking 11529 median feeds with no added out-arcs. The ten-vertex clone control's 288 median orders and unique feed were independently counted by exact subset dynamic programming. These audits verify identities and attack the proofs; they do not prove the remaining all-deficient completion-budget existence.

Next global targets. The exact weighted threshold may now be sought over this alternative family, with only protection-forced arcs fixed. Alternatively, the fixed maximal completion has eliminated all added first neighbors at every median feed, isolating the obstruction to released reverse convenient directions or bad directions which create multiple far heads. A subsequent argument must control those heads without assuming a globally compatible packet selection or claiming an arc-weighted feed theorem that the previous controls refute.

S34 Balanced forced-family feeds must expose a boundary defect

All original predicates refer to a finite nonempty oriented graph D . For a vertex f , let $\mathcal{M}(f)$ be its missing partners, $\mu(f) = |\mathcal{M}(f)|$, let $P(f) = \{p : f \in U_D(p)\}$, and put

$$C_f = \partial P(f), \quad Q_f = N_D^{--}(f).$$

Let T_* be a score-maximizing protection-forced reference completion, as in the preceding theorem. Every global median of T_* extends original protection, places every released added arc forward, and has an original first neighborhood at its feed.

Lemma S34.1 (A forced incoming partner lies outside the far-in boundary). *If a direction $r \rightarrow f$ on an original missing pair is protection-forced, then*

$$r \in (Q_f \setminus C_f) \cap \mathcal{M}(f).$$

This assertion needs neither minimality nor all-deficiency.

Proof. Choose a forcing witness w with $r \triangleleft w$ and original $w \rightarrow f$. Original protection includes $r \rightarrow w$, so $r \rightarrow w \rightarrow f$ is an original two-step path. Since rf is missing, it is exact and $r \in Q_f$. If $r \in C_f$, some $p \in P(f)$ points originally to r . But f is far from p , so this predecessor witnesses nonconvenience of $r \rightarrow f$, contradicting convenience of every forced direction. Thus $r \notin C_f$. $\square$

Theorem S34.2 (The exact balanced-feed obstruction). *Let f be any global median feed of T_* with $g_D(f) > 0$. Then $t_D(f) \geq g_D(f)$. If equality holds, every original missing pair incident to f is completed by a protection-forced direction into f , and*

$$\emptyset \neq \mathcal{M}(f) \subseteq Q_f \setminus C_f.$$

In particular

$$C_f = Q_f \implies t_D(f) \geq g_D(f) + 1.$$

There is no residual, source-support, or minimality assumption.

Proof. The maximum theorem gives $K_f = \emptyset$. Let A_f be the genuinely new original-far exact second heads. Protection preservation puts $A_f \subseteq \mathcal{F}_D(f)$, and old exact second paths persist, so

$$N_{T_*}^{++}(f) = N_D^{++}(f) \dot{\cup} A_f, \quad g_{T_*}(f) = g_D(f) - |A_f|.$$

Feed SNP gives $g_D(f) \leq |A_f| \leq t_D(f)$.

Suppose equality $t_D(f) = g_D(f) > 0$ holds. Then $A_f = \mathcal{F}_D(f)$ and $g_{T_*}(f) = 0$. For any global median L ending at f , the strong feed inequalities give

$$|N_{T_*}^+(f)| \leq |G_L| \leq |N_{T_*}^{++}(f)| = |N_{T_*}^+(f)|,$$

hence

$$G_L = N_{T_*}^{++}(f) = N_D^{++}(f) \dot{\cup} \mathcal{F}_D(f).$$

Every missing partner belongs to this union: it is neither an original first neighbor nor a protected far target, because original protection requires adjacency. Thus $\mathcal{M}(f) \subseteq G_L$.

Also $\mu(f) > 0$. Indeed a whole vertex is adjacent to every other vertex and all its far targets are protected: if y is far from it, then $y \rightarrow f$; any original $z \rightarrow y$ must have $z \rightarrow f$, since the alternative $f \rightarrow z$ would reach y in two steps. Therefore a whole vertex has $t_D(f) = 0$, contrary to the positive equality.

Every missing partner is completed inward since $K_f = \emptyset$. If any such incoming arc $r \rightarrow f$ were released, global sedimentation of L would place f before $r \in G_L$ while remaining a global median of the same T_* . The released arc would become backward, contradicting all-median rigidity. Every incoming missing direction is therefore forced. Apply Lemma S34.1 to obtain the stated nonempty containment. If $C_f = Q_f$, equality is impossible; since both budgets are integral and $t_D(f) \geq g_D(f)$, the strict-surplus conclusion follows. $\square$

Corollary S34.3 (Every minimal-counterexample forced feed has nonunit residual). *Suppose D is strongly connected, all-deficient, and minimal under deletion of arbitrary nonempty arc sets on its fixed vertex set. Every global median feed f of T_* has*

$$\eta(f) \geq 2.$$

This conclusion does not assert that f is an actual bad-direction failure source.

Proof. All original gaps are positive. If $\eta(f) = 0$, the reviewed zero-residual protection theorem gives $t_D(f) = 0$, contradicting $t_D(f) \geq g_D(f) > 0$. If $\eta(f) = 1$, the unit budget gives $t_D(f) \leq 1$, so the same inequality forces $g_D(f) = t_D(f) = 1$. The full-boundary lemma has precisely the hypotheses $\eta(f) = 1, t_D(f) = 1$; it does not require actual BAD-source status. It gives $C_f = Q_f$, contradicting Theorem S34.2. Equivalently, the residual identity directly gives $|P(f)| = 2\mu(f) - 1$, hence $\mu(f) \geq 1$ and $P(f) \neq \emptyset$ before applying the unit packet bounds. Residual nonnegativity leaves only $\eta(f) \geq 2$. $\square$

Scope. The full-boundary strict-surplus theorem is an arbitrary-graph result; only its residual corollary uses the minimal-counterexample foundation. A forced-family feed may acquire its far heads through released reversals of unique convenient GOOD directions. It need not be an actual BAD source, and no complete SNC proof follows here.

S35 A balanced minimal forced-family feed has no far-in source

Here D is finite, nonempty, strongly connected, all-deficient and minimal under deletion of arbitrary nonempty arc sets on its fixed vertex set. Let T_* be a maximum protection-forced reference completion. Fix any global median feed f satisfying the *balanced* condition $t_D(f) = g_D(f)$. Put $p = |P(f)|$, $\mu = \mu(f)$, $g = g_D(f)$, $Q = N_D^-(f)$, and $C = \partial P(f)$.

Proposition S35.1 (The balanced-feed empty far-in refinement). *One has $\mu \geq 1$ and $P(f) = \emptyset$. The only remaining numerical shapes are*

$$(g, t_D(f), p, \eta(f)) \in \{(1, 1, 0, 2\mu), (2, 2, 0, 2\mu + 1)\}.$$

Thus every other original vertex reaches f within at most two steps. This is an incoming-reachability statement, not an assertion that f has the original second-neighborhood property or is an actual BAD source.

Proof. The general balanced-feed theorem S34.2 gives $\mu \geq 1$ and $\mathcal{M}(f) \subseteq Q \setminus C$. The strong deletion consequence S3.2 gives $g \in \{1, 2\}$. The residual identity is

$$p = 2\mu + g - 1 - \eta(f).$$

If $P(f) = \emptyset$, use only this identity: it gives the first branch when $g = 1$ and the second when $g = 2$. The nonempty- P packet estimates below are *not* applied in this case.

Assume $P(f) \neq \emptyset$. In the established partner-packet notation let $J = Q \setminus C$, $B = J \cap \mathcal{M}(f)$, $k = |B|$, and $s = |J \cap N_D^+(f)|$. The containment above implies $B = \mathcal{M}(f)$ and $k = \mu$; the target-bundle partition gives $|J| = k + s = \mu + s$. Lemma S8.9, whose stated hypothesis includes $P(f) \neq \emptyset$, gives a genuine set L with

$$0 \leq |L| \leq \begin{cases} \eta(f) - 2\mu, & s = 0, \\ \eta(f) - g - 2\mu, & s > 0. \end{cases}$$

Equivalently these two lower bounds on η follow directly from the original target-bundle refinements $\eta \geq 2|J|$ for $s = 0$, and $\eta \geq g + 2(|J| - s)$ for $s > 0$.

If $s > 0$, substituting $\eta \geq g + 2\mu$ into the residual identity gives $p \leq -1$, impossible. Hence $s = 0$ and $\eta \geq 2\mu$, so $1 \leq p \leq g - 1$. Since $g \in \{1, 2\}$, this forces $g = 2$, $p = 1$, and then $\eta = 2\mu$. Also $|J| = \mu$ and $\mathcal{M}(f) \subseteq J$, so $J = \mathcal{M}(f)$.

Let $P(f) = \{z\}$. A member of $P(f)$ cannot point to f , and $P(f)$ is disjoint from Q . As every missing partner is in Q , the vertex z is not a missing partner either. Therefore the original direction is $f \rightarrow z$. Since $P(f)$ is a singleton, $C = N_D^+(z)$. No member of C can point to f , for that would give z a two-step path to f . Nor can a member of C be missing from f , since $\mathcal{M}(f) = Q \setminus C$ is disjoint from C . Consequently $C \subseteq N_D^+(f)$. The target-bundle partition

$$V(D) = \{f\} \dot{\cup} N_D^-(f) \dot{\cup} P(f) \dot{\cup} C \dot{\cup} (Q \setminus C)$$

and $Q \setminus C = \mathcal{M}(f)$ exclude all other first neighbors, so

$$N_D^+(f) = \{z\} \dot{\cup} C = \{z\} \dot{\cup} N_D^+(z).$$

Each $z \rightarrow c$ with $c \in C$ is original, so no $c \in C$ points to z . Delete just the original arc $f \rightarrow z$, and call the result D' . The removed intermediate z had only heads in C , still original first neighbors of f , so removing it destroys no old exact second target of f . The removed first target z cannot become exact second, since no retained first intermediate $c \in C$ points to z . Thus

$$N_{D'}^{++}(f) = N_D^{++}(f), \quad g_{D'}(f) = g_D(f) - 1 = 1.$$

Every other vertex has unchanged first neighborhood and can only lose second targets. Hence D' is still all-deficient, contradicting arc-set deletion minimality. This excludes the only possible nonempty- P branch, so $P(f) = \emptyset$ and the two stated formulas follow. $\square$

Corollary S35.2 (Nonempty far-in forces strict head surplus). *In the same minimal-counterexample class, every global median feed f of T_* with $P(f) \neq \emptyset$ satisfies*

$$t_D(f) \geq g_D(f) + 1.$$

Proof. The general feed inequality gives $t_D(f) \geq g_D(f) > 0$. Equality would imply $P(f) = \emptyset$ by the proposition. Integrality gives the strict-surplus bound. $\square$

What remains unresolved. The proposition does not exclude either of its two remaining branches. In the $g = 1$ balanced branch, $\eta(f) = 2\mu(f) \geq 2$, but there may still be arbitrarily many forced incoming missing partners and reversed unique-GOOD second arcs. In the $g = 2$ balanced branch, $\eta(f) = 2\mu(f) + 1 \geq 3$; this branch is also unresolved. Feeds with strict head surplus $t_D(f) > g_D(f)$ are outside this refinement, and their far-in sets need not be small. In particular this is not a universal claim $P(f) = \emptyset$ for every maximal forced-family feed, nor a reduction of the full SNC to finitely many graph orders or residual allocations.

S36 Iterated protection closure and a strict original far-head surplus

All graphs are finite, nonempty, loopless and without directed two-cycles. First and second neighborhoods always mean directed distance exactly one and exactly two. For any graph H , protection means

$$a \triangleleft_H w \iff a \rightarrow w \in A(H), \quad N_H^-(a) \subseteq N_H^-(w) \setminus \{a\}.$$

The protected vertex a is far from w . A missing direction $a \rightarrow b$ is forced in H when some w satisfies $a \triangleleft_H w$ and $w \rightarrow b \in A(H)$. As previously proved, a forced direction is uniquely convenient in H : every predecessor of a already reaches b within two steps of H .

S36.1 A finite closure which preserves all protections already obtained

Starting at $Q_0 = D$, add *all* directions currently forced in Q_i to form Q_{i+1} . Stop when none remain, and call the final graph Q . No arc is removed. Every nonterminal round fills at least one missing pair, so there are at most $M(D)$ nonterminal rounds. The construction is deterministic and never attempts to add two opposite directions.

Lemma S36.1 (Monotone protection and a stable partial median). *Every protection of Q_i survives in Q_{i+1} . Consequently every added arc has a tail which protects some vertex in the final graph Q . In Q , every protected pair has both incoming and outgoing neighborhood nesting. Every global median order of Q therefore extends its protection partial order.*

Proof. Consistency and convenience hold in each current graph by the forced direction lemma; no minimality or deficiency is used. Fix $y \triangleleft_{Q_i} x$ and a newly incoming arc $a \rightarrow y$. There is a witness $a \triangleleft_{Q_i} w$, $w \rightarrow y$ in Q_i . Protection of y implies $w \rightarrow x$ in Q_i . If ax is missing, the same witness forces $a \rightarrow x$ in this round. An old $x \rightarrow a$ would imply $x \rightarrow w$, contradicting $w \rightarrow x$. Thus the new incoming arc cannot destroy the required nesting at y, x .

At termination, if $y \triangleleft_Q x$ and $x \rightarrow z$ in Q , then $z \rightarrow y$ would contradict incoming nesting, whereas a missing yz would still force $y \rightarrow z$. Therefore $y \rightarrow z$ is already in Q . This proves outgoing nesting. Swapping a reversed pair x, y in a vertex order gains the mutual arc $y \rightarrow x$, and loses no intermediate incoming or outgoing contribution. Missing arcs contribute zero, so this argument applies to the incomplete graph Q . Every global median order of Q places y before x . $\square$

S36.2 Maximization after closure, not during its construction

Orient every remaining missing pair of Q forward in one linear extension of its protection order. Since there are no forced pairs left, the previously proved protection-preserving reference theorem applies with an empty forced set. Each resulting tournament preserves every protection of Q , hence every protection of every earlier Q_i . Call the still-missing pairs *free pairs* in this subsection.

Choose a tournament T_* maximizing its ordinary global median score over *all* these reference extensions. No assumption is made that an intermediate Q_i is all-deficient or arc-minimal.

Lemma S36.2 (All-median rigidity and exact original feed neighborhoods). *Every global median order of T_* extends protection in Q and places every free added arc forward. If f is any of its median feeds, then*

$$N_{T_*}^+(f) = N_Q^+(f) = N_D^+(f), \quad N_Q^{++}(f) = N_D^{++}(f).$$

Moreover, writing r for the number of free pairs,

$$\max_T \text{medscore}(T) = \text{medscore}(Q) + r.$$

The possible feeds across all maximizing completions are precisely the global median feeds of the fixed partial graph Q .

Proof. Tournament protection gives domination, so the two-position swap forces every median order to extend protection. If a free added arc were backward in any median of a maximizing T_* , using that median as a new reference would flip all backward free arcs forward without changing any Q arc. This strictly increases the score, a contradiction. Closure under *all* protection extensions is essential here.

A feed cannot have an outgoing free arc because it would be backward. Nor can it be the tail of any arc added during closure: that tail has a protection successor which survives in Q and T_* , and must occur later in every median order. Thus no added first arc exists at f , at any stage of the construction.

Induct on the closure stages to obtain exact second-neighborhood equality. Any new path from f uses an original first arc $f \rightarrow a$. If its second arc $a \rightarrow b$ was added in the current round, its convenience in the *previous* graph means that this predecessor f already reached b within two steps there. A first target remains first throughout, so no new exact second head occurs. Old exact second paths persist because the first neighborhood is unchanged. The induction proves the equality with D . It does not assert that every closure arc is convenient in D .

Each Q median extends protection, and completing all free pairs forward adds exactly r to its score. Conversely any tournament/order pair scores at most $\text{medscore}(Q) + r$. Equality requires both components to attain their maxima. This also gives the feed equivalence. $\square$

S36.3 An original positive-gap feed cannot exhaust its far budget

Let $\mathcal{F}_D(f)$ denote the originally far heads not protected in D , let $t_D(f) = |\mathcal{F}_D(f)|$, and let $\mathcal{M}_Q(f)$ denote the missing partners of f in the terminal partial graph Q . Put $g_H(f) = |N_H^+(f)| - |N_H^{++}(f)|$ and

$$A_f = N_{T_*}^{++}(f) \setminus N_D^{++}(f).$$

The ordinary strong feed theorem and its global equality sedimentation are those of Havet–Thomasse; no arc-weighted feed theorem is used.

Theorem S36.3 (Strict surplus, without a minimal-counterexample hypothesis). *For any median feed f of T_* with $g_D(f) > 0$, one has*

$$\mathcal{M}_Q(f) \neq \emptyset, \quad t_D(f) \geq g_D(f) + 1.$$

More precisely, if $g_{T_}(f) = 0$, then*

$$\emptyset \neq \mathcal{M}_Q(f) \subseteq U_{T_*}(f) \cap (\mathcal{F}_D(f) \setminus A_f), \quad t_D(f) \geq g_D(f) + |\mathcal{M}_Q(f)|.$$

This is an arbitrary-graph theorem: it needs neither all-deficiency of D , strong connectivity, residual assumptions, nor arc minimality.

Proof. Protection in D survives. The exact first set is original and original exact second paths persist, so

$$N_{T_*}^{++}(f) = N_D^{++}(f) \dot{\cup} A_f, \quad A_f \subseteq \mathcal{F}_D(f), \quad g_{T_*}(f) = g_D(f) - |A_f| \leq 0.$$

The last inequality is the tournament feed theorem.

If $\mathcal{M}_Q(f)$ were empty, f would be whole in Q . Every far head y of a whole vertex in an oriented graph is protected: $y \rightarrow f$; if $z \rightarrow y$ then $f \rightarrow z$ would give $f \rightarrow z \rightarrow y$, so adjacency at f forces $z \rightarrow f$. Thus $y \triangleleft_Q f$. Since all these protections survive in T_* , f acquires no further second heads. The original first and second sets then give $g_{T_*}(f) = g_D(f) > 0$, a contradiction. Therefore $\mathcal{M}_Q(f)$ is nonempty.

Suppose $g_{T_*}(f) = 0$. For any median order L ending at f , the strong feed inequalities imply $G_L = N_{T_*}^{++}(f)$ and $|G_L| = |N_{T_*}^+(f)|$. If a remaining missing partner b belonged to this second set, the completed direction would be the free incoming arc $b \rightarrow f$, since f has no added first arc. Global sedimentation puts f before every vertex of G_L while retaining a global median order of the *same* T_* . That free arc becomes backward, contradicting all-median rigidity.

Every partner in $\mathcal{M}_Q(f)$ is consequently far in T_* . It was also an original missing partner, hence cannot have been an original protected head, and belongs to $\mathcal{F}_D(f) \setminus A_f$. Also $|A_f| =$

$g_D(f)$ in the equality case. The two disjoint subsets A_f and $\mathcal{M}_Q(f)$ give the sharper budget inequality.

If $g_{T_*}(f) < 0$, integrality gives $|A_f| \geq g_D(f) + 1$, which also proves the strict-surplus bound. $\square$

Corollary S36.4 (An unbounded sufficient structural condition). *If every deficient vertex v of an oriented graph D satisfies $t_D(v) \leq g_D(v)$, then D has a Seymour vertex.*

Proof. Otherwise all original gaps are positive and any feed of the construction contradicts the strict-surplus theorem. $\square$

Corollary S36.5 (Nonunit residual at every closure feed). *If D belongs to the previously strongly connected arbitrary- arc-set-deletion-minimal all-deficient class, every median feed of T_* has $\eta \geq 2$. If $t_D(f) \leq 2$ at such a feed, necessarily $g_D(f) = 1$ and $t_D(f) = 2$.*

Proof. The zero and unit residual budgets are $t = 0$ and $t \leq 1$ respectively, while the strict-surplus theorem gives $t \geq g_D + 1 \geq 2$. For $t \leq 2$ the same inequalities force $g_D = 1, t = 2$. $\square$

Comparison and scope. The two balanced branches of the preceding *one-round* forced-family analysis are now absent for the *iterated-closure* maximizing family. This does not retroactively claim that they are impossible in every old one-round completion. The new terminal graph need not remain a minimal counterexample, and its arcs need not all be convenient relative to D . No minimality-based packet bound is applied to an intermediate graph. The strict-surplus theorem still permits arbitrarily large original unprotected far budgets. It neither proves the complete SNC nor supplies an actual all-deficient graph. External novelty and peer review remain outstanding.

S37 Weak protection steps force a directed cycle at every zero-gap feed

Use the preceding iterated closure definitions. In addition to adding all currently FORCED directions, allow the following *single-arc* step in a current graph H :

$$\begin{aligned} &ab \text{ missing,} \quad N_H^-(a) \subseteq N_H^-(b), \\ &b \text{ is maximal in the current protection order; add } a \rightarrow b. \end{aligned}$$

Maximal means there is no original-current relation $b \triangleleft_H x$. This step is called weak protection. Recompute protection and forced directions after each weak step: simultaneous arbitrary weak choices are *not* permitted. Continue until neither a forced direction nor an eligible weak step remains; any deterministic tie rule may be used. Termination requires at most $M(D)$ added pairs. Denote the terminal graph by Q^w .

Lemma S37.1 (The single weak step is convenient and protection-monotone). *Every weak step preserves all current protections and creates $a \triangleleft b$. Its new direction is convenient in the graph before the step.*

Proof. Only the incoming neighborhood of b changes. An old relation $b \triangleleft_H x$ could be destroyed by the new predecessor, but maximality excludes every such relation. An old relation $y \triangleleft_H b$ can only become easier to satisfy when its right vertex gains a predecessor; every other relation is unchanged. After the addition, $N^-(a) \subseteq N^-(b) \setminus \{a\}$, so the new arc protects b . For convenience, every old predecessor of a already points directly to b by the assumed containment. The graph remains oriented because ab was missing. $\square$

At the terminal graph no forced missing direction remains, so current protection has both incoming and outgoing nesting. Complete all its remaining missing pairs forward in one current-protection extension, and maximize the ordinary global median score over all such extensions. Call a maximizing tournament T_*^w .

The preceding closure proof applies without change: every added arc has a tail with a protection successor which survives to the end, and every step was convenient relative to its preceding graph. Thus, at every global median feed f ,

$$N_{T_*^w}^+(f) = N_{Q^w}^+(f) = N_D^+(f), \quad N_{Q^w}^{++}(f) = N_D^{++}(f).$$

Every median respects terminal protection and makes all still-free added arcs forward. No intermediate graph is assumed all-deficient, strongly connected or minimal.

Theorem S37.2 (The terminal missing partners of a zero-gap feed contain a cycle). *Let f be any median feed of T_*^w such that $g_D(f) > 0$ and $g_{T_*^w}(f) = 0$. Then*

$$R = \mathcal{M}_{Q^w}(f) \neq \emptyset, \quad \delta^-(Q^w[R]) \geq 1.$$

In fact the original induced graph $D[R]$ already contains a directed cycle. In particular a cycle of length at least three exists, and

$$|R| \geq 3, \quad t_D(f) \geq g_D(f) + |R| \geq g_D(f) + 3.$$

Neither minimality nor actual BAD failure-source status is required.

Proof. The preceding arbitrary-closure proof gives $R \neq \emptyset$ and shows that every member of R is far from f in the completed tournament: otherwise it belongs to the equality good set, and global sedimentation would reverse its free incoming arc to f . Also $R \subseteq \mathcal{F}_D(f) \setminus A_f$ and $|A_f| = g_D(f)$, where A_f is the set of new original-far second heads. This gives $t_D(f) \geq g_D(f) + |R|$.

Fix $r \in R$. The feed f is maximal in terminal protection, since every median order extends that order. There is no eligible weak $r \rightarrow f$ step, so

$$N_{Q^w}^-(r) \not\subseteq N_{Q^w}^-(f).$$

Choose a witness p in the difference. Since r is far from f in the tournament and $p \rightarrow r$ is retained, the alternative $f \rightarrow p$ would give $f \rightarrow p \rightarrow r$. Therefore $p \rightarrow f$ in the tournament. It was not an arc of the terminal graph by choice of p , and the opposite direction could not have been an arc there either. Thus pf is a remaining missing pair, so $p \in R$. Also $p \neq r$, because the graph is loopless. Hence every $r \in R$ has a predecessor inside $Q^w[R]$. Following predecessors in a finite graph produces a directed cycle. Loops and directed two-cycles are forbidden, so at least three distinct vertices occur.

It remains to lift cycle existence back to $D[R]$. Hold this final set R fixed throughout the earlier construction. At a FORCED step which adds an internal arc $a \rightarrow b$ of R , let $a \triangleleft_H w$ and $w \rightarrow b$ be its preceding-graph witness. Since b is far from f in the final tournament, the retained arc $w \rightarrow b$ implies $w \rightarrow f$ there. If wf were adjacent in the terminal graph, its direction would already be $w \rightarrow f$. The retained relation $a \triangleleft w$ would then force the still-missing $a \rightarrow f$, contradicting terminality. Thus $w \in R$. The new internal arc is merely a shortcut of the old internal path $a \rightarrow w \rightarrow b$. Adding all such shortcuts cannot make an acyclic induced graph cyclic.

At a single WEAK step adding an internal $a \rightarrow b$, suppose an acyclic previous induced graph would become cyclic. Then it previously had a path $b \rightsquigarrow p \rightarrow a$ inside R . Its length

cannot be one since ab was missing. The old containment $N_H^-(a) \subseteq N_H^-(b)$ gives $p \rightarrow b$ already, so the previous graph itself had the cycle $b \rightsquigarrow p \rightarrow b$, a contradiction. Steps outside R do not alter its induced arcs. Hence an acyclic $D[R]$ would stay acyclic throughout, contrary to the terminal cycle just proved. The original cycle exists. $\square$

Corollary S37.3 (Low-budget feeds are strictly, not equally, Seymour). *At a feed with $g_D(f) > 0$, either*

$$g_{T_*^w}(f) \leq -1, \quad \text{or} \quad t_D(f) \geq g_D(f) + 3.$$

In particular if $t_D(f) \leq g_D(f) + 2$, the completed feed has strict SNP. If $t_D(f) \leq 3$, a zero completed gap is impossible.

Proof. The feed theorem gives $g_{T_*^w}(f) \leq 0$. The theorem excludes the equality case under either displayed budget restriction. $\square$

Scope and remaining obstruction. The stronger +3 budget is asserted only for a *zero completed gap*. If the completed gap is negative, the general strict original surplus remains $t_D \geq g_D + 1$, not $g_D + 3$. A feed may have strict SNP only because it acquired two or more original-far second heads; its original gap can still be positive. No full SNC proof follows. Although a cycle is now proved to exist in the original graph $D[R]$, its vertices need not be actual BAD sources in D . It must not be counted as three additional original positive-residual vertices without a separate argument. Positive indegree is proved for the terminal induced graph, not for every vertex of the original induced graph.

S38 A weak-closure zero-gap feed must have residual at least four

Let D be strongly connected, all-deficient, and minimal under deletion of arbitrary nonempty arc sets on its fixed vertices. All sets in this section are computed in the original D . For any vertex v put

$$S_v = \mathcal{M}_D(v) \cap U_D(v), \quad g = g_D(v), \quad \mu = \mu_D(v), \quad p = |P_D(v)|.$$

If $p > 0$, use the packet with $J = N_D^{--}(v) \setminus \partial P_D(v)$, $B = J \cap \mathcal{M}_D(v)$, $k = |B|$, $s = |J \cap N_D^+(v)|$, and

$$E = (\mathcal{M}_D(v) \setminus B) \dot{\cup} \partial^2 P_D(v), \quad L = E \setminus N_{\widehat{D}}^{++}(v),$$

where $\widehat{D}$ deletes the s original arcs from v to J . The packet estimates are

$$|L| \leq \eta_D(v) - 2k - g\mathbf{1}_{s>0}, \quad \bigcup_{r \in \mathcal{M}_D(v) \setminus B} (N_D^+(r) \cap U_D(v)) \subseteq L.$$

Lemma S38.1 (Original missing far heads and the nonexceptional budget). *If $p > 0$, then*

$$|S_v| \leq k + |L| \leq \eta_D(v) - k - g\mathbf{1}_{s>0}.$$

If $p = 0$, the residual identity gives

$$\eta_D(v) = 2\mu + g - 1 \geq 2|S_v|.$$

Moreover, when $p > 0$ and $k = 0$,

$$\mathcal{F}_D(v) \subseteq L, \quad t_D(v) \leq \eta_D(v) - g\mathbf{1}_{s>0}.$$

Proof. Every nonexceptional missing far vertex itself belongs to $\mathcal{M}_D(v) \setminus B \subseteq E$. An original far head cannot become reachable by deleting arcs, so it is not in $N_D^{++}(v)$ and belongs to L . At most k original missing far vertices can be exceptional, proving the first bound. When $p = 0$, use only the residual identity and $g \geq 1$; no nonempty- P packet is applied.

Suppose $p > 0$ and $k = 0$. An unprotected far head y is either originally missing from v , or an original incoming neighbor of v . The missing case was just proved. In the incoming case failure of protection supplies an original predecessor $r \rightarrow y$ which does not point to v . The witness is neither v (that would contradict the far relation) nor y (there are no loops). An original $v \rightarrow r$ would give $v \rightarrow r \rightarrow y$, also impossible. Therefore r is an original missing partner of v . Since $k = 0$, it is nonexceptional; the shared head budget places y in L . These cases exhaust $\mathcal{F}_D(v)$. $\square$

Theorem S38.2 (Residual-four lower bound at a zero completed gap). *Let f be a median feed of the maximum weak-closure completion with $g_D(f) > 0$ and zero completed gap. Write $m = |\mathcal{M}_{Q^w}(f)| \geq 3$ for its remaining terminal missing degree. Then*

$$\eta_D(f) \geq m + 1 \geq 4.$$

The more precise original-graph bounds are

$$\begin{aligned} p = 0 : \quad & \eta_D(f) \geq 2m + g_D(f) - 1, \\ p > 0, k = 0 : \quad & \eta_D(f) \geq g_D(f) + m + g_D(f)\mathbf{1}_{s>0}, \\ p > 0, k > 0 : \quad & \eta_D(f) \geq m + k + g_D(f)\mathbf{1}_{s>0}. \end{aligned}$$

Proof. The weak-closure zero-gap theorem gives $m \geq 3$, $\mathcal{M}_{Q^w}(f) \subseteq S_f$, and $t_D(f) \geq g_D(f) + m$. Apply the preceding original-graph packet lemma. If $p = 0$, $\mu_D(f) \geq |S_f| \geq m$ gives the first bound, which is at least six. If $p > 0$ and $k = 0$, use the far-head bound and $t_D(f) \geq g_D(f) + m$. If $k > 0$, use the missing-far bound. Each expression is at least $m + 1$: the first has $m \geq 3$, the second has $g_D(f) \geq 1$, and the third has $k \geq 1$. Thus the unified bound is $\eta_D(f) \geq m + 1 \geq 4$. No closure graph has been assumed minimal or all-deficient. $\square$

Corollary S38.3 (The residual-four equality shape). *If such a feed has $\eta_D(f) = 4$, necessarily $P_D(f) \neq \emptyset$, and*

$$m = 3, \quad k = s = 0, \quad g_D(f) = 1, \quad t_D(f) = 4.$$

In this necessary equality shape, the original packet also satisfies

$$\begin{aligned} \mu_D(f) &\geq 3, & |P_D(f)| &= 2\mu_D(f) - 4 \geq 2, \\ N_D^-(f) &= \partial P_D(f), & L &= \mathcal{F}_D(f), & |L| &= 4. \end{aligned}$$

and $|\partial^2 P_D(f)| = |\partial P_D(f)| - 1$.

Proof. The previous bounds exclude $p = 0$ and $s > 0$ at residual four and force $m = 3$ with $k \leq 1$. When $k = 0$, $g + m \leq t_D(f) \leq 4$ forces $g = 1$ and $t_D(f) = 4$. When $k = 1$, the missing-far bound gives $3 \leq |S_f| \leq 4 - 1 = 3$, so S_f is exactly the three terminal partners. The bound $|L| \leq 4 - 2 = 2$ forces the unique exceptional partner b to belong to S_f ; the other two partners already belong to L . The original-cycle lifting conclusion of Theorem S37.2 gives a directed cycle in $D[S_f]$. With only three vertices this is an original directed triangle. The predecessor $r \rightarrow b$ of b on that triangle is a nonexceptional original missing partner. Since b is an original far head, the shared partner-head budget places b in L as well. Thus

$|L| \geq 3$, contradicting $|L| \leq 2$. The $k = 1$ case is impossible, leaving exactly the stated necessary shape. The residual identity gives the formula for $|P_D(f)|$, and $\mu_D(f) \geq m = 3$. The original packet partition has $|N_D^{--}(f) \setminus \partial P_D(f)| = k + s = 0$. Because $k = 0$, the inclusion $\mathcal{F}_D(f) \subseteq L$ and its four-element size saturate $|L| \leq \eta_D(f) = 4$. Here $\widehat{D} = D$, the envelope is contained in $E = \mathcal{M}_D(f) \dot{\cup} \partial^2 P_D(f)$, and $|E| = d_D^{++}(f) + 4$. The residual identity now gives $|\partial^2 P_D(f)| = d_D^{++}(f) + 4 - \mu_D(f) = |\partial P_D(f)| - 1$. $\square$

Scope. This residual bound concerns zero completed gap, not all median feeds. Strictly negative completed gaps remain possible at residual two or three. The residual-four shape is a necessary condition, not a model known to exist. The original partner cycle is justified by the cited lifting theorem, not by assuming closure-added directions are original. Its vertices need not be actual BAD failure sources and cannot be counted as additional positive-residual vertices without a separate proof.

S39 Remaining missing degree is charged to original head surplus

Let Q be the terminal iterated protection closure of an arbitrary finite nonempty oriented graph D , and let T_* be a maximum-score reference completion of Q . Every remaining missing pair of Q is free, and every median of T_* puts all free arcs forward. Let f be a median feed with $g = g_D(f) > 0$, $t = t_D(f)$ and $\mu_Q = |\mathcal{M}_Q(f)|$. Its first set is original, and its completed second set is

$$N_{T_*}^{++}(f) = N_D^{++}(f) \dot{\cup} A_f, \quad A_f \subseteq \mathcal{F}_D(f).$$

Put $c = |A_f|$ and let G_L be the good vertices of a median order L ending at f .

Lemma S39.1 (Every newly reached original-far head is good). *One has $A_f \subseteq G_L$ in every median order ending at f . Writing $b_L = |N_D^{++}(f) \setminus G_L|$ and $s_L = |G_L| - |N_{T_*}^+(f)|$, the exact identity is*

$$s_L = c - g - b_L \geq 0.$$

Proof. A new head y has a path $f \rightarrow a \rightarrow y$ with original first arc. The second arc cannot belong to Q , whose exact second neighborhood at the feed is still original. Thus it is a free completion direction, which is forward in every median order. Its first intermediate a occurs before y , so y is good. Consequently the good set consists of all c new heads and $d_D^{++}(f) - b_L$ old heads. Subtract the original first count $d_D^{++}(f) + g$ to obtain the identity. The weak inequality is the strong feed theorem. $\square$

Proposition S39.2 (The stronger equality budget). *If $|G_L| = |N_{T_*}^+(f)|$, then*

$$\mu_Q \leq t - g.$$

This includes negative completed gap as well as zero completed gap. No arc-minimality, source-support or residual assumption is needed.

Proof. If $r \in \mathcal{M}_Q(f) \cap G_L$, its completed direction is the free incoming arc $r \rightarrow f$, because the feed has no added first arc. Equality permits global sedimentation. It places f before r while preserving the median score, making this free arc backward in another median of the same T_* . All-median rigidity excludes this. Therefore

$$|\mathcal{M}_Q(f) \cap N_{T_*}^{++}(f)| \leq |N_{T_*}^{++}(f) \setminus G_L| = d_D^{++}(f) + c - d_D^+(f) = c - g.$$

In fact every such remaining partner in the second set is an original *nongood* second neighbor: the new-head lemma excludes all of A_f from this nongood class. Every remaining missing partner not in the completed second set is far in T_* , since it is not a completed first neighbor either. It was originally missing, hence cannot be originally protected, and belongs to $\mathcal{F}_D(f) \setminus A_f$. Consequently

$$|\mathcal{M}_Q(f) \cap U_{T_*}(f)| \leq t - c.$$

The two classes partition $\mathcal{M}_Q(f)$. Their sum is at most $(c - g) + (t - c) = t - g$, proving the result. $\square$

Corollary S39.3 (Large remaining degree forces stable medians). *If $\mu_Q > t - g$, every median order ending at f has strict strong-feed slack $|G_L| > |N_{T_*}^+(f)|$. In particular it is stable under the Havet–Thomasse sedimentation operation. If $t = 2$, the strict original-surplus theorem gives $g = 1$; strong-feed equality then forces $\mu_Q = 1$.*

Proof. The strong feed theorem always gives the weak inequality. Equality contradicts the proposition, so all such orders have strict inequality. For $t = 2$, positive original gap and $t \geq g + 1$ give $g = 1$. The proposition gives $\mu_Q \leq 1$, while terminal whole-feed exclusion gives $\mu_Q \geq 1$. $\square$

Remaining scope. The budget does not apply automatically to medians with strict good slack. It does not assert that every median order has equality, nor that every remaining missing partner is far when the completed gap is negative. It separates the two possible global obstructions rather than resolving either one in arbitrary size.

S40 Degree localization and an arbitrary-order low-far-budget class

Write $h_D(v)$ for the number of originally protected far heads of v , and $t_D(v)$ for its originally unprotected far heads. In every finite oriented graph of order n ,

$$t_D(v) - g_D(v) = n - 1 - 2d_D^+(v) - h_D(v).$$

This follows by partitioning all other vertices into exact first, exact second and far targets, and splitting the far targets into protected and unprotected heads.

Corollary S40.1 (Adjusted-degree localization). *At every median feed f of the iterated closure maximizing completion with $g_D(f) > 0$,*

$$2d_D^+(f) + h_D(f) \leq n - 2.$$

If $\delta = \delta^+(D)$ and $q = n - 2\delta$, then

$$0 \leq d_D^+(f) - \delta \leq \left\lfloor \frac{q - 2 - h_D(f)}{2} \right\rfloor.$$

In particular when $n = 2\delta + 3$, every such feed has $d_D^+(f) = \delta$ and $h_D(f) \leq 1$.

Proof. The strict original surplus gives $t_D(f) - g_D(f) \geq 1$. Substitute the identity and rearrange. The last assertion uses that the nonnegative integer $d_D^+(f) - \delta$ is at most $\lfloor (1 - h_D(f))/2 \rfloor$. $\square$

Corollary S40.2 (At most one unprotected far head at each vertex). *Every finite nonempty oriented graph with $t_D(v) \leq 1$ at every vertex has a Seymour vertex. No graph-order, minimality, missing-pair or cross-witness-flexibility hypothesis is required.*

Proof. If all vertices were deficient, any feed of the iterated-closure maximizing completion would have $t_D(f) \geq g_D(f) + 1 \geq 2$. $\square$

Scope. These statements are consequences of the all-order closure proof, not finite enumeration. The degree localization is conditional on original positive gap at the chosen feed. It does not assert that every graph of order $2\delta + 3$ has a minimum-outdegree Seymour vertex. Novelty relative to the extensive existing SNC literature remains to be assessed.

S41 Exact outgoing-star deletion constraints in the original graph

For an original oriented graph D and a vertex f , put $O = N_D^+(f)$, $B = N_D^{++}(f)$, and $g = g_D(f)$. For every $b \in B$ let

$$I_b = N_D^-(b) \cap O$$

be its nonempty set of original first-neighborhood witnesses. For $\emptyset \neq S \subseteq O$, delete precisely the original arcs $f \rightarrow S$, obtaining D_S , and define

$$L(S) = \{b \in B : I_b \subseteq S\}, \quad R(S) = S \cap \Gamma_D(O \setminus S).$$

Thus $L(S)$ is the set of lost old second targets, while $R(S)$ is the set of removed first targets which become exact second targets.

Lemma S41.1 (Exact loss and retargeting formula). *For every original oriented D ,*

$$N_{D_S}^{++}(f) = (B \setminus L(S)) \dot{\cup} R(S),$$

and hence

$$g_{D_S}(f) = g - |S| + |L(S)| - |R(S)|.$$

If D is all-deficient and minimal under deletion of arbitrary nonempty arc sets on its fixed vertex set, then

$$|L(S)| \leq |S| + |R(S)| - g \quad \text{for every } \emptyset \neq S \subseteq O. \quad (\text{OD1})$$

Proof. The retained first set is $O \setminus S$. An old second target b remains reached exactly when I_b meets that retained set. No originally far target can become reached when arcs are deleted. The only possible new exact second targets are removed first targets in S , and precisely those reached from a retained intermediate form $R(S)$. These two classes are disjoint, proving the exact identity.

Every vertex other than f has unchanged first neighborhood and can only lose second targets. Thus any Seymour vertex forced by minimality of the nonempty deletion must be f , giving $g_{D_S}(f) \leq 0$. Substitute the exact gap formula. $\square$

Corollary S41.2 (Every proper retained set is externally expanding). *In the same minimal class, every proper subset $X \subsetneq O$ satisfies*

$$|\Gamma_D(X) \setminus X| \geq |X|, \quad |\Gamma_D(O) \setminus O| = |O| - g < |O|.$$

Thus O is inclusion-minimal among its subsets that fail this external expansion inequality.

Proof. For $S = O \setminus X$, the retained first set is X . No member of X points to f , because each original $f \rightarrow x$ excludes its reverse. Therefore $N_{D_S}^{++}(f) = \Gamma_D(X) \setminus X$. Minimality gives the first inequality. With $X = O$, the same external boundary is precisely B , giving the strict deficit. The empty retained set has both sides zero and causes no exception. $\square$

Corollary S41.3 (Gap two has no private old second heads). *If $g_D(f) = 2$ in the original minimal class, then for every $a \in O$:*

1. *Some other original first vertex points to a .*
2. *No old second target has a as its unique original first witness.*
3. *Deleting just $f \rightarrow a$ preserves all old exact second targets and adds precisely a as an exact second target. The new gap at f is zero.*

Consequently

$$\delta^-(D[O]) \geq 1, \quad |I_b| \geq 2 \quad (b \in B).$$

The induced original first-neighborhood graph contains a directed cycle of length at least three. For $g_D(f) = 1$, the singleton constraint is instead $|L(\{a\})| \leq |R(\{a\})| \leq 1$; it does not force a cycle.

Proof. For $S = \{a\}$, the retargeting set has size zero or one and the loss set has nonnegative size. At gap two, (OD1) gives $0 \leq |L(S)| \leq |R(S)| - 1 \leq 0$. Thus $|R(S)| = 1$ and $L(S)$ is empty. The exact identities prove all three assertions. A head with a unique first witness would belong to that singleton loss set, so every I_b has at least two vertices. Following predecessors in $D[O]$ produces a cycle; orientedness excludes lengths one and two. The gap-one bound is the same formula with $g = 1$. $\square$

Use at closure feeds. These inequalities apply to the original first set of every final closure feed, since that set has been proved unchanged. They constrain original two-step witnesses, not new second heads added by closure or completion. No intermediate closure graph is assumed arc-minimal, and deleting its newly added arcs is not justified by original minimality. The conditions are necessary, not sufficient to construct a graph with all vertices deficient.

S42 A flexible pair between two uncontrolled actual sources

All predicates refer to the original finite nonempty oriented graph D . Let A be its actual bad-direction failure sources. Suppose that $A \setminus \{u, v\}$ has unprotected far-head budget at most one, and every cross-witness pair between two sources of this set is flexible (bad or double-convenient good). No budget restriction is made on u, v or inactive vertices. Assume uv itself is a flexible original missing pair.

For every linear extension σ of original protection, complete every flexible pair forward in σ and fix uniquely convenient good directions. Maximize the unweighted forward score over this *complete family of all protection extensions* and all orders. Every median order of a maximizing member extends original protection and puts every flexible added direction forward. Indeed, any backward flexible pair could be made forward by re-referencing to that median order, while leaving original and fixed directions unchanged.

Theorem S42.1 (The head of the two-exception flexible pair is a feed). *Suppose D is all-deficient. In any score-maximizing reference completion T in which uv is directed $u \rightarrow v$,*

a global median order maximizing $\text{pos}(v)$ among median orders of this fixed T has feed v . Consequently v has no successor in original protection. The theorem applies in arbitrary finite size.

Proof. Let L be the indicated median order and f its feed. A feed outside A is impossible: reverse all its added out-arcs inward, preserving global optimality. Its first neighborhood is original, and any new far second target would require a bad directional failure at f or a nonconvenient chosen good direction. Both are impossible, so feed SNP would transfer to the originally deficient vertex.

If $f = u$, the flexible direction $u \rightarrow v$ would be backward in L , contradicting maximum-reference rigidity. Suppose that $f \in A \setminus \{u, v\}$. The universal simultaneous envelope and its $t(f) \leq 1$ force

$$K_f = \emptyset, \quad g_D(f) = t(f) = 1, \quad g_T(f) = 0, \quad G_L = N_T^{++}(f) = N_D^{++}(f) \cup \{y\},$$

where y is its sole unprotected original far head. Its new head uses an original first arc and an added bad second direction. Choose an opposite witness r . The original missing pair fr has $r \neq y$, so $r \in N_D^{++}(f) \subseteq G_L$.

If $r \notin \{u, v\}$, its cross-witness pair fr is flexible by hypothesis and has completed direction $r \rightarrow f$ because K_f is empty. Sedimentation makes this direction backward in another global median order of T , contradicting maximum-reference rigidity. If $r = v$, sedimentation strictly increases the position of v , contradicting the selected median order.

It remains that $r = u$. If v is not an original protected far target of f , it is an original first target, an original exact second target, or the sole unprotected original far head y . Thus $v \in N_T^+(f) \cup G_L$, and sedimentation increases its position, again contradicting the selected order. If instead $v \triangleleft f$, then v remains far from f in T and is a bad vertex of L . Sedimentation places v before the feed f and $u \in G_L$ after it. It therefore puts v before u , making the fixed tournament's flexible arc $u \rightarrow v$ backward in another median order. This too contradicts maximum-reference rigidity.

All cases except $f = v$ are impossible. Every median order extends original protection, so its last position at v forbids every original relation $v \triangleleft q$. $\square$

Corollary S42.2 (Two nonmaximal exceptions with a flexible pair). *Under the same budget and cross-witness-flexibility hypotheses, if both u and v have successors in original protection and uv is flexible, D has a Seymour vertex.*

Proof. In any maximum-reference completion the flexible pair has a head. The theorem forces that head to have no original protection successor, contradicting either of the two asserted relations. $\square$

Corollary S42.3 (Two-heavy minimal counterexample reduction). *In the strong arbitrary-arc-set-deletion-minimal all-deficient class, suppose there are exactly two nonunit actual sources u, v , all other actual sources having residual one. If uv is bad or double-convenient good, at least one of u, v is maximal in the original protection order. More precisely, the head of uv in any maximum-reference completion is maximal and can be chosen as a median feed of that completion. Inactive residuals are unrestricted.*

Proof. Unit budgets and the unit-unit cross-witness-tail badness give the two hypotheses on other actual sources. Apply the theorem. $\square$

S42.1 The good-tail obstruction forced by a protected unit shadow

Lemma S42.4 (A protected unit shadow with a separate incoming path). *Suppose f is an actual unit source in the minimal class, uv and vf are original missing pairs, v is an opposite witness to f for some bad direction, v reaches f at exact distance two, and $u \triangleleft f$. Then $v \rightarrow u$ is nonconvenient. In particular, if uv is good, its only convenient direction is $u \rightarrow v$.*

Proof. The independently assumed incoming path and the full unit packet give $v \in N_D^-(f) = \partial P(f)$. Hence some $p \in P(f)$ points originally to v . Protection $u \triangleleft f$ implies that any vertex reaching u within two steps reaches f within two steps: an original predecessor of u also points to f , and an incoming two-step path into u can replace its last arc by one into f . Thus $p \in P(u)$. The original arc $p \rightarrow v$ witnesses nonconvenience of $v \rightarrow u$. Goodness forces the other direction to be convenient. $\square$

Rejected attempted strengthening. The opposite-witness property alone gives $f \rightarrow v$ at exact distance two, and does *not* imply $v \in N_D^-(f)$. A protected unit shadow alone does not establish the reverse convenience failure without a separate original path $v \rightsquigarrow f$, which is an explicit extra hypothesis of the last lemma. No such reachability is assumed in the main theorem.

Remaining gap. This does not exclude a two-heavy minimal counterexample with a maximal heavy source. The controlling completion may give its heavy feed more than one newly reached far head, so unit-feed sedimentation equality need not hold. Uniquely convenient heavy tails are outside the flexible pair theorem. No claim that every allocation $(m_1, m_2, 1^k)$ is excluded is made.

S43 A flexible exception clique has a selected terminal heavy feed

Let D be a finite nonempty oriented graph, A its original actual bad-direction failure sources, and let $H \neq \emptyset$ be an exception set. Suppose every source in $A \setminus H$ has at most one unprotected far head, and every cross-witness pair between two such sources is flexible. Inactive budgets are unrestricted. Use the complete family of all canonical protection-reference completions, fixing uniquely convenient good directions and orienting all other missing pairs forward in the reference. Let T be a member of maximum median score. All its global medians extend original protection and put all flexible added arcs forward.

Proposition S43.1 (An inward flexible exception star). *Suppose some $h \in H$ receives a flexible added arc $r \rightarrow h$ from every $r \in H \setminus \{h\}$. If D is all-deficient, a median of the fixed tournament T maximizing the position of h has feed h . In particular h has no original protection successor.*

Proof. An inactive feed transfers its tournament SNP to D after reversing all added outgoing feed arcs inward, a contradiction. A feed $r \in H \setminus \{h\}$ would have its flexible outgoing arc $r \rightarrow h$ backward, contrary to all-median rigidity.

For a remaining feed $f \in A \setminus H$, its low budget and the simultaneous envelope force no added first neighbor, original gap and far budget one, completed gap zero, and $G_L = N_D^{++}(f) \cup \{y\}$ for its sole unprotected far head y . Choose an opposite witness r for a bad second direction introducing y . The witnesses are missing, $r \neq y$, so $r \in G_L$. If $r \notin H$, the

assumed flexible cross-witness pair fr is incoming at f and becomes backward after global sedimentation, contradicting rigidity.

If $r \in H$, first suppose h is not an originally protected far target of f . It is then an original first target, an original second target, or y , and hence belongs to $N_T^+(f) \cup G_L$. Sedimentation strictly increases its position, a contradiction. Otherwise $h \triangleleft f$, so h is a bad median vertex and $r \neq h$ is good. Sedimentation puts h before r , making the flexible direction $r \rightarrow h$ backward in another median of T . This contradicts rigidity. Only feed h remains, and the last position forbids an original protection successor at h . $\square$

Corollary S43.2 (Arbitrary-size flexible missing cliques). *If every pair of distinct exception vertices in H is an original flexible missing pair, their directions in every maximum-reference completion form a transitive tournament. Its terminal vertex can be chosen as a median feed and has no original protection successor. Consequently, if each exception vertex has an original protection successor, D has a Seymour vertex.*

Proof. All flexible pairs follow one reference order, so their restriction to H is transitive. Its terminal vertex receives a flexible added arc from every other exception. Apply the proposition. $\square$

Scope. This extends the two-exception head theorem without a bound on $|H|$, but does not force the terminal vertex's completed gap to be zero or bound its multiple new far heads. The assertion is about a selected median of the fixed maximizing completion, not every median. The clique hypothesis concerns original missing pairs with flexible directions, not merely edges of the cross-witness graph. It must not be inferred solely from the definition of heavy failure sources.

S44 Original partner paths at general strong-feed equality

Let Q be a terminal weak protection closure of a finite nonempty oriented graph D , using current FORCED rounds and safe single WEAK steps. Let T be a maximum-score reference completion of Q , and let L be a global median order of T with feed f . Suppose $g_D(f) > 0$ and the strong feed inequality is an equality:

$$|G_L| = |N_T^+(f)|.$$

All terminal remaining directions are free and forward in every median. The original first set $O = N_D^+(f)$ and exact second set at every closure stage are unchanged. Write

$$R = \mathcal{M}_Q(f), \quad R_2 = R \cap N_D^{++}(f), \quad R_\infty = R \cap U_T(f).$$

Equality sedimentation and free-arc rigidity exclude $R \cap G_L$. Every newly reached original-far head belongs to G_L , since its second arc is free and forward. Consequently

$$R = R_2 \dot{\cup} R_\infty, \quad R_2 \subseteq N_D^{++}(f) \setminus G_L, \quad R_\infty \subseteq \mathcal{M}_D(f) \cap U_D(f). \quad (\text{WP1})$$

Theorem S44.1 (An original seed-path or unpaid-partner cycle). *Let X consist of the vertices reachable from R_2 in the original induced graph $D[R]$, including the seed vertices themselves, and put $U = R_\infty \setminus X$. Then either $U = \emptyset$, or the original graph $D[U]$ contains a directed cycle of length at least three. Moreover, weak closure creates no new reachability from R_2 into R_∞ within the fixed partner set R :*

$$\text{Reach}_{Q[R]}(R_2) = \text{Reach}_{D[R]}(R_2). \quad (\text{WP2})$$

This statement needs no minimality or residual hypothesis.

Proof. Hold the final set R and its partition fixed throughout the earlier closure stages. Consider a FORCED step adding an internal arc $a \rightarrow b$ with $b \in R_\infty$, and let $a \triangleleft_H w$, $w \rightarrow b$ be its preceding-graph witness. Since b is far from f in the final tournament, $w \rightarrow b$ forces $w \rightarrow f$ there. If wf were adjacent in Q , its direction would already be $w \rightarrow f$. The retained relation $a \triangleleft w$ would then force the still-missing $a \rightarrow f$ at termination. Thus $w \in R$.

Suppose a was already reachable from the seeds in $H[R]$. If $a \notin R_2$, a seed path has a final old arc $p \rightarrow a$ in R ; incoming protection nesting gives $p \rightarrow w$ in H , so b was already reached by the old path ending $p \rightarrow w \rightarrow b$. If $a \in R_2$, choose an original first intermediate $o \in O$ with $o \rightarrow a$. Nesting gives $o \rightarrow w$ in H . As wf stays missing, w is an exact second neighbor of f in this stage and hence an original exact second neighbor. Thus $w \in R_2$ and its old arc $w \rightarrow b$ already reaches b from a seed.

For a WEAK step adding internal $a \rightarrow b$ with $b \in R_\infty$, the old containment is $N_H^-(a) \subseteq N_H^-(b)$. If a seed path to a has positive length and last predecessor p , the old arc $p \rightarrow b$ already reaches b . If $a \in R_2$, its original first predecessor $o \rightarrow a$ would force $o \rightarrow b$ before the step, contradicting that b is far from f in the retained final tournament. So this case cannot occur. An added arc into R_2 cannot add seed reachability because its head is already a seed. Steps outside R do not alter the induced graph. Induction proves (WP2); simultaneous FORCED additions use their common preceding graph and obey the same argument.

For $r \in U$, the feed is maximal in terminal protection and rf remains missing. Absence of the eligible WEAK direction $r \rightarrow f$ provides $p \in N_Q^-(r) \setminus N_Q^-(f)$. Since r is far in T , $f \rightarrow p$ in T would produce a two-step path to r . Thus $p \rightarrow f$ in T , but not in Q , so $p \in R$. It cannot be reached from R_2 in $Q[R]$, or its arc $p \rightarrow r$ would reach r too. By (WP2), $p \in U$. Therefore every vertex of $Q[U]$ has an incoming arc inside U , and if U is nonempty the finite oriented graph $Q[U]$ has a cycle.

To lift it to $D[U]$, consider an internal FORCED arc $a \rightarrow b$ of U . The argument above supplies a witness $w \in R$. If $w \in X$, the old arc $w \rightarrow b$ reaches b from the seeds, contrary to $b \in U$. Hence $w \in U$: the new arc shortcuts an old internal path $a \rightarrow w \rightarrow b$ and cannot make an acyclic induced graph cyclic. For a single WEAK internal addition, a new cycle would have an old path $b \rightsquigarrow p \rightarrow a$. Old predecessor containment gives $p \rightarrow b$ as well, so the old graph already had a cycle. Steps outside U do not alter its induced arcs. An acyclic $D[U]$ would therefore remain acyclic, contradicting the terminal cycle. Loops and directed two-cycles are absent, so its length is at least three. $\square$

S44.1 Residual two and three localize the nongood obstruction

Now assume the original D is strongly connected, all-deficient and minimal under arbitrary nonempty arc-set deletion. All packet variables in what follows are computed in D . Write $\eta = \eta_D(f)$, $g = g_D(f)$, $t = t_D(f)$ and $p = |P_D(f)|$. For $p > 0$ use the original partner packet with exceptional missing set B , count k , deleted-outgoing count s , and remainder L_D :

$$|L_D| \leq \eta - 2k - g\mathbf{1}_{s>0}, \quad \mathcal{M}_D(f) \cap U_D(f) \subseteq B \cup L_D,$$

and the original shared far-head budget for every nonexceptional missing partner. An exceptional far partner may belong to the packet remainder through $\partial^2 P_D(f)$. No general exclusion of that possibility is assumed: a nonexceptional missing predecessor pointing to an exceptional far partner merely places that head in L_D and charges one unit of the actual remainder budget.

Corollary S44.2 (Low residual forces an original nongood-second seed). *Under strong feed equality, if $\eta \in \{2, 3\}$, then*

$$R_2 \neq \emptyset, \quad R_\infty \subseteq \text{Reach}_{D[R]}(R_2).$$

More specifically, at residual two $R_\infty = \emptyset$. At residual three $|R_\infty| \leq 1$; if it consists of a vertex r , some $q \in R_2$ has the original arc $q \rightarrow r$. At residual three the sole far partner is not asserted nonexceptional.

Proof. If $p = 0$, the residual identity $\eta = 2\mu_D(f) + g - 1$, with $g \in \{1, 2\}$ and $\eta \in \{2, 3\}$, gives $\mu_D(f) = 1$. Hence $|R| \leq 1$. If $p > 0$, nonnegativity of the packet remainder gives $k \leq 1$. When $k = 0$, the original far-head packet gives $t \leq \eta \leq 3$, and the strong-equality missing-degree bound gives $|R| \leq t - g \leq 2$. When $k = 1$, the original missing-far budget gives $|R_\infty| \leq \eta - k \leq 2$. Thus in every branch the unreachable set U of the theorem has at most two vertices; it cannot contain the forced original cycle. Therefore U is empty. Since R is nonempty, R_2 must be nonempty as well.

At residual two, the $p = 0$ or $k = 0$ bounds give $|R| \leq 1$, so the nonempty seed set consumes R and R_∞ is empty. For $k = 1$, $L_D = \emptyset$. Every originally missing far vertex is consequently the sole exceptional partner b , if one exists. If $b \in R_\infty$, its original seed path has a final arc $q \rightarrow b$ from another partner, necessarily nonexceptional. The original shared head budget would place b in $L_D = \emptyset$, a contradiction. So R_∞ is empty in this branch too.

At residual three, the $p = 0$ branch gives no far partner and the $k = 0$ bound $|R| \leq 2$ gives at most one, because there is a seed. If $k = 1$ and there were two far partners, at least one is a nonexceptional missing far head and itself belongs to L_D , whose size is at most $3 - 2 = 1$. The other must therefore be the unique exceptional partner b . Its original seed path has a last predecessor distinct from b , hence nonexceptional. The shared head budget places b in L_D too, giving two distinct heads in a one-element budget, a contradiction. Thus there is at most one far partner in this branch as well; that partner may be exceptional. With only one far partner r , a simple original path from the seeds to r has its last predecessor in R_2 , giving the claimed original arc $q \rightarrow r$. $\square$

Corollary S44.3 (The nonexceptional residual-three far-partner shape). *If $\eta = 3$, $p > 0$, $k = 0$ and $R_\infty \neq \emptyset$, then*

$$g = 1, \quad t = 3, \quad s = 0, \quad |R| = 2, \quad |R_2| = |R_\infty| = 1, \quad |A_f| = 2, \quad g_T(f) = -1.$$

Writing $R_2 = \{q\}$ and $R_\infty = \{r\}$, the original arc $q \rightarrow r$ exists. The full original packet also satisfies $N_D^-(f) = \partial P_D(f)$ and $L_D = \mathcal{F}_D(f)$ with $|L_D| = 3$.

Proof. There is a seed and a far partner, so $|R| \geq 2$. The bounds $|R| \leq t - g \leq 3 - g$ force $g = 1$, $t = 3$, $|R| = 2$, and also $s = 0$ in the stronger packet bound $t \leq 3 - g\mathbf{1}_{s>0}$. The far partner is not a new head, so $|A_f| \leq t - 1 = 2$. The old nongood seed consumes one nongood old second head; strong equality gives $|A_f| - g \geq 1$, so $|A_f| = 2$ and completed gap is $g - |A_f| = -1$. The preceding corollary gives $q \rightarrow r$. The partition has exceptional/outgoing defect $k + s = 0$, so the incoming packet is full. The inclusion $\mathcal{F}_D(f) \subseteq L_D$ and the three-element far budget saturate $|L_D| \leq 3$. $\square$

Corollary S44.4 (The exceptional-count-one far-partner shape). *If $\eta = 3$, $p > 0$, $k = 1$ and $R_\infty \neq \emptyset$, then $s = 0$, $|L_D| = 1$ and the sole terminal far partner r belongs to L_D . It may be exceptional or nonexceptional, and some original nongood-second terminal partner q has $q \rightarrow r$ in D .*

Proof. The preceding classification gives the original arc $q \rightarrow r$. If r is nonexceptional, its original missing-far status puts it in L_D directly. If r is exceptional, it is the unique exceptional partner and $q \neq r$ is nonexceptional; the shared head budget again puts r in L_D . Hence $|L_D| \geq 1$, while the packet estimate gives $|L_D| \leq 1 - g\mathbf{1}_{s>0}$. It follows that $s = 0$ and $L_D = \{r\}$. $\square$

Scope and remaining obstruction. The seed vertices are original nongood *second* targets, not original far targets and not necessarily actual BAD sources. These results do not eliminate the residual-two all-second trap or the residual-three second-to-far path. In particular they do not assert an original cycle on all remaining partners at negative completed gap. They isolate the precise original path that replaces the zero-gap unpaid-partner cycle, without applying minimality to an altered graph.

A corrected packet-layer assumption. An earlier draft incorrectly excluded exceptional far heads from $\partial^2 P_D(f)$ and hence from E . That exclusion is not used in this corrected proof. The strong six-vertex geometry control [34, 4, 24, 16, 1, 2] at $f = 0$ has

$$P = \{1, 5\}, \quad C = \{2\}, \quad B = \{3\}, \quad \partial^2 P = \{3, 4\}, \quad E = \{2, 3, 4\}, \quad L = \{3, 4\}.$$

The exceptional missing far head 3 therefore does belong to the external second layer and to L . Other vertices have nonpositive gaps, so this control does not meet MIN and its numerical packet budget is not applied. It verifies the geometric scope limit only; it is not an SNC counterexample. The corrected residual-three conclusion allows its sole terminal far partner to be exceptional and charges it to the one-element remainder rather than forbidding it.

S45 A near-whole feed exposes its unprotected heads in an original third layer

Let D be strongly connected, all-deficient and minimal under arbitrary nonempty arc-set deletion. Let Q be a terminal weak protection closure and T_* a maximum reference completion, as previously defined. Fix a global median feed f and suppose $\mu_D(f) = 1$; write r for its sole original missing partner. All sets below, except those explicitly indexed by Q or T_* , are original sets in D .

Lemma S45.1 (A sole remaining partner is an old second target). *One has $\mathcal{M}_Q(f) = \{r\}$ and $r \in N_D^{++}(f)$.*

Proof. The original feed gap is positive. The closure theorem excludes a terminal whole feed, while $\mathcal{M}_Q(f) \subseteq \mathcal{M}_D(f)$. Thus r is the sole remaining missing partner. The feed is maximal in terminal protection, so absence of the weak $r \rightarrow f$ step supplies $p \in N_Q^-(r) \setminus N_Q^-(f)$. The vertices p, f, r are distinct. Every pair pf is already adjacent in Q , because r is the only missing partner; the direction cannot be $p \rightarrow f$, so $f \rightarrow p$ in Q . Its first neighborhood is original, so $f \rightarrow p$ is original. The retained path $f \rightarrow p \rightarrow r$ proves $r \in N_Q^{++}(f)$; exact original second equality at feeds gives $r \in N_D^{++}(f)$. $\square$

Proposition S45.2 (Global original boundary charges every unprotected head). *Assume additionally $P_D(f) = \emptyset$. Put $d = d_D^+(f)$, $g = g_D(f)$, $t = t_D(f)$ and $O = N_D^+(f)$. Then*

$$\{f\} \dot{\cup} \mathcal{F}_D(f) \subseteq \partial_D^2 O, \quad d \geq g + t + 2, \quad n \geq g + 3t + 5.$$

In particular at $(\eta_D(f), g) = (2, 1)$ one has $t \geq 2$ and $d \geq t + 3$; at $(\eta_D(f), g) = (3, 2)$ one has $t \geq 3$ and $d \geq t + 4$. These are original-global-minimality constraints, not conditions imposed on an altered closure graph.

Proof. The sole missing partner is an original exact second target by the lemma. Every unprotected far head y is therefore an original incoming neighbor of f . Failure of protection supplies a predecessor $p \rightarrow y$ which does not point to f . It is not f , and cannot be an original first neighbor of f , since that would reach y in two steps. All other vertices except r are adjacent to f , so this predecessor must be r . Thus $r \rightarrow y$ originally for every $y \in \mathcal{F}_D(f)$.

Since $\partial_D O = N_D^{++}(f)$ contains r , and no far head is in $O \cup N_D^{++}(f)$, these heads lie in $\partial_D^2 O$. Also choose any $a \in O$, which is nonempty. The hypothesis $P_D(f)$ empty says a reaches f within two steps. It cannot point directly to f , because $f \rightarrow a$ is original. Hence some original $a \rightarrow w \rightarrow f$ exists. The vertex w is an original exact second target of f and points to f . Thus $f \in \partial_D^2 O$ too.

The original external-boundary minimality lemma now gives

$$t + 1 \leq |\partial_D^2 O| < |\partial_D O| = d - g,$$

so integrality yields $d \geq g + t + 2$. The total original far degree is $n - 1 - d - (d - g) = n - 1 - 2d + g$, which is at least t . Therefore $n \geq 1 + 2d - g + t \geq g + 3t + 5$. Finally $P_D(f)$ empty and $\mu_D(f) = 1$ give $\eta_D(f) = 2 + g - 1 = g + 1$, while strict original closure surplus gives $t \geq g + 1$. The two numerical cases follow. $\square$

Why the previous local control is insufficient. The ten-vertex stable control has original first set $O = \{1, 2, 3, 4\}$, original boundary $\{5, 6, 7\}$, and external second boundary $\{0, 8, 9\}$. Their sizes are equal. Hence it violates the strict original-global boundary inequality even though all outgoing-star deletions at its feed are locally critical. This pinpoints a concrete global constraint missing from that local model. The present degree bound still leaves arbitrarily large feasible parameter ranges; it is not an exclusion of the stable low-residual branch or a full SNC proof.

S46 A convenient step from an already nonmaximal protected tail

All predicates in this section refer to the *current* graph H . Recall that $a \triangleleft_H w$ means $a \rightarrow w$ and $N_H^-(a) \subseteq N_H^-(w) \setminus \{a\}$.

Lemma S46.1 (A larger safe single-step operator). *Suppose ab is missing, b has no current protection successor, a has some current protection successor w , and the direction $a \rightarrow b$ is convenient in H . Adding only $a \rightarrow b$ preserves every current protection. Its tail still has a protection successor, and the new arc is convenient relative to the graph before its addition.*

Proof. Only the incoming neighborhood of b changes. A protected relation whose protected head is b cannot exist, by its maximality. A relation whose target is b becomes easier to satisfy because the target's incoming set increases. Every other incoming-neighborhood comparison is unchanged. Thus every old protection survives, including $a \triangleleft_H w$. Convenience before addition is an explicit assumption, not inferred from protection or from the original graph. The graph remains oriented because the pair was missing. $\square$

One may add these steps to the previously FORCED/WEAK closure, performing them singly and recomputing protection after each step. At most the original missing-pair count many nonterminal additions are possible. Every added arc has a tail with a protection successor which survives to termination. If all FORCED operations are also exhausted, terminal protection has both incoming and outgoing nesting.

Corollary S46.2 (The exact original feed invariants survive). *Complete the terminal partial graph Q by all remaining directions forward in a protection extension and maximize global median score over all such extensions. At every feed f ,*

$$N_T^+(f) = N_Q^+(f) = N_D^+(f), \quad N_Q^{++}(f) = N_D^{++}(f).$$

Every original protected far target stays protected and far. Every new original-far second head in T is good in every median order ending at f . In particular the previously strict-original-surplus and strong-equality missing-degree proofs continue to apply.

Proof. Original protections survive every safe step and the final completion. Every added arc's tail has a surviving protection successor and cannot be last in a median order. The feed consequently has no added first arc at any stage. A new second direction at that feed uses an original first step and an arc convenient in its preceding graph, so it cannot create an exact second head not previously present. Induction gives the exact original second set in Q . At the final maximizing completion all remaining free arcs are forward in every median, by the all-reference closure argument. A newly reached original-far head therefore has a forward free second arc from an original first neighbor, making it good. The earlier surplus and equality-degree arguments use exactly these invariants and free-arc rigidity. $\square$

Terminal obstruction. At a terminal feed f , if r is still missing from f and has a current protection successor, the reverse direction $r \rightarrow f$ must be *currently* nonconvenient; otherwise the new safe operator could add it. Original convenience is insufficient for this claim, because closure may have added new predecessors of r .

Important scope limit. The FORCED/WEAK original-partner cycle and seed-path lifting proofs do not automatically apply to this enlarged operator. A convenient direction need not shortcut an old two-step path *inside* the fixed partner set, and no such shortcut is asserted here. In particular the residual-four zero-gap equality shape which used an original partner triangle cannot be transferred without a separate lifting proof. The operator preserves the stated feed invariants; it has not been shown to settle any remaining multiple-heavy counterexample class.

Part IV

Weighted transport and amplification

S47 A dense amplification that preserves the local obstruction

For any oriented graph D with n vertices, form $\mathcal{C}(D)$ from three disjoint copies D_0, D_1, D_2 , adding every arc from D_i to D_{i+1} cyclically. There are no digons. The complete cyclic connections make $\mathcal{C}(D)$ strongly connected even if D was not. This is the three-copy specialization of the cycle blow-up mechanism used by Espuny Diaz, Girao, Granet and Kronenberg [2] in their counterexample reductions; the exact invariants below are recorded explicitly.

Lemma S47.1 (Exact invariants of cyclic amplification). *For a copy x_i of $x \in D$, all first, exact second, and far relations satisfy*

$$\begin{aligned} N_{\mathcal{C}(D)}^+(x_i) &= N_D^+(x)_i \dot{\cup} V(D_{i+1}), \\ N_{\mathcal{C}(D)}^{++}(x_i) &= N_D^{++}(x)_i \dot{\cup} V(D_{i+2}), \\ P_{\mathcal{C}(D)}(x_i) &= P_D(x)_i, \quad \mathcal{F}_{\mathcal{C}(D)}(x_i) = \mathcal{F}_D(x)_i. \end{aligned}$$

Thus g, μ, η , and the actual bad failure-source status are preserved pointwise. The global quantities obey

$$n' = 3n, \quad \delta' = n + \delta, \quad M' = 3M, \quad R' = 3R, \quad n' - 2\delta' = n - 2\delta.$$

In particular a counterexample is amplified to a strongly connected counterexample without decreasing any of its original vertex gaps.

Proof. An internal first step introduces only internal original second targets or vertices of the next copy, already first targets. A first step into the next copy reaches the entire following copy. It cannot return to the original copy within two steps. These facts prove the first two identities. The reverse exact-second degree likewise increases by n . All vertices outside the starting copy are reached in one or two steps, so far and far-in relations stay internal. Every internal incoming neighborhood gets the same previous-copy set, preserving protection.

All missing pairs stay inside copies. An outside predecessor of one endpoint of a missing pair already reaches its other endpoint in one or two steps; it is not a new directional failure witness. Therefore the old failure sets and actual source status are copied exactly. The stated degree, residual, and missing-pair formulas follow. $\square$

Corollary S47.2 (An asymptotically dense global target). *If SNC were proved for every oriented graph with $\delta^+(D) > (1/2 - \varepsilon)|V(D)|$, for one fixed $\varepsilon > 0$, then SNC would hold for all finite oriented graphs. More generally, it suffices to prove that for every fixed defect $q = n - 2\delta$, counterexamples of that defect have bounded order.*

Proof. Iterating the lemma gives $n_k = 3^k n$ and $\delta_k = (n_k - q)/2$, with the same fixed q . The ratio $\delta_k/n_k = 1/2 - q/(2n_k)$ tends to one half from below, and any initial counterexample persists at unbounded order. It would eventually violate either proposed theorem, yielding the contradiction. $\square$

This is a genuine global reduction but is not an easier theorem already proved here. Crucially, it does not preserve strong arc-set minimality: if D is all-deficient, deleting *all*

cross-copy arcs leaves three all-deficient copies. Minimizing arcs can therefore discard the high minimum degree. A bound proved only for arc-minimal graphs cannot simply be applied to the dense amplifications. Likewise the pointwise residual is unchanged; the amplification is not a reduction to pure-unit residuals.

S48 The exact weighted transfer problem

Let N_D have entry $+1$ for an exact second target, -1 for a first target, and zero otherwise. Seacrest's Farkas alternative [9] relates an expansion weighting to a weighted counterexample in the reverse graph. It is equivalent to SNC in general, not a proof of either alternative. For tournaments, Fisher's losing-density result supplies a nonzero $\lambda \geq 0$ with $N_T^T \lambda \geq 0$.

In any protection-preserving completion define A_x to be the newly reached originally far second heads. Original second paths persist unless their endpoints have become first neighbors. Hence the exact gap difference is

$$g_T(x) - g_D(x) = 2|K_x| - c_x, \quad c_x = |K_x \cap \mathcal{F}_D(x)| + |A_x| \leq t(x).$$

The two cost sets are disjoint. If a tournament losing density could be chosen with

$$\sum_x \lambda_x (2|K_x| - c_x) \geq 0,$$

then all-deficiency of D would contradict

$$0 \geq \sum_x \lambda_x g_T(x) = \sum_x \lambda_x g_D(x) + \sum_x \lambda_x (2|K_x| - c_x) > 0.$$

This is a global mass-transport target, allowing different vertices to compensate one another. Existence of a suitable completion and losing density has not been proved. The looser replacement $c_x \leq t(x)$ can discard exactly the slack needed even in an ordinary directed four-cycle.

For an arbitrary incomplete oriented graph, an equilibrium of its skew adjacency matrix alone does not suffice: the positive five-vertex control weights in the companion file balance incoming and outgoing weights but fail second-neighborhood expansion. This is not a tournament control: Fisher's tournament losing densities do satisfy reverse expansion, and the follow-up density-collapse lemma proves equality of the two normalized feasible sets. The completion, rather than an extra density degree of freedom, is the remaining choice in that formulation. Nor does the existential arc-weighted tournament theorem authorize using an arbitrary arc-weighted median feed: the canonical six-vertex control has a verified such feed with arc-first weight 74 and arc-second weight 38. These are failures of proposed proof shortcuts, not conjecture counterexamples.

S49 The missing global step

The two promising concrete tasks are now: (i) a quantitative extension of maximal reference completion that handles a nonunit actual feed and its multiple inverse witnesses; or (ii) a completion/losing-density selection proving the exact weighted mass balance above. Simultaneous packet control is a third sufficient route, but it must establish independence and acyclic reference priorities, not assume them separately for every target.

No terminating reduction from arbitrary mixed residuals to the excluded pure-unit class has been obtained. The preceding amplified graphs show why density growth, vertex-weight reformulation, or arc minimization alone does not provide that reduction.

S50 The tournament expansion polytope has no extra density freedom

For a tournament T , write A_T for its adjacency matrix, $B_T = A_T - A_T^T$, and F_T for the matrix of original far relations: $(F_T)_{xy} = 1$ precisely when $x \neq y$ and x does not reach y within two steps. Let N_T have entry $+1$ on exact second targets, -1 on first targets, and zero elsewhere. Define

$$\mathcal{P}(T) = \{\lambda \geq 0 : \mathbf{1}^T \lambda = 1, N_T^T \lambda \geq 0\}, \quad \mathcal{L}(T) = \{\lambda \geq 0 : \mathbf{1}^T \lambda = 1, B_T \lambda \geq 0\}.$$

The second set is exactly the set of Fisher's *losing densities*. The first set should initially be called the reverse expansion-density polytope, rather than treating its defining inequality as the definition of a skew losing density.

Lemma S50.1 (Density collapse in tournaments). *For every tournament T , $\mathcal{P}(T) = \mathcal{L}(T)$. In particular the expansion-density formulation does not provide an additional choice of density beyond the usual tournament losing density.*

Proof. Every nonfirst vertex of a tournament is an incoming vertex; it is either an exact second target or a far target. Consequently

$$N_T = A_T^T - A_T - F_T, \quad N_T^T \lambda = B_T \lambda - F_T^T \lambda.$$

Since $F_T^T \lambda \geq 0$, membership in $\mathcal{P}(T)$ implies $B_T \lambda \geq 0$ and hence membership in $\mathcal{L}(T)$. Conversely Fisher's tournament Lemma 2 states that, for a losing density, the mass of exact second predecessors of each vertex is at least the mass of its first predecessors. This is $N_T^T \lambda \geq 0$. $\square$

The standard uniqueness theorem for a tournament game's equilibrium then implies that these two normalized sets are a singleton. Uniqueness is not needed for the obstruction proofs below. The equality itself, however, prevents an apparent "optimize over a larger expansion polytope" escape from a bad Fisher losing density. In an arbitrary oriented graph missing pairs contribute another matrix term, so the first implication is no longer available by the same argument.

Lemma S50.2 (Protected dominators carry no losing mass). *If $y \triangleleft x$ is protected in a tournament and $\lambda \in \mathcal{L}(T)$, then $\lambda_y = 0$. Consequently, in every protection-preserving completion of D , its losing density is supported on maximal elements of the original protection poset.*

Proof. Protection implies $y \rightarrow x$, and every outneighbor of x is an outneighbor of y . Thus the y row of B_T dominates the x row entrywise, with a difference of one in both the x and y columns. If $\lambda_y > 0$, skew-symmetry and $B_T \lambda \geq 0$ imply $(B_T \lambda)_y = 0$: indeed all terms of $\lambda^T B_T \lambda = 0$ are nonnegative. But then

$$0 = (B_T \lambda)_y \geq (B_T \lambda)_x + \lambda_x + \lambda_y > 0,$$

a contradiction. Original protected relations survive the admissible completion, so any original vertex below another in that poset has zero mass. This is a support reduction, not a conclusion that all remaining sources have unit budget. $\square$

Primary source. D. C. Fisher [4], *Squaring a tournament: A proof of Dean's conjecture*, Journal of Graph Theory 23 (1996), 43–48, Section 3, Lemma 2. An author-paper scan is available at <https://www.math.ru.nl/OpenGraphProblems/TimV/%5BDavid%20Fisher%20Proof%20for%20tournaments.pdf>. Fisher's Section 2 also records the earlier uniqueness theorem for tournament winning densities, which applies to losing densities by reversing the tournament.

S51 Exact duality and a weaker sufficient transfer threshold

For a protection-preserving completion T of D , put $\Delta_x = g_T(x) - g_D(x) = 2|K_x| - c_x$, where c_x is the exact new-head cost from the preceding draft. For fixed T let

$$B(D, T) = \max_{\lambda \in \mathcal{P}(T)} \lambda^T \Delta.$$

The polytope is nonempty by Fisher's theorem and is compact. Ordinary finite-dimensional linear-programming duality gives the exact formula

$$B(D, T) = \min_{w \geq 0} \max_x (\Delta_x + (N_T w)_x). \quad (\text{S51.1})$$

Indeed a dual upper bound a has $w \geq 0$ and $\Delta + N_T w \leq a\mathbf{1}$. Multiplication by any feasible λ yields $\lambda^T \Delta \leq a$ because $\lambda^T N_T w \geq 0$. The usual LP dual gives equality and attainment. Thus an exact obstruction can be certified using a single nonnegative *head-weight vector*, without numerical trust in an LP solver.

There is also a useful correction to the earlier target threshold. If D is all-deficient, every integer gap is at least one. A normalized $\lambda \in \mathcal{P}(T)$ would already yield a contradiction from

$$\lambda^T \Delta > -1, \quad (\text{S51.2})$$

since $\lambda^T g_T = \lambda^T g_D + \lambda^T \Delta > 0$, whereas $\lambda^T g_T = -\mathbf{1}^T N_T^T \lambda \leq 0$. The old sufficient target $\lambda^T \Delta \geq 0$ is strictly stronger than necessary. In the minimal class $g_D \in \{1, 2\}$, writing $G_2 = \{x : g_D(x) = 2\}$ gives the still more accurate threshold

$$\lambda^T (\Delta + \mathbf{1}_{G_2}) > -1.$$

These are sufficient contradiction criteria, not existence theorems. No unrestricted completion guarantee is supplied by LP duality alone.

S52 An all-order obstruction to automatic reference-budget balance

Consider the six-vertex graph with out masks

$$(0, 49, 51, 6, 9, 8).$$

Equivalently its outgoing sets are

$$\emptyset, \quad \{0, 4, 5\}, \quad \{0, 1, 4, 5\}, \quad \{1, 2\}, \quad \{0, 3\}, \quad \{3\}.$$

For any integer $k \geq 1$, replace vertex a by an independent clone class C_a of size k , and replace each original arc by the complete bipartite orientation between its two classes. Call the resulting oriented graph D_k . Its pointwise gaps on $C_0, \dots, C_5$ are

$$0, \quad 2k, \quad 3k, \quad -k, \quad 0, \quad -k.$$

It has Seymour vertices and is *not* all-deficient.

Every within-class missing pair is convenient in both directions. Between distinct classes the only missing pairs are C_0C_3 , C_0C_5 , and C_4C_5 . The first is uniquely conveniently oriented $C_0 \rightarrow C_3$, the second $C_5 \rightarrow C_0$, and the third is convenient in both directions. Hence there are no bad missing pairs at all. The original protection relations include $C_4 \triangleleft C_0$ and $C_2 \triangleleft C_1$, with the notation interpreted pointwise.

A canonical completion T obtained from any linear extension of original protection therefore has transitive orders on each of C_0, C_1, C_2, C_3 and on the entire union $C_4 \cup C_5$; the two uniquely convenient interclass directions are fixed. Conversely all of these internal orders can be realized by a protection reference. After clone relabeling, the only distinct choices are the $\binom{2k}{k}$ binary interleavings of C_4 and C_5 .

Theorem S52.1 (Exact negative optimum for every reference completion). *For every $k \geq 1$ and every protection-reference canonical completion T of D_k ,*

$$B(D_k, T) = -\frac{k}{3}.$$

This statement quantifies over every feasible expansion density, not only one arbitrarily selected density.

Proof. Give every vertex of $C_0, \dots, C_5$ the respective head weight

$$w = (4, 13/3, 0, 11/3, 0, 0).$$

For $x \in C_a$, $a \leq 3$, write $s_a(x)$ for the number of later vertices in its class. For $x \in C_4 \cup C_5$ write $s_W(x)$ for the number of later vertices in the transitive order on that union. The ranges are $0 \leq s_a \leq k-1$ and $0 \leq s_W \leq 2k-1$. Direct first- and exact-second-neighborhood calculations give

class of x	Δ_x	$(N_T w)_x$
C_0	$s_0 - k$	$2k/3 - 4s_0$
C_1	s_1	$-k/3 - 13s_1/3$
C_2	s_2	$-14k/3$
C_3	s_3	$-k/3 - 11s_3/3$
C_4	s_W	$-10k/3$
C_5	$k + s_W$	$-10k/3$

For example the first targets of C_5 are all of C_3, C_0 plus the later vertices of $C_4 \cup C_5$; its exact second targets are exactly $C_1 \cup C_2$. Added arcs inside the union are all forward, so they do not create exact second targets earlier in its order. The other five rows follow in the same way. Each row of the table satisfies

$$\Delta_x + (N_T w)_x \leq -k/3.$$

For C_2, C_4, C_5 the stronger upper bounds are respectively $-11k/3 - 1$, $-4k/3 - 1$, and $-k/3 - 1$. This common dual vector proves $B(D_k, T) \leq -k/3$.

For the matching lower bound, let v_0, v_1, v_3 be the last vertices in C_0, C_1, C_3 respectively and put weight $1/3$ on each, zero elsewhere. Their exact signed-neighborhood rows sum to zero:

$$N_T(v_0, \cdot) + N_T(v_1, \cdot) + N_T(v_3, \cdot) = 0.$$

Indeed their interclass first sets are respectively C_3 , $C_0 \cup C_4 \cup C_5$, and $C_1 \cup C_2$, while their second sets are $C_1 \cup C_2$, C_3 , and $C_0 \cup C_4 \cup C_5$. They have no remaining within-class first vertices. Hence this density satisfies $N_T^T \lambda = 0$. Its three correction values are $-k, 0, 0$, so its budget is exactly $-k/3$. $\square$

In particular $k \geq 4$ refutes even an unconditional existence claim with the weaker threshold $\lambda^T \Delta > -1$. These graphs are already covered by the good-missing-edge SNC theorem. Thus completion budget balance is a *sufficient but not necessary* certificate for SNC on arbitrary graphs.

S53 The negative budget survives strong dense amplification

Apply the cyclic three-copy amplification from the companion draft $m \geq 1$ times to D_k , obtaining $D_{k,m}$. Every missing pair still lies inside a base copy and is good, so every canonical reference completion independently chooses a completion of D_k in each base copy. All original gaps and all completion corrections stay the same pointwise. Put the preceding w in every base copy. For every vertex the added full next-copy first contribution to $N_T w$ is canceled by the equally weighted full following-copy second contribution. The same cancellation applies at every amplification level. Thus $\Delta + N_T w \leq -(k/3)\mathbf{1}$ remains valid.

Conversely put equal total density on every base copy and use the attaining three-point density inside each. Its external signed column contributions cancel and the internal contributions are zero. Therefore for every protection-reference completion T of $D_{k,m}$,

$$B(D_{k,m}, T) = -k/3.$$

These original graphs are strongly connected and have

$$n = 6k3^m, \quad \delta^+(D_{k,m}) = \frac{n - 6k}{2}, \quad \frac{\delta^+}{n} \rightarrow \frac{1}{2}.$$

No fixed dense threshold below $1/2$, strong connectivity, the absence of bad missing pairs, or optimization among maximal reference completions alone can force a nonnegative budget or the weaker > -1 budget threshold on arbitrary SNC graphs. This does not refute a theorem *under the all-deficient arc-minimal hypothesis*: the family violates all-deficiency and also the minimal-class gap bound $g_D \leq 2$ for most of its vertices when k is large.

S54 Median rigidity is not equilibrium-support rigidity

One cannot combine maximal-reference median rigidity with the assertion that every arc between positive losing-density vertices becomes backward in some median order. The six-vertex tournament with out masks

$$(42, 60, 57, 16, 33, 8)$$

has losing density $(0, 0, 0, 1, 1, 1)/3$: its skew residual, after multiplication by three, is $(1, 3, 3, 0, 0, 0)$. Its unique global median order is

$$(1, 2, 4, 0, 5, 3),$$

with thirteen forward arcs. Thus $5 \rightarrow 3$ lies between two positive density vertices and is forward in every median order. This is only a control against that generic support shortcut, not an assertion that a maximal completion arising from a minimal counterexample can realize this particular configuration.

Independent verification boundary. The companion Node program `verify_transport_obstruction_family.js` uses exact integer arithmetic with the scaled head weights $(12, 13, 0, 11, 0, 0)$. It checks all 98 canonical quotient types for $k = 1, 2, 3, 4$, all 224 independently chosen three-copy types for $k = 1, 2$, nine iterated amplifications through order 216, and all 720 orders of the final median control. These checks confirm the supplied all- k proof; they are not a search for an SNC counterexample or a proof that the counterexample-specific selection step is impossible.

Remaining global step after these repairs. The mathematically relevant remaining choice is the completion: in a tournament the normalized Fisher density itself is fixed. A successful transfer theorem may target the strict > -1 threshold (including the $g_D = 2$ mass correction), but it must use the true all-deficient arc-minimal structure. Generic strong or asymptotically dense properties do not supply that theorem. The explicit common dual also identifies what must be ruled out: a nonnegative head-weight vector whose signed neighborhood gain simultaneously offsets every admissible completion's keeper gain by a negative constant.

Part V

Auxiliary results at the dense odd boundary

All results in this part assume $n = 2\delta + 3$ with the actual minimum outdegree δ , as well as each theorem's explicit hypotheses. Its capacity surplus σ and core K are distinct from η and actual failure-source support.

S55 Whole minimum-degree vertices at order $2\delta + 3$

Throughout this section, D is strongly connected, arc-minimal and all-deficient, $n = 2\delta + 3$, and $\delta = \delta^+(D)$. Let z count whole minimum-outdegree vertices, and let h count vertices with $g(x) = 2$. We will prove

$$z \geq h + 7. \quad (\text{S55.1})$$

S55.1 Degree balance at the first odd boundary

Corollary S55.1. *At $n = 2\delta + 3$, let r_i count vertices with outdegree $\delta + i$. Then $i \in \{0, 1, 2\}$ and*

$$r_0 = M + r_2 \geq \delta + 5 + r_2.$$

Thus a strict majority of the vertices have minimum outdegree.

Proof. The gap bound gives $2d^+(x) - g(x) \leq n - 1$ and $g(x) \leq 2$, so all outdegrees are at most $\delta + 2$. Counting arcs gives $M = r_0 - r_2$. Use the uniform margin in Theorem S20.1. $\square$

For each target x , use T_x, P_x from the foundational target-packet section and write

$$e_x = d^+(x) - \delta, \quad q_x = |T_x| - \delta = e_x + \mu(x), \quad p_x = |P_x|.$$

Counting outgoing arcs from P_x gives, whenever $p_x > 0$,

$$\delta p_x + \sum_{v \in P_x} e_v \leq \binom{p_x}{2} + p_x(\delta + q_x - p_x). \quad (\text{S55.2})$$

In particular, $q_x = 0$ implies $p_x = 0$, and $q_x > 0$ implies $p_x \leq 2q_x - 1$. Define

$$\sigma_x = \begin{cases} 2q_x - 1 - p_x, & q_x > 0, \\ 0, & q_x = 0. \end{cases}$$

Let $K = \{x : \sigma_x > 0\}$, $k = |K|$, and $\Sigma = \sum_x \sigma_x$. A target in K with $\sigma_x = 1$ is called a *unit-slack target*; a target with $\sigma_x \geq 2$ is called a *higher-slack target*.

Put $U(v) = V(D) \setminus (\{v\} \cup N^+(v) \cup N^{++}(v))$. Lemma S3.2 gives

$$|U(v)| = 2 - 2e_v + g(v), \quad e_v \in \{0, 1, 2\}, \quad (\text{S55.3})$$

and $e_v = 2$ forces $g(v) = 2$ and $U(v) = \emptyset$. If r_i counts excess- i vertices, then $M = r_0 - r_2$. Counting far source-target pairs and total degrees gives

$$\sum_x p_x = 2M + n + h, \quad \sum_x q_x = n + M.$$

Since the $q_x = 0$ targets are precisely the z whole minimum-degree vertices,

$$\Sigma = z - h. \quad (\text{S55.4})$$

Lemma S55.2. *A zero-slack target has exactly one of the following forms:*

$$q_x = 0, \quad e_x = \mu(x) = p_x = 0; \quad \text{or} \quad q_x = 1, \quad e_x = 0, \quad \mu(x) = p_x = 1.$$

In the second case, the unique vertex of P_x has minimum outdegree. Every source is far from at most one zero-slack target.

Proof. Suppose $q_x > 0$ and $\sigma_x = 0$. Equality holds throughout (S55.2): all vertices of P_x have outdegree δ , $D[P_x]$ is a regular tournament, and P_x completely dominates $R_x = T_x \setminus P_x$. If $q_x \geq 2$, then $|P_x| = 2q_x - 1 \geq 3$ and P_x has an internal arc, contradicting Lemma S7.1. Hence $q_x = p_x = 1$. The alternative $e_x = 1, \mu(x) = 0$ contradicts Proposition S5.2, so $e_x = 0, \mu(x) = 1$.

If $P_x = \{v\}$, capacity equality gives $T_x = \{v\} \dot{\cup} N^+(v)$. For any other target $y \in U(v)$, we have $y \notin T_x$, so $y \rightarrow x$. Two zero-slack far targets of the same source would thus form a digon. $\square$

It follows that every nonminimum-degree vertex belongs to K . Every minimum-degree vertex has three or four far targets by (S55.3); at most one is outside K , so at least two lie in K . In particular, K is nonempty.

S55.2 Protected predecessors and higher-slack targets

Lemma S55.3. *Every unit-slack target x has $e_x \leq 1$. If $p_x > 0$, it has a vertex $y \in K$ such that*

$$y \rightarrow x, \quad q_y > q_x, \quad P_x \cup \{x\} \subseteq P_y.$$

Proof. For $q_x > 0$, the residual identity reads $\eta(x) = \sigma_x + g(x) - 2e_x$. Thus $\sigma_x = 1$ and $\eta(x) \geq 0$ imply $2e_x \leq 3$.

If $p_x > 0$, then $q_x \geq 2$, $p_x = 2q_x - 2$, and $|R_x| = \delta - q_x + 2$. Omitting even one vertex of R_x from ∂P_x would bound the total outdegree of P_x by $(\delta - \frac{1}{2})p_x < \delta p_x$. Hence $\partial P_x = R_x$. Now $|W_x| = |R_x|$ for $W_x = N^-(x)$, and $X_x = \partial^2 P_x \subseteq W_x$ has size less than $|R_x|$. Choose $y \in W_x \setminus X_x$. No arc from $T_x = P_x \cup R_x$ points to y , so

$$N^-(y) \subseteq N^-(x) \setminus \{y\}.$$

Consequently $q_y > q_x$, and x cannot reach y in at most two steps. Nor can any vertex of P_x reach y : an arc to y would give a two-step path to x , and a two-step path to y would pass through $N^-(y) \subseteq N^-(x)$. This proves the inclusion. Since $q_y > q_x \geq 2$, Lemma S55.2 also gives $y \in K$. $\square$

Lemma S55.4. *There are at least two higher-slack targets.*

Proof. Call a target active when $p_x > 0$. Starting from a unit-slack target with $p_x > 0$, iterate Lemma S55.3. The strictly increasing q values force termination at a higher-slack target. Such a starting target exists unless the required higher-slack targets already exist, because minimum-degree sources have far targets in K .

Suppose that the only higher-slack target is c . Every active unit-slack target x then satisfies $P_x \cup \{x\} \subseteq P_c$. A minimum-degree source has at least two far targets in K , so one differs from c ; it follows that the source lies in P_c . Every nonminimum-degree vertex $v \neq c$ is unit-slack and has $e_v = 1$, hence $|U(v)| = g(v) \geq 1$. It cannot be far from a zero-slack target, whose far sources have minimum degree. Its far target is therefore either c or an

active unit-slack target; in both cases $v \in P_c$. Thus $P_c = V(D) \setminus \{c\}$, which makes the indegree of c zero, contradicting strong connectivity. The case of no higher-slack target is already excluded by the strictly increasing chains. $\square$

Every unit-slack vertex has an incoming arc from K . For $p_x > 0$ this is Lemma S55.3. If $p_x = 0$, then $q_x = 1$. When $e_x = 1, \mu(x) = 0$, a far target in K exists and is adjacent to x , so points to x . When $e_x = 0, \mu(x) = 1$, at least two far targets lie in K , at most one is missing from x , and an adjacent one again points to x . Distinct unit-slack vertices give arcs with distinct heads.

S55.3 The missing-pair balance inside K

Theorem S55.5. *In the setting of this section,*

$$z \geq h + 7.$$

Moreover $k \geq 5$ and $k \leq \lfloor (n - h - 2)/2 \rfloor$.

Proof. Let ℓ be the number of unit-slack targets and $H = k - \ell$ the number of higher-slack targets. Let A_K count arcs inside K , let m count minimum-degree vertices in K , let t count gap-two vertices with excess at most one, and let M_{out} count missing pairs with both endpoints outside K . Thus $h = r_2 + t$.

By Lemma S55.2, outside K there are z whole minimum-degree vertices and $L = n - k - z$ minimum-degree vertices of missing degree one. Their missing-degree sum is $L = M(K, V \setminus K) + 2M_{\text{out}}$. As $r_0 = n - k + m$ and $M = r_0 - r_2$, the number of missing pairs inside K equals

$$M_K = M - L + M_{\text{out}} = m - r_2 + z + M_{\text{out}} = \Sigma + m + t + M_{\text{out}}.$$

Therefore

$$\binom{k}{2} = \Sigma + m + t + A_K + M_{\text{out}}. \quad (\text{S55.5})$$

The preceding incoming-arc argument gives $A_K \geq \ell$, while $\Sigma + \ell \geq 2k$. Hence $\binom{k}{2} \geq 2k$, and $k \geq 5$ since $k > 0$. Lemma S55.4 gives $H \geq 2$, so $\Sigma \geq k + H \geq 7$. Equation (S55.4) proves $z \geq h + 7$. Finally, $z \leq n - k$ and $\Sigma \geq k + 2$ give $n \geq 2k + h + 2$, proving the upper bound on k . $\square$

In particular, at least seven minimum-degree vertices are adjacent to every other vertex. This does not exclude the order $2\delta + 3$ itself; it supplies additional structure that any such minimal counterexample must possess.

S56 An even-order capacity identity

For any oriented graph H , set $b_H(v) = d_H^+(v) - 1 - d_H^{++}(v)$ and $B(H) = \sum_v b_H(v)$. These quantities can be negative in a vertex-deleted graph. At order $n' = 2\alpha + 2$ with actual minimum outdegree α , write $q'(x) = |T'(x)| - \alpha$ and define the capacity surplus $\sigma'(x) = 2q'(x) - 1 - |P'(x)|$ when $q' > 0$, and zero when $q' = 0$. Let z' count whole vertices of outdegree α . Then

$$B(H) + n' = z' - \sum_x \sigma'(x).$$

Indeed $q' = 0$ implies $P' = \emptyset$ by the same capacity count; these vertices are exactly the z' perfect vertices. Summing q' gives $n'(\alpha + 1) - \sum d^+$, while summing the far-in sets gives

$n'(n' - 1) - \sum d^+ - \sum d^{++}$. Substitution proves the identity. No counterexample hypothesis on H is used.

S56.1 Excluding a five-vertex positive-surplus core

Throughout this subsection, D is a strongly connected counterexample which is minimal under deletion of nonempty sets of arcs, and $n = 2\delta + 3$. Write

$$e(v) = d^+(v) - \delta, \quad g(v) = d^+(v) - d^{++}(v), \quad b(v) = g(v) - 1.$$

Thus $b(v) \geq 0$. For a target x , put

$$T(x) = V \setminus N^-[x], \quad P(x) = \{u : x \in U(u)\}, \quad p(x) = |P(x)|, \quad q(x) = e(x) + \mu(x).$$

For $q(x) > 0$ define $\sigma(x) = 2q(x) - 1 - p(x)$, and put $\sigma(x) = 0$ when $q(x) = 0$. Let

$$K = \{x : \sigma(x) > 0\}, \quad k = |K|, \quad S = \sum_x \sigma(x).$$

Call $\sigma = 1$ targets light and $\sigma \geq 2$ targets heavy. We use the previously established core identity

$$\binom{k}{2} = S + m + t + A_K + M_{\text{out}}, \quad (\text{S56.1})$$

where m is the number of minimum-degree vertices in K , t counts gap-two vertices of excess at most one, A_K is the number of arcs induced by K , and M_{out} is the number of missing pairs with both endpoints in $V \setminus K$.

Lemma S56.1 (Minimum predecessors of heavy targets). *Every target s with $\sigma(s) \geq 2$ has an in-neighbour of out-degree δ .*

Proof. Suppose otherwise. Then $D' = D - s$ has order $n' = 2\delta + 2$ and all its out-degrees are at least δ : minimum-degree vertices lose no outgoing arc, and every other vertex loses at most one. The oriented arc-count bound gives minimum out-degree at most $\lfloor (n' - 1)/2 \rfloor = \delta$, so it is exactly δ . The fixed-target budget identity in D' gives

$$B' + n' = z' - \sum_x \sigma'(x) \leq n', \quad B' = \sum_{v \in V(D')} b_{D'}(v),$$

since the capacity bound gives $\sigma'(x) \geq 0$ and $z' \leq n'$. Thus $B' \leq 0$.

Deleting s can only remove exact second neighbours of the other vertices. If $E_s(v)$ denotes those lost second neighbours other than s , the exact transfer identity is

$$B' = B - b(s) + (d^{--}(s) - d^-(s)) + \sum_{v \neq s} |E_s(v)|.$$

The first term $B - b(s)$ and the final sum are nonnegative. Furthermore

$$d^{--}(s) - d^-(s) = 2q(s) - 2 - p(s) = \sigma(s) - 1 \geq 1.$$

Consequently $B' \geq 1$, a contradiction. $\square$

Lemma S56.2 (A weighted five-vertex relation). *Let R be an oriented graph with positive integer vertex weights $w_1, \dots, w_H$ whose sum is five. If*

$$\sum_{j \in N_R^+(i)} w_j \geq 2 \quad \text{for every } i,$$

then $H = 5$, all weights equal one, and R is a regular tournament.

Proof. There is no possibility with $H = 1$ or $H = 2$. With $H = 3$, positive out-degree at every vertex forces a directed triangle. Each successor weight would then be at least two, exceeding the total weight five. With $H = 4$, the weights are 2, 1, 1, 1. The weight-two vertex must point to two singleton vertices. Neither can point back to it; each must therefore point to both remaining singleton vertices, giving a digon. Finally, with $H = 5$ every weight is one. The minimum out-degree two uses all ten possible arcs, and every out-degree is two. $\square$

Theorem S56.3. *The positive-surplus core satisfies $k \geq 6$. In particular,*

$$z \geq h + 8.$$

Proof. An all-deficient graph has minimum outdegree at least two, so the outside tournament used below is nonempty. The preceding core bound gives $k \geq 5$. Suppose $k = 5$. The established light-target incidence bound $A_K \geq L$, where L is the number of light targets, and $S \geq k + H$, where H is the number of heavy targets, make every inequality in (S56.1) an equality. In particular,

$$m = t = M_{\text{out}} = 0, \quad \sigma(x) \in \{1, 2\} \ (x \in K), \quad A_K = L.$$

Thus every core vertex is nonminimum-degree; every outside vertex has degree δ and is a zero-surplus target; and the outside graph is a tournament on $2\delta - 2$ vertices.

Every light target is nonminimum-degree and, by the previously established degree-surplus bounds, has $e = 1$, $g = 1$, and $|U(x)| = 1$. A light target with $q = 1$, $p = 0$ would therefore have $\mu = 0$ and be adjacent to all four other core vertices. This would give $A_K \geq 4$, whereas $A_K = L = 5 - H \leq 3$ by $H \geq 2$. Hence every light target is active, with $p > 0$.

The active-light protected-source lemma supplies, for each light target x , a parent $y \in K$ satisfying

$$y \rightarrow x, \quad N^-(y) \subseteq N^-(x) \setminus \{y\}.$$

Since $q(y) > q(x)$, the chosen parent edges form a forest. They have distinct heads and number L , so they are all the core arcs. A parent y cannot itself be light: its own parent z would also satisfy $N^-(z) \subseteq N^-(x)$ and $z \rightarrow x$. The protected containments would make both y and z unreachable from x within two steps, contradicting $|U(x)| = 1$. Thus the forest consists of stars rooted at the heavy targets. Let $C_1, \dots, C_H$ be these star basins, with roots $r_1, \dots, r_H$. Put $w_i = |C_i|$, so $\sum_i w_i = 5$.

Every minimum-degree vertex has at least two unreachable core vertices, by the tight-target demand lemma. Protected containment has two consequences. First, a minimum vertex u with $u \rightarrow r_i$ points to every vertex of C_i . Second, if u reaches r_j within two steps, it reaches every vertex of C_j within two steps. Both statements follow by iterating the containment along a protected tree.

Define a directed relation on the heavy roots by

$$i \rightarrow j \iff \text{there exists a degree-}\delta \text{ vertex } u \text{ with } u \rightarrow r_i \text{ and } r_j \in U(u).$$

Every heavy root has such a minimum predecessor by the preceding lemma. The predecessor lies outside K , and its at least two unreachable core vertices lie in basins other than its own. Therefore

$$\sum_{j \in N_R^+(i)} w_j \geq 2.$$

The relation has no digon. Indeed, witnesses $u \rightarrow r_i, r_j \in U(u)$ and $v \rightarrow r_j, r_i \in U(v)$ forbid both $u \rightarrow v$ and $v \rightarrow u$, since either arc would give a forbidden two-step path. But u and v are distinct vertices of the outside tournament.

The weighted-relation lemma now forces $H = 5$, all C_i to be singletons, and the relation to be a regular tournament. If an outside vertex u points to r_i , its unreachable core set contains at least two vertices and is contained in $N_R^+(i)$, which has exactly two vertices. It must equal $N_R^+(i)$. If u pointed to two roots, their relation out-neighbourhoods would therefore be equal. Distinct vertices of a tournament cannot have equal out-neighbourhoods: their mutual arc distinguishes them. Hence every outside vertex has at most one out-neighbour in K .

Every outside vertex consequently has out-degree at least $\delta - 1$ in the outside tournament. Its average out-degree, however, is $(2\delta - 3)/2 = \delta - 3/2$, a contradiction. This excludes $k = 5$.

Finally, the established $H \geq 2$ gives

$$z - h = S \geq k + H \geq 8,$$

as claimed. $\square$

S57 A sharper lower bound on minimum-degree predecessors

Throughout this section, D is an oriented graph with no Seymour vertex, minimum outdegree δ , and order $n = 2\delta + 3$. All second neighborhoods are defined by directed distance exactly two. Put

$$a(v) = d^+(v) - \delta, \quad b(v) = d^+(v) - 1 - d^{++}(v), \quad L = \{v : a(v) = 0\}.$$

Thus $a(v), b(v) \geq 0$. A vertex is called *perfect* here if it has outdegree δ and is adjacent to every other vertex; write $\mathcal{P}$ for the set of perfect vertices. For $s \in V(D)$ define

$$c(s) = |N^-(s) \cap L|, \quad c_0(s) = |N^-(s) \cap \mathcal{P}|, \quad \Phi(s) = d^{--}(s) - d^-(s).$$

The incoming second neighborhood defining $d^{--}(s)$ also uses exact directed distance two.

Theorem S57.1. *Every vertex s of D satisfies*

$$9c(s) + 6c_0(s) \geq \delta + 1 + 3\Phi(s). \quad (\text{S57.1})$$

In particular, when $\delta \geq 3$,

$$c(s) \geq \max \left\{ 1, \left\lceil \frac{\delta - 5}{15} \right\rceil \right\}. \quad (\text{S57.2})$$

The source-restricted deletion identity and the charging argument needed for this estimate are derived below. In particular it does not assume that $D - s$ is a counterexample or that $D - s$ is arc-minimal.

S57.1 The source-restricted budget

Fix s , abbreviate $c = c(s)$ and $c_0 = c_0(s)$, and set

$$D' = D - s, \quad Z = N^-(s) \cap L, \quad W = V(D') \setminus Z.$$

Write d' for outdegree in D' , and put $a'(v) = d'(v) - \delta$. The vertices of Z have $a' = -1$, while all vertices of W have $a' \geq 0$. For a target $x \in V(D')$ define

$$T'(x) = V(D') \setminus (\{x\} \cup N_{D'}^-(x)), \quad q'(x) = |T'(x)| - \delta,$$

$$P'_W(x) = \{v \in W : x \notin \{v\} \cup N_{D'}^+(v) \cup N_{D'}^{++}(v)\}.$$

Let $\mathcal{X} = \{x : P'_W(x) \neq \emptyset\}$ be the set of targets, and for $x \in \mathcal{X}$ put

$$p_x = |P'_W(x)|, \quad \Delta_x = 2q'(x) - 1 - p_x.$$

Every source in $P'_W(x)$ has outdegree at least δ and sends all its outgoing arcs into $T'(x)$. The usual capacity count therefore gives $\Delta_x \geq 0$ and the exact defect identity

$$\sum_{v \in P'_W(x)} (a'(v) + \ell_v) + M_{P'_W(x)} = \frac{p_x \Delta_x}{2}, \quad (\text{S57.3})$$

where ℓ_v counts the vertices of $T'(x) \setminus P'_W(x)$ not dominated by v , and $M_{P'_W(x)}$ counts missing pairs inside the source set. Denote

$$N = |\mathcal{X}|, \quad \mathcal{D} = \sum_{x \in \mathcal{X}} \Delta_x.$$

If $q'(x) = -1$, then $x \in Z \cap \mathcal{P}$ and x is not a target. For a nontarget not in $Z \cap \mathcal{P}$, $q'(x) \geq 0$. Define the following nonnegative budget terms:

$$t_2 = 2 \sum_{x \notin \mathcal{X}, x \notin Z \cap \mathcal{P}} q'(x), \quad t_3 = \sum_{v \in W} b(v),$$

$$E_s(v) = (N_D^{++}(v) \setminus \{s\}) \setminus N_{D'}^{++}(v), \quad t_4 = \sum_{v \in W} |E_s(v)|.$$

The source-restricted deletion identity states

$$N + \mathcal{D} + t_2 + t_3 + t_4 = \beta := 3c - \Phi(s) + 2c_0. \quad (\text{S57.4})$$

For completeness, this follows by double counting $\sum_{v \in W} |U_{D'}(v)| = \sum_x |P'_W(x)|$, using $|U_{D'}(v)| = 2 - 2a'(v) + b_{D'}(v)$, and substituting the exact transfer

$$b_{D'}(v) = b(v) - [v \rightarrow s] + [s \in N_D^{++}(v)] + |E_s(v)|.$$

Here $[\cdot]$ is the indicator of the indicated event. Among minimum-degree vertices, precisely the c vertices of Z point to s ; each perfect member of Z contributes the removed nontarget term $2q' = -2$. These are the terms $3c$ and $2c_0$ in (S57.4).

Partition W as

$$\Pi = \{v \in W : q'(v) = 0\}, \quad Y = W \setminus \Pi, \quad y = |Y|.$$

Every vertex of Π has outdegree δ in D' and is adjacent to all vertices of D' . Thus every missing pair of D' lies inside $Y \cup Z$. Also each target in Y contributes one to N , and each nontarget in Y contributes at least two to t_2 . Consequently

$$y \leq N + \frac{t_2}{2}. \quad (\text{S57.5})$$

Let M' count missing pairs of D' and let $A'_Y = \sum_{v \in Y} a'(v)$. Since $|V(D')| = 2\delta + 2$, an exact arc count gives

$$M' = \delta + 1 + c - A'_Y. \quad (\text{S57.6})$$

S57.2 The integer improvement in the charging argument

Call a target x heavy if $|P'_W(x) \cap \Pi| \geq p_x/2$, and light otherwise. For a light target put $r_x = |P'_W(x) \cap Y|$. The strict inequality $r_x > p_x/2$ and integrality give $p_x \leq 2r_x - 1$. Hence

$$q'(x) = \frac{1 + \Delta_x + p_x}{2} \leq r_x + \left\lfloor \frac{\Delta_x}{2} \right\rfloor. \quad (\text{S57.7})$$

The cancellation of the additional constant in this inequality is useful: a light target with zero defect needs no extra target charge.

Assign each missing pair of D' to the first applicable category below. Writing $\mu'(x)$ for the missing degree in D' , note that $\mu'(x) \leq q'(x) + [x \in Z]$.

1. If an endpoint is not a target, charge the pair to such an endpoint. This costs at most $t_2/2$, together with one possible additional unit for each such endpoint in Z . A vertex in $Z \cap \mathcal{P}$ receives no charge, since its missing degree is zero.
2. If both endpoints are targets and an endpoint is light, charge to a light endpoint. By (S57.7), its charges can be paid by r_x distinct source–target pairs (v, x) with $v \in Y$, at most $\lfloor \Delta_x/2 \rfloor$ defect units, and one possible additional unit if $x \in Z$.
3. Otherwise both endpoints are heavy targets. If $u \in P'_W(x)$ dominates x' and $v \in P'_W(x')$ dominates x , then u, v are nonadjacent: neither can point to the other without creating a two-step path to its own far target. Since vertices of Π are adjacent to every vertex, such a mutual configuration cannot use a Π -source at both ends. Orient the names of the endpoints so that no Π -source of x dominates x' . If $x' \in P'_W(x)$, then $x' \in Y$ and charge the pair to (x', x) . Otherwise $x' \in T'(x) \setminus P'_W(x)$ is omitted by every Π -source of x . Each such omitted endpoint uses at least $p_x/2$ units in the left side of (S57.3), so at most Δ_x such pairs can be charged to x .

All source–target pairs charged in the second and third categories are distinct: in the second the target is light, while in the third it is heavy. Their total number is at most $\sum_{v \in Y} |U_{D'}(v)|$. Defect charges total at most $\mathcal{D}$, and the additional Z units from the first and second categories total at most c . We conclude that

$$M' \leq \mathcal{D} + \frac{t_2}{2} + \sum_{v \in Y} |U_{D'}(v)| + c. \quad (\text{S57.8})$$

S57.3 Proof of the general bound

For $v \in Y$ put $j_v = [v \rightarrow s]$. With $A_Y = \sum_{v \in Y} a(v)$ and $J_Y = \sum_{v \in Y} j_v$, one has $A'_Y = A_Y - J_Y$. Deleting s removes it from the old far set if it belonged to that set, and adds precisely the lost second neighbors $E_s(v)$. Thus, writing $B_Y = \sum_{v \in Y} b(v)$ and $F_Y = \sum_{v \in Y} [s \in U_D(v)]$, the exact identity is

$$\sum_{v \in Y} |U_{D'}(v)| = 3y - 2A_Y + B_Y - F_Y + \sum_{v \in Y} |E_s(v)|. \quad (\text{S57.9})$$

Since $A_Y \geq A'_Y$, $B_Y \leq t_3$, and $\sum_Y |E_s(v)| \leq t_4$, this yields

$$\sum_{v \in Y} |U_{D'}(v)| \leq 3y - 2A'_Y + t_3 + t_4.$$

Substituting this into (S57.8) and using (S57.6), we obtain

$$\delta + 1 + A'_Y \leq \mathcal{D} + \frac{t_2}{2} + 3y + t_3 + t_4 \leq 3N + \mathcal{D} + 2t_2 + t_3 + t_4 \leq 3\beta.$$

Dropping $A'_Y \geq 0$ and using (S57.4) proves (S57.1).

For every target of the original graph, the capacity inequality gives $\Phi(s) \geq -2$: it equals -2 for a perfect vertex, and equals $\sigma(s) - 1 \geq -1$ when $q(s) > 0$, where

$$q(s) = a(s) + \mu(s), \quad p(s) = |\{v : s \in U_D(v)\}|, \quad \sigma(s) = 2q(s) - 1 - p(s).$$

As $c_0 \leq c$, (S57.1) gives $15c \geq \delta - 5$. For $\delta \geq 6$ this bound also forces $c \geq 1$. The cases $3 \leq \delta \leq 5$ have no counterexample by the established minimum-outdegree-six theorem of Kaneko and Locke [8]. This completes the proof of Theorem S57.1.

S57.4 A density-sensitive refinement

Assume now that D is strongly connected and minimal under deletion of arbitrary nonempty sets of arcs. Let M count its missing pairs. The pointwise second-neighborhood imbalance theorem proved earlier gives nonnegative integer residuals

$$\eta(v) = \mu(v) - 1 + d^{--}(v) - d^{++}(v), \quad \sum_v \eta(v) = R := 2M - n.$$

Theorem S57.2. *Under these additional assumptions every vertex s satisfies*

$$6c(s) + 4c_0(s) \geq \delta + 1 - R + 2\Phi(s). \quad (\text{S57.10})$$

Consequently

$$c(s) \geq \left\lceil \frac{3\delta - 2M}{10} \right\rceil. \quad (\text{S57.11})$$

Proof. Let $m_Y = |Y \cap L|$, where Y is defined above. Every minimum-degree vertex of Y is nonwhole in the original graph. Indeed, a whole minimum-degree vertex pointing to s belongs to Z ; if it does not point to s , it remains whole of degree δ in D' and belongs to Π . For a nonwhole minimum-degree vertex v , the original capacity inequality gives

$$\eta(v) = \sigma(v) + g(v) \geq 1, \quad g(v) = d^+(v) - d^{++}(v) \geq 1.$$

Hence $m_Y \leq R$.

Every original nonminimum-degree vertex of Y has $a(v) \geq 1$, so $A_Y \geq y - m_Y$. Combining the exact identities (S57.6) and (S57.9) with (S57.8) gives

$$\delta + 1 \leq \mathcal{D} + \frac{t_2}{2} + 3y - A_Y - J_Y + t_3 + t_4 \leq \mathcal{D} + \frac{t_2}{2} + 2y + m_Y + t_3 + t_4.$$

Using (S57.5), this is at most

$$2N + \mathcal{D} + \frac{3t_2}{2} + t_3 + t_4 + m_Y \leq 2\beta + R.$$

Substitute (S57.4) to obtain (S57.10). Finally $c_0 \leq c$ and $\Phi(s) \geq -2$ imply

$$10c \geq \delta - 3 - R = 3\delta - 2M,$$

where the last equality uses $n = 2\delta + 3$. This proves (S57.11). $\square$

S57.5 Target refinements and examples

For $q(s) > 0$, $\Phi(s) = \sigma(s) - 1$, so the two refined estimates imply

$$c(s) \geq \left\lceil \frac{\delta + 3\sigma(s) - 2}{15} \right\rceil, \quad c(s) \geq \left\lceil \frac{3\delta + 2 - 2M + 2\sigma(s)}{10} \right\rceil$$

under their respective hypotheses. These may be combined with $c(s) \geq 1$ and with the stronger inequalities retaining $c_0(s)$.

For example, at $\delta = 30$, the general bound forces at least two minimum-degree predecessors. No claim of a literature-wide optimal constant is made.

As an updated numerical example, at $(n, \delta, M) = (21, 9, 14)$ the residual is seven. A heavy target has $\Phi \geq 1$, and hence $9c + 6c_0 \geq 13$. A single nonwhole minimum-degree predecessor is therefore impossible. This is a conditional structural calculation, not an assertion that a counterexample exists at that order.

Lemma S57.3. *Let D be any oriented graph of order $2\delta + 3$ with minimum outdegree δ . Suppose $S \subsetneq V(D)$ has cardinality $2t + 1$ and $D' = D - S$ has minimum outdegree at least $\delta - t$. Then this minimum outdegree is exactly $\delta' = \delta - t$, and the order of D' is $n' = 2\delta' + 2$. For $v \notin S$ put*

$$j_S(v) = |N_D^+(v) \cap S|, \quad E_S(v) = (N_D^{++}(v) \setminus S) \setminus N_{D'}^{++}(v),$$

and put

$$\Phi(S) = \sum_{v \notin S} (|N_D^{++}(v) \cap S| - j_S(v)).$$

If $B_{\text{out}} = \sum_{v \notin S} b_D(v)$, then

$$B(D') = B_{\text{out}} + \Phi(S) + \sum_{v \notin S} |E_S(v)|. \quad (\text{S57.12})$$

Let z' be the number of vertices of D' adjacent to every other vertex and having outdegree δ' . With σ' denoting its nonnegative fixed-target capacity surpluses relative to δ' , one has

$$z' = n' + B_{\text{out}} + \Phi(S) + \sum_{v \notin S} |E_S(v)| + \sum_x \sigma'(x). \quad (\text{S57.13})$$

Furthermore, writing A_S for the number of arcs inside S and Q_S for the number of ordered pairs (u, v) of vertices in S with $v \in N_D^{++}(u)$, one has

$$\Phi(S) = \sum_{s \in S} (d_D^-(s) - d_{D'}^-(s)) + A_S - Q_S. \quad (\text{S57.14})$$

Proof. The oriented arc-count bound at order $2(\delta - t) + 2$ makes the minimum outdegree at most $\delta - t$, proving equality. Deletion of vertices does not change an old direct adjacency between two surviving vertices. Consequently

$$b_{D'}(v) = b_D(v) - j_S(v) + |N_D^{++}(v) \cap S| + |E_S(v)|.$$

Summation gives (S57.12). The fixed-target budget identity at order $2\delta' + 2$ is $B(D') + n' = z' - \sum_x \sigma'(x)$; this gives (S57.13). Finally, summing $d_D^-(s) - d_{D'}^-(s)$ over S counts the same incoming terms as $\Phi(S)$, together with the internal contribution $Q_S - A_S$. This proves (S57.14). $\square$

Lemma S57.4. *If an oriented graph R on three or four vertices has positive outdegree at every vertex, it contains three vertices whose outneighborhoods in R are pairwise disjoint.*

Proof. Complete R to a tournament; this preserves positive outdegree, and any disjointness proved in the completion holds in R . On three vertices the completion must be a directed triangle. On four vertices, if some vertex has outdegree three, the remaining three vertices form a directed triangle and their outneighborhoods are disjoint. Otherwise the tournament has score multiset $(1, 1, 2, 2)$. Let a, b have outdegree one and orient their mutual arc as $a \rightarrow b$. If the other vertices are c, d , then both point to a , while b points to exactly one of them, say c . Thus $d \rightarrow b$. The arc $d \rightarrow c$ would give d outdegree three, so $c \rightarrow d$. The outneighborhoods of a, b, c are respectively $\{b\}, \{c\}, \{a, d\}$, which are pairwise disjoint. $\square$

S58 Odd heavy-set capacity at arbitrary residual

Let D be a strongly connected all-deficient oriented graph, minimal under deletion of arbitrary nonempty sets of arcs, of order $n = 2\delta + 3$ and minimum outdegree δ . Let $e(v) = d_D^+(v) - \delta$ and $g(v) = d_D^+(v) - d_D^{++}(v)$. Single-arc minimality gives $g(v) \in \{1, 2\}$; the non-negativity of $|U_D(v)| = 2 - 2e(v) + g(v)$ gives $e(v) \in \{0, 1, 2\}$, and excess two forces gap two. Define target surpluses $\sigma(v)$ as before and call a vertex heavy when $\sigma(v) \geq 2$.

Theorem S58.1 (Odd heavy-set capacity). *Let $S \subsetneq V(D)$ consist of $m = 2t + 1$ heavy vertices, where $t \geq 1$. Suppose $D' = D - S$ has minimum outdegree at least $\delta - t$. Let A_S be the number of arcs inside S , and let Q_S count the ordered pairs (u, v) in $S \times S$ with $v \in N_D^{++}(u)$, using distance exactly two. Put*

$$\gamma_S = \sum_{s \in S} \sigma(s) - 2m \geq 0, \quad B_{\text{out}} = \sum_{v \notin S} (g(v) - 1).$$

Call a vertex of D' perfect if it is adjacent to every other vertex and has outdegree $\delta - t$. Let a count the nonperfect vertices. Then

$$a \leq 4t^2 - 1 - 2A_S - \gamma_S - B_{\text{out}}, \tag{S58.1}$$

and

$$\delta - t + 1 \leq (t + 1)a + \min\{a, B_{\text{out}}\} + \binom{a}{2}. \tag{S58.2}$$

In particular,

$$\delta \leq t - 1 + t(2t + 1)(4t^2 - 1). \tag{S58.3}$$

Proof. The oriented arc-count bound shows that the actual minimum outdegree of D' is exactly $\delta' = \delta - t$, at order $n' = 2\delta' + 2$. No counterexample or minimality assumption is made about D' .

For each heavy s , the original fixed-target identity gives $\Phi(s) = d_D^{--}(s) - d_D^-(s) = \sigma(s) - 1$. By Lemma S57.3, the exact transfer is

$$z' = n' + B_{\text{out}} + \Phi(S) + \sum_{v \notin S} |E_S(v)| + \sum_x \sigma'(x),$$

where $z' = n' - a$, the loss terms and the capacity surpluses σ' are nonnegative, and

$$\Phi(S) = \sum_{s \in S} \Phi(s) + A_S - Q_S = m + \gamma_S + A_S - Q_S.$$

Each internal arc excludes its own ordered pair from exact distance two. Hence $Q_S \leq m(m-1) - A_S$, giving

$$a \leq Q_S - A_S - m - \gamma_S - B_{\text{out}} \leq m(m-2) - 2A_S - \gamma_S - B_{\text{out}}.$$

As $m(m-2) = 4t^2 - 1$, this proves (S58.1).

All missing pairs of D' are confined to its a nonperfect vertices, so $M(D') \leq \binom{a}{2}$. The exact degree excess at its even order is

$$\sum_{v \in V(D')} (d_{D'}^+(v) - \delta') = \delta - t + 1 - M(D').$$

A surviving vertex of original excess at most one contributes at most $t+1$ to this sum. Original excess-two vertices can contribute one additional unit, and their number is at most B_{out} , since each has gap two. Only nonperfect vertices contribute any excess. The sum is therefore at most $(t+1)a + \min\{a, B_{\text{out}}\}$, proving (S58.2).

For the coarse estimate put $L_0 = 4t^2 - 1$. The first bound implies $a + B_{\text{out}} \leq L_0$. Thus

$$(t+1)a + \min\{a, B_{\text{out}}\} + \binom{a}{2} \leq ta + L_0 + \binom{a}{2} \leq (t+1)L_0 + \binom{L_0}{2}.$$

Substitution and simplification give $\delta \leq t - 1 + t(2t+1)(4t^2 - 1)$. $\square$

Corollary S58.2 (Heavy-triple obstruction). *If $\delta \geq 10$, no set S of three heavy vertices satisfies $\delta^+(D - S) \geq \delta - 1$. Equivalently, for every three heavy vertices, there exists a surviving minimum-degree vertex pointing to at least two of them, or a surviving excess-one vertex pointing to all three.*

Proof. Set $t = 1$ in (S58.3), which gives $\delta \leq 9$ whenever the deletion condition holds. Its failure means that some $v \notin S$ has $|N_D^+(v) \cap S| \geq e(v) + 2$. Since $|S| = 3$, excess two cannot cause this failure; the two remaining possibilities are exactly those stated. $\square$

Corollary S58.3 (Saturation of a valid heavy triple). *Suppose $\delta \geq 7$ and a heavy triple S satisfies $\delta^+(D - S) \geq \delta - 1$. Then $7 \leq \delta \leq 9$, every vertex outside S has gap one, and all of the following equalities hold:*

$$\begin{aligned} A_S &= 0, & \sigma(s) &= 2 \ (s \in S), & Q_S &= 6, & z' &= 2\delta - 3, \\ E_S(v) &= \emptyset \ (v \notin S), & \sigma'(x) &= 0 \ (x \in V(D - S)), & B(D - S) &= -3. \end{aligned}$$

In particular, S is pairwise nonadjacent and every ordered pair of its distinct vertices is connected by an exact two-step path in D .

Proof. For a triple, write $B = B_{\text{out}}$ and $D_0 = 2A_S + \gamma_S$. The capacity inequalities become

$$a \leq 3 - B - D_0, \quad \delta \leq 2a + \min\{a, B\} + \binom{a}{2}.$$

If $B = 1$, the right side is at most six; if $B = 2$, it is at most three; and if $B = 3$, it is zero. Larger B is impossible. Thus $\delta \geq 7$ forces $B = 0$. If $D_0 \geq 1$, then $a \leq 2$ and the degree bound is at most five, again impossible. Consequently $A_S = \gamma_S = 0$, and $a = 3$ because $a \leq 2$ would still give $\delta \leq 5$. The coarse estimate gives $\delta \leq 9$.

Now $\Phi(S) = 3 - Q_S \geq -3$, and the exact perfect-vertex transfer with $a = 3$ reads

$$-3 = 3 - Q_S + \sum_{v \notin S} |E_S(v)| + \sum_x \sigma'(x).$$

Its nonnegative terms force $Q_S = 6$ and both sums to vanish. The remaining assertions follow from $B = 0$, $\gamma_S = 0$, and the exact transfer for $B(D - S)$. $\square$

Corollary S58.4 (Covering the heavy triples). *Let $\delta \geq 10$, let $\mathcal{H}$ be the set of heavy vertices, and put $H = |\mathcal{H}|$ and $a_v = |N_D^+(v) \cap \mathcal{H}|$. Then*

$$\binom{H}{3} \leq \sum_{e(v)=0} \left[\binom{a_v}{2} (H - a_v - [v \in \mathcal{H}]) + \binom{a_v}{3} \right] + \sum_{e(v)=1} \binom{a_v}{3}.$$

Proof. Every heavy triple has a witness outside it by Corollary S58.2. For a minimum-degree vertex v , the number of triples avoiding v with exactly two dominated vertices is $\binom{a_v}{2}(H - a_v - [v \in \mathcal{H}])$, and the number with three dominated vertices is $\binom{a_v}{3}$. For an excess-one vertex, only the latter triples qualify. The union bound over all possible witnesses gives the inequality. $\square$

S59 Protected basins beyond the first residual layer

Let D be a strongly connected all-deficient oriented graph, minimal under deletion of arbitrary nonempty sets of arcs, with $n = 2\delta + 3$. Retain the target surpluses σ , the positive core K , its size k , and the heavy targets $\mathcal{H}$, $H = |\mathcal{H}|$. Let m count minimum-degree vertices in K , and let L count the nonwhole vertices outside K . The zero-surplus classification gives the following dichotomy outside K : a vertex is either a whole minimum-degree vertex, or a minimum-degree vertex with $\mu = q = p = 1$. Put

$$R = 2M - n = \sum_v \eta(v), \quad \eta(v) = 2\mu(v) + g(v) - 1 - p(v) \geq 0.$$

For $v \in K$ of minimum degree one has $\eta(v) = \sigma(v) + g(v) \geq 2$. An outside nonwhole vertex has $\eta(v) = g(v) \geq 1$. Consequently

$$2m + L \leq R. \tag{S59.1}$$

Let B be the set of inactive minimum-degree unit-surplus targets:

$$B = \{x \in K : e(x) = 0, \sigma(x) = 1, p(x) = 0\}, \quad b = |B|.$$

Each such target has $q = \mu = 1$ and $\eta(x) = \sigma(x) + g(x) \geq 2$. Importantly, $p(x) = 0$ means that no vertex has x in its far set.

Lemma S59.1 (Certificates outside the exceptional set). *The set $K \setminus B$ can be partitioned into basins $B_1, \dots, B_H$, one for each heavy root r_i , with*

$$r_i \rightarrow x, \quad N^-(r_i) \subseteq N^-(x) \quad (x \in B_i \setminus \{r_i\}).$$

In particular, an external vertex pointing to r_i points to every vertex of B_i ; reaching r_i within two steps reaches every vertex of B_i within two steps.

Proof. An active unit target has a protected parent of strictly larger q , with an arc to the child and an incoming-neighborhood containment. Iteration must end at a heavy target; all such certificates compose transitively.

An inactive unit target satisfies $q = 1$. If it is not in B , it has excess one and missing degree zero. It is whole, has a nonempty far set, and cannot be far from a zero-surplus target, whose T -set has at most $\delta + 1$ vertices while the closed outneighborhood of this source has $\delta + 2$ vertices. Choose an active far target $y \in K$. Wholeness and the absence of a two-step path to y give $y \rightarrow x$ and $N^-(y) \subseteq N^-(x)$. Compose this certificate with

the heavy-ancestor certificate of y , if y is itself unit. Such a chain never enters B , whose vertices have $q = 1$ and no far source. Assigning each nonheavy vertex one heavy certificate gives the partition. The two stated reachability properties follow directly from the incoming-neighborhood containment. $\square$

Theorem S59.2 (Good minimum-degree witnesses). *Suppose every heavy root has more than $m + 2L$ minimum-degree in-neighbors. Then there is an oriented relation on the H roots with basin weights $w_i = |B_i|$, whose weighted outdegree is at least three at every root. Writing $k' = k - b = \sum_i w_i$, one obtains*

$$\sum_i w_i^2 \leq (k')^2 - 6k'. \quad (\text{S59.2})$$

In particular $H \geq 3$ and $k' \geq 7$. The same conclusion holds if every heavy root has more than L whole minimum-degree in-neighbors.

Proof. Exclude three kinds of minimum-degree predecessors of a heavy root: those inside K (at most m); nonwhole predecessors outside K (at most L); and whole outside predecessors whose far set contains a nonwhole target outside K (at most L). The last count uses $p = 1$ at each such outside target. The union has cardinality at most $m + 2L$.

Thus choose a remaining predecessor u_i for every root r_i . It is a whole minimum-degree vertex outside K with no outside far target. Since targets in B have no far sources, its far set lies in $K \setminus B$ and has size 3 or 4. Define $i \rightarrow j$ whenever some whole minimum-degree vertex u outside K , with no far target outside K , satisfies $u \rightarrow r_i$ and $r_j \in U_D(u)$. The chosen u_i guarantees such a witness for each demanded outgoing basin. There is no loop since $u \rightarrow r_i$. There is no digon: witnesses for opposite relations would be distinct and nonadjacent, as either direction between them would create a forbidden two-step path. But both witnesses are whole. The basin certificate makes each far core vertex imply that its root is far, so the weighted outdegree at i is at least three.

Multiplying these demands by w_i and summing gives

$$3k' \leq \sum_{i \rightarrow j} w_i w_j \leq \frac{(k')^2 - \sum_i w_i^2}{2},$$

which is (S59.2). Positive outdegree in an oriented relation forces $H \geq 3$; positivity and integrality of the weights give $\sum_i w_i^2 \geq k'$, hence $k' \geq 7$. If whole minimum-degree predecessors are counted directly, only the at most L sources far from an outside nonwhole target need to be excluded, proving the alternate sufficient condition. $\square$

Corollary S59.3 (A degree-residual sufficient condition). *The conclusions of Theorem S59.2 hold whenever*

$$\delta > 10(m + 2L) + R - 3.$$

In particular, the simpler condition $\delta > 21R - 3$ suffices.

Proof. For a heavy target s , $\Phi(s) = \sigma(s) - 1 \geq 1$. The density-sensitive predecessor bound gives

$$10c(s) \geq \delta + 1 - R + 2\Phi(s) \geq \delta + 3 - R.$$

The first hypothesis thus forces $c(s) > m + 2L$ for every heavy target. Finally (S59.1) implies $m + 2L \leq 2R$, giving the simpler condition. $\square$

S60 Capacity toward the heavy targets

Let D be strongly connected, all-deficient, and minimal under deletion of arbitrary nonempty sets of arcs. Assume $n = 2\delta + 3$ and $\delta \geq 6$. Let K be its positive-surplus core, let $\mathcal{H}$ be the set of heavy vertices ($\sigma \geq 2$), and put $H = |\mathcal{H}|$. The established general result gives $H \geq 2$. Use the certified basins $B_1, \dots, B_H$ of Lemma S59.1, and the exceptional set

$$B = \{x \in K : e(x) = 0, \sigma(x) = 1, p(x) = 0\}.$$

No source is far from a vertex of B . Every other core vertex has a heavy-root certificate. A source pointing to a root points to its entire basin; a source reaching a root within two steps reaches its entire basin within two steps. Also, every minimum-degree source has at least two far vertices in K , all certified: its far set has size three or four and contains at most one zero-surplus target. Every heavy root has a minimum-degree predecessor.

Lemma S60.1 (Minimum-degree capacity toward the heavy set). *For every minimum-degree vertex v ,*

$$|N_D^+(v) \cap \mathcal{H}| \leq H - 2.$$

Proof. If v is itself a heavy root and points to every other heavy root, it points to its own children and to all the other basins. It would have no far certified core target, a contradiction. Thus consider a minimum-degree vertex v that is not heavy. Suppose it points to at least $H - 1$ heavy roots. It cannot point to all H , since that would directly reach every certified core target. Let r be the unique heavy root not dominated by v .

Every far core target of v is in the basin of r . The certificate therefore makes r far from v . Choose any minimum-degree predecessor $w \rightarrow r$. Vertex w points to the basin of r and has a far certified core target elsewhere, hence some root $s \neq r$ is far from w . As $v \rightarrow s$, neither $v \rightarrow w$ nor $w \rightarrow v$ is possible: the former would let v reach r , and the latter would let w reach s within two steps.

If v is whole, this nonadjacency is impossible. If $\mu(v) = 1$, the same reasoning for every minimum-degree predecessor of r confines that entire predecessor set to the unique nonneighbor of v . It is nonempty, so $c(r) = 1$; its member is nonwhole, so $c_0(r) = 0$. The refined predecessor bound gives

$$9c(r) + 6c_0(r) \geq \delta + 1 + 3\Phi(r) \geq \delta + 4,$$

contradicting $\delta \geq 6$.

These cases cover every minimum-degree vertex outside K and every inactive minimum-degree unit target. In the remaining case v is an active minimum-degree unit target and has a heavy-root certificate. Its certified root cannot be one of the roots it dominates, so it must be r . The containment $N^-(r) \subseteq N^-(v)$ makes $w \rightarrow v$, contradicting the preceding nonadjacency (equivalently, $w \rightarrow v \rightarrow s$ reaches a far root). All possible types are excluded. $\square$

Theorem S60.2 (Capacity for every vertex). *Every vertex v of D satisfies*

$$|N_D^+(v) \cap \mathcal{H}| \leq H - 2. \tag{S60.1}$$

In particular $H \geq 3$. If $\delta \geq 10$, then $H \geq 4$ and

$$z \geq h + 10,$$

where z counts whole minimum-degree vertices and h counts gap-two vertices.

Proof. Minimum-degree vertices were handled above. If a heavy vertex v points to all the other heavy roots, a minimum predecessor $w \rightarrow v$ reaches its own root directly and every other root within two steps through v . It therefore reaches every certified core target within two steps, contradicting the minimum-source far demand.

The remaining vertices are nonminimum unit targets, and all have a heavy-root certificate, say with root r . The arc $r \rightarrow v$ prevents v from pointing back to r . If it pointed to every other heavy root, any minimum predecessor $w \rightarrow r$ would point to v by $N^-(r) \subseteq N^-(v)$, and would again reach every root and certified core target within two steps. This contradiction proves (S60.1).

If $H = 2$, the minimum-degree capacity is zero, contrary to the existence of a minimum-degree predecessor for every heavy root. Thus $H \geq 3$. If $H = 3$, delete the set S of all three heavy roots. Each surviving minimum-degree vertex loses at most one outgoing arc by (S60.1). Every surviving nonminimum vertex is unit and has excess one, so its outdegree after deletion is also at least $\delta - 1$. All excess-two vertices are heavy and have been deleted. Hence $\delta^+(D - S) \geq \delta - 1$. The heavy-triple obstruction in Corollary S58.2 contradicts $\delta \geq 10$. Consequently $H \geq 4$ in that range. The general core bound gives $k \geq 6$, while the fixed-target budget gives $z - h = \sum_x \sigma(x) \geq k + H$. This yields $z \geq h + 10$. $\square$

Corollary S60.3 (A pair covering when there are four heavy roots). *Suppose $\delta \geq 10$ and $H = 4$. Form a simple graph J on $\mathcal{H}$, joining two heavy roots if some minimum-degree vertex outside that pair points to both. Every three-element subset of $\mathcal{H}$ contains an edge of J ; equivalently, the independence number of J is at most two. More explicitly, with $a_v = |N_D^+(v) \cap \mathcal{H}|$, one has*

$$4 \leq \sum_{e(v)=0, a_v=2} (2 - [v \in \mathcal{H}]).$$

Proof. The heavy-triple obstruction requires an outside minimum-degree vertex pointing to at least two members of each triple, or an outside excess-one vertex pointing to all three. The latter alternative is ruled out by (S60.1), which limits every heavy outdegree to two. Thus every triple contains a pair dominated by a minimum-degree source outside that triple. Such a source has $a_v = 2$ and covers two triples if it is not heavy, and one triple if it is itself heavy; summation gives the explicit inequality. $\square$

S61 A source-restricted odd deletion with exceptional minima

Let D be strongly connected, all-deficient and minimal under deletion of arbitrary nonempty sets of arcs, with $n = 2\delta + 3$. Use $e(v) = d_D^+(v) - \delta$, $b(v) = g(v) - 1 \geq 0$, and the heavy-target notation established earlier.

Lemma S61.1 (A three-vertex deletion with d exceptional sources). *Let S consist of three heavy targets. Suppose every surviving minimum-degree vertex sends at most two arcs into S , and every surviving nonminimum-degree vertex has outdegree at least $\delta - 1$ after deletion of S . Let*

$$Z = \{v \notin S : e(v) = 0, |N_D^+(v) \cap S| = 2\}, \quad d = |Z|.$$

Then

$$\delta \leq 20d + 12. \tag{S61.1}$$

If $\delta \geq 10$, then $d \geq 1$.

Proof. Put $D' = D - S$, $n' = 2\delta$, and fix the counting baseline $\alpha = \delta - 1$. Set $W = V(D') \setminus Z$. The vertices of Z have outdegree $\alpha - 1$, while every vertex of W has outdegree at least α . When $d > 0$, the actual minimum outdegree of D' is $\delta - 2$; we do not apply a full-graph minimum-degree theorem at baseline α . All capacity arguments below use only the source set W .

Write $a'(v) = d_{D'}^+(v) - \alpha$, let $\mu'(v)$ be the missing degree in D' , and define

$$T'(x) = V(D') \setminus N_{D'}^-[x], \quad q'(x) = |T'(x)| - \alpha = a'(x) + \mu'(x),$$

$$P'_W(x) = \{v \in W : x \in U_{D'}(v)\}, \quad \mathcal{X} = \{x : P'_W(x) \neq \emptyset\}, \quad N = |\mathcal{X}|.$$

For $x \in \mathcal{X}$ put $p_x = |P'_W(x)|$ and $\Delta_x = 2q'(x) - 1 - p_x$. The source-restricted capacity count gives $\Delta_x \geq 0$. Let $\mathcal{D} = \sum_{x \in \mathcal{X}} \Delta_x$. The only negative q' values are $q' = -1$ at

$$Z_0 = \{z \in Z : \mu'(z) = 0\}, \quad z_0 = |Z_0|,$$

and these vertices are never targets. Define the nonnegative quantities

$$t_2 = 2 \sum_{x \notin \mathcal{X}, x \notin Z_0} q'(x), \quad t_3 = \sum_{v \in W} b(v),$$

$$E_S(v) = (N_D^{++}(v) \setminus S) \setminus N_{D'}^{++}(v), \quad t_4 = \sum_{v \in W} |E_S(v)|.$$

Let $\Phi(S)$ be the exact deletion imbalance of Lemma S57.3, and put $Q_Z = \sum_{v \in Z} |S \cap N_D^{++}(v)|$. Each vertex of Z directly dominates two of the three deleted vertices, so $0 \leq Q_Z \leq d$.

The exact restricted budget is

$$N + \mathcal{D} + t_2 + t_3 + t_4 = \beta := 2d - \Phi(S) + 2z_0 + Q_Z. \quad (\text{S61.2})$$

To derive it, put $A' = \sum_{v \in V(D')} a'(v)$ and $B'_W = \sum_{v \in W} b_{D'}(v)$. The even-order arc count gives $\sum_x q'(x) = n' - A'$. Since $a' = -1$ on Z , double counting the far source–target pairs with sources in W gives

$$N + \mathcal{D} + 2 \sum_{x \notin \mathcal{X}} q'(x) + B'_W = 4d.$$

The exact transfer on W is

$$B'_W = t_3 + \Phi(S) - Q_Z + 2d + t_4.$$

Move the $-2z_0$ nontarget contribution to the other side to obtain (S61.2). For the heavy triple, $\Phi(S) \geq -3$ by its exact internal-pair formula. Hence

$$\beta \leq 5d + 3. \quad (\text{S61.3})$$

Partition the nonexceptional sources into

$$\Pi = \{v \in W : q'(v) = 0\}, \quad Y = W \setminus \Pi, \quad y = |Y|.$$

Vertices of Π are whole of outdegree α in D' ; all missing pairs of D' lie inside $Y \cup Z$. Moreover $y \leq N + t_2/2$. The integer light-target charging argument of Section S57, applied with sources W and baseline α , gives

$$M(D') \leq \mathcal{D} + \frac{t_2}{2} + \sum_{v \in Y} |U_{D'}(v)| + d. \quad (\text{S61.4})$$

The additional d accounts for the term $[x \in Z]$ in $\mu'(x) \leq q'(x) + [x \in Z]$. The vertices of Z_0 have missing degree zero and incur no missing-pair charge. No full-graph capacity assertion is used in this charging step.

Write

$$A_Y = \sum_{v \in Y} e(v), \quad J_Y = \sum_{v \in Y} |N_D^+(v) \cap S|, \quad A'_Y = \sum_{v \in Y} a'(v).$$

Then $A'_Y = A_Y + y - J_Y$. The exact degree count in D' gives

$$M(D') = \delta + d - A'_Y.$$

The far-set transfer, retaining the original excess, gives

$$\sum_{v \in Y} |U_{D'}(v)| = 3y - 2A_Y + \sum_Y b(v) - \sum_Y |S \cap U_D(v)| + \sum_Y |E_S(v)| \leq 3y - 2A_Y + t_3 + t_4.$$

Substitution into (S61.4) yields

$$\delta + A_Y + J_Y \leq \mathcal{D} + \frac{t_2}{2} + 4y + t_3 + t_4 \leq 4N + \mathcal{D} + \frac{5t_2}{2} + t_3 + t_4 \leq 4\beta.$$

In fact the retained version is

$$\delta + A_Y + J_Y \leq 4\beta - 3\mathcal{D} - \frac{3t_2}{2} - 3t_3 - 3t_4.$$

Dropping the nonnegative terms and using (S61.3) proves (S61.1). If $d = 0$, the deletion is admissible for the heavy-triple obstruction, which rules out $\delta \geq 10$. $\square$

Theorem S61.2 (A residual-dependent restriction on four heavy roots). *Assume $\delta \geq 10$ and $H = 4$. Let m count minimum-degree vertices in K , let L count nonwhole vertices outside K , and put $E = m + 2L$. Then $E \geq 1$ and*

$$\delta \leq 20(m + 2L) + 12 \leq 40R + 12. \quad (\text{S61.5})$$

Consequently

$$M \geq \left\lceil \frac{81\delta + 108}{80} \right\rceil \quad \text{when } H = 4.$$

Proof. For a heavy root, the general predecessor bound gives $15c \geq \delta + 4$. If some heavy root has $c \leq E$, then $\delta \leq 15E - 4 \leq 20E + 12$.

Otherwise every heavy root has more than E minimum-degree predecessors. Excluding the at most E bad witnesses described in Theorem S59.2 leaves good whole minimum-degree predecessors at every root. Use the *full* oriented relation witnessed by all such good sources: $i \rightarrow j$ if a good source points to r_i and is far from r_j . It has positive weighted outdegree at every root and no digon, since both witnesses of a putative digon would be whole.

By Lemma S57.4, choose three roots S whose outneighborhoods in this relation are pairwise disjoint. A good minimum-degree source cannot point to two chosen roots: its nonempty far certified core set would place some far root in both outneighborhoods. Thus only bad minimum sources can lose two arcs into S , and their number d is at most E . The universal heavy-capacity bound at $H = 4$ limits every vertex to two outgoing arcs into the full heavy set. All other minimum sources lose at most one arc, and nonminimum sources retain outdegree at least $\delta - 1$. Lemma S61.1 therefore gives $\delta \leq 20d + 12 \leq 20E + 12$. For $\delta \geq 10$ that lemma also gives $d \geq 1$. If $E = 0$, the first case is impossible and the second would have $d = 0$, so $E \geq 1$.

Finally $2m + L \leq R$ implies $m + 2L \leq 2R$. This proves (S61.5). Substitute $R = 2M - (2\delta + 3)$ and rearrange to obtain $80M \geq 81\delta + 108$, giving the stated integer bound. $\square$

Part VI

Verification, attribution, and retained evidence

S62 Reproducibility and evidence inventory

The certificate tables for cores of orders one through five are supplied in the common verification directory. Run the two independent Node.js verifiers there to check all 59 809 cores and all 462 404 nonempty pattern inequalities. Including the empty pattern gives 522 213. The checker uses exact integer arithmetic and does not require the solver used to find forty of the weights. The complete primary verifier is printed below; the second implementation and table hashes are included in the source archive.

```

1  "use strict";
2  // Solver-independent complete verification of the generated integer witnesses.
3  // This reconstructs incoming patterns using outgoing rows, not generator helpers.
4  const fs = require("fs");
5  const path = require("path");
6  function requireThat(ok, message) { if (!ok) throw new Error(message); }
7  function decode(n, code) {
8    const rows = Array.from({length:n}, ()=>Array(n).fill(false));
9    for (let a=0;a<n;a++) for (let b=a+1;b<n;b++) {
10      const trit=code%3; code=Math.floor(code/3);
11      if(trit===1) rows[a][b]=true;
12      if(trit===2) rows[b][a]=true;
13    }
14    return rows;
15  }
16  const results=[];
17  for (const n of [1,2,3,4,5]) {
18    const source=path.join(__dirname,`incoming_core_certificates_n${n}.json`);
19    const artifact=JSON.parse(fs.readFileSync(source,"utf8"));
20    const total=3**((n*(n-1)/2));
21    requireThat(artifact.n===n && artifact.certificates.length===total,`Incomplete order-${n} certificate coverage`);
22    let inequalities=0,largest=0;
23    for(let code=0;code<total;code++) {
24      const adj=decode(n,code), w=artifact.certificates[code];
25      requireThat(w.length===n&&w.every(x=>Number.isSafeInteger(x)&&x>=0)&&w.some(x=>x>0)&&Number.isSafeInteger(w.reduce((s,x)=>s+x,0)),`Bad vector at ${n}/${code}`);
26      largest=Math.max(largest,...w);
27      for(let mask=1;mask<(1<<n);mask++) {
28        let hasCommonOutneighbor=false;
29        for(let j=0;j<n;j++) {
30          let allPointToJ=true;
31          for(let i=0;i<n;i++) if((mask&(1<<i))&&!adj[i][j]) allPointToJ=false;
32          if(allPointToJ) hasCommonOutneighbor=true;
33        }
34        if(!hasCommonOutneighbor) continue;
35        let inside=0,externalIncoming=0;
36        for(let i=0;i<n;i++) {
37          if(mask&(1<<i)) inside+=w[i];
38          else {
39            let pointsIntoPattern=false;
40            for(let j=0;j<n;j++) if((mask&(1<<j))&&adj[i][j]) pointsIntoPattern=true;

```

```

41         if(pointsIntoPattern) externalIncoming+=w[i];
42     }
43 }
44     requireThat(externalIncoming>=inside,`Inequality failed at n=${n},code=${code},P=${
mask}`);
45     inequalities++;
46 }
47 }
48 results.push({n,graphs:total,inequalities,maximumIntegerWeight:largest,status:"PASS"});
49 }
50 console.log(JSON.stringify({status:"PASS",orders:results,graphs:results.reduce((s,r)=>s+r
.graphs,0),inequalities:results.reduce((s,r)=>s+r.inequalities,0),arithmetic:"Exact
safe integers only; no solver, numerical tolerance, or imported graph code.",scope:"
Complete certificate verification for every labeled oriented CORE of orders one
through five. The transfer to arbitrary-order graphs is a separate mathematical
theorem."},null,2));

```

Listing 1: Complete integer certificate checker.

S63 Historical computations and source attribution

The formal claims of this manuscript are the proved statements in the article and this supplement. Earlier bounded searches, solver timeouts, and local branch logs are indexed in the accompanying coverage inventory. The archived minimum-degree-seven work contains two specified branches, with 30 and 610 leaf records respectively; these counts do not imply a proof of the entire minimum-degree-seven case. Solver logs and reproducible source are not automatically formal unsatisfiability proof objects.

An imported excess-measurement manuscript attributed to Daiki Tahara is external related material. Its definitions and search reports are not presented as original theorems of this article. Likewise, unverified repository and preprint claims from the exploratory notes are not used as unproved premises in the present proof chain. The source-restricted deletion identity needed for the dense-boundary predecessor estimates is derived explicitly in its own section.

The whole conjecture remains unresolved by this work. In particular, strictly negative completed gaps, universal pattern-certificate existence, and arbitrary compatible heavy-source choices are open steps in these arguments. Exact control graphs verify failures of stronger shortcuts; each such graph has Seymour vertices elsewhere and is not a counterexample.

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